
Simplified and Generalized Masked Diffusion for Discrete Data

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Abstract

Masked (or absorbing) diffusion is actively explored as an alternative to autoregressive models for generative modeling of discrete data. However, existing work in this area has been hindered by unnecessarily complex model formulations and unclear relationships between different perspectives, leading to suboptimal parameterization, training objectives, and ad hoc adjustments to counteract these issues. In this work, we aim to provide a simple and general framework that unlocks the full potential of masked diffusion models. We show that the continuous-time variational objective of masked diffusion models is a simple weighted integral of cross-entropy losses. Our framework also enables training generalized masked diffusion models with state-dependent masking schedules. When evaluated by perplexity, our models trained on OpenWebText surpass prior diffusion language models at GPT-2 scale and demonstrate superior performance on 4 out of 5 zero-shot language modeling tasks. Furthermore, our models vastly outperform previous discrete diffusion models on pixel-level image modeling, achieving 2.75 (CIFAR-10) and 3.40 (ImageNet 64×64) bits per dimension that are better than autoregressive models of similar sizes. Our code is available at <https://github.com/google-deepmind/md4>.

1 Introduction

Since their inception [1, 2, 3], diffusion models have emerged as the workhorse for generative media, achieving state-of-the-art in tasks such as image synthesis [4, 5, 6], audio [7, 8] and video generation [9, 10, 11, 12, 13]. The majority of existing successes are for continuous state space diffusions. While diffusion models have been extended to discrete state spaces [1, 14, 15] and have been successfully applied to applications ranging from graph generation [16], text-to-sound generation [17] or protein design [18], they remain not as widely used as their continuous counterparts as they are not competitive with autoregressive models in important domains such as text modeling. This has motivated the development of continuous space diffusion models where the discrete data are embedded in the Euclidean space [19, 20, 21, 22, 23] or the simplex [24, 25, 26, 27, 28]. We believe that one of the reasons for the limited success of discrete diffusions is that they have been hindered by fairly complex formulations and training objectives. This paper is a step towards closing this gap.

In this work, we focus on “masked” (or “absorbing”) diffusions, a discrete diffusion formulation first presented by Austin et al. [14], and later explored by the literature from various perspectives [29, 30, 31, 32]. We follow here a continuous-time framework which has proven very useful to improve the training and understanding of continuous state space diffusions [see e.g., 3, 33, 34]. We make several technical contributions which simplify the training of these models and improve significantly their performance. Our contributions are as follows:

- Using elementary arguments, we establish several properties for the forward process induced by this model and its corresponding time reversal, improving our understanding of this model class.

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离散数据的简化和广义掩蔽扩散

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抽象的

人们正在积极探索掩蔽（或吸收）扩散作为离散数据生成建模的自回归模型的替代方案。然而，该领域的现有工作受到不必要的复杂模型公式和不同视角之间不明确关系的阻碍，导致参数化、培训目标和临时调整来解决这些问题。在这项工作中，我们的目标是提供一个简单而通用的框架，以释放掩蔽扩散模型的全部潜力。我们证明了掩蔽扩散模型的连续时间变分目标是交叉熵损失的简单加权积分。我们的框架还可以使用状态相关的掩蔽时间表来训练广义掩蔽扩散模型。通过困惑度进行评估时，我们在 Open WebText 上训练的模型在 GPT-2 规模上超越了先前的扩散语言模型，并且在 5 个零样本语言建模任务中的 4 个上表现出了卓越的性能。此外，我们的模型在像素级图像建模方面远远优于以前的离散扩散模型，每维达到 2.75 (CIFAR-10) 和 3.40 (ImageNet 64×64) 位，优于类似大小的自回归模型。我们的代码可在 <https://github.com/google-deepmind/md4> 获取。

1 简介

自诞生以来[1,2,3]，扩散模型已成为生成媒体的主力，在图像合成[4,5,6]、音频[7,8]和视频生成[9,10,11,12,13]等任务中实现了最先进的技术。大多数现有的成功都是针对连续状态空间扩散。虽然扩散模型已扩展到离散状态空间[1,14,15]，并已成功应用于图形生成[16]、文本到声音生成[17]或蛋白质设计[18]等应用，但它们仍然没有像连续模型那样广泛使用，因为它们在文本建模等重要领域与自回归模型不具有竞争力。这推动了连续空间扩散模型的发展，其中离散数据嵌入欧几里德空间 [19,20,21,22,23] 或单纯形 [24,25,26,27,28] 中。我们认为，离散扩散成功有限的原因之一是它们受到相当复杂的公式和培训目标的阻碍。本文是朝着缩小这一差距迈出的一步。

在这项工作中，我们关注“掩蔽”（或“吸收”）扩散，这是由 Austin 等人首次提出的离散扩散公式。[14]，后来通过文献从不同的角度进行了探索[29,30,31,32]。我们在这里遵循一个连续时间框架，该框架已被证明对于改善连续状态空间扩散的训练和理解非常有用[参见例如 3、33、34]。我们做出了多项技术贡献，简化了这些模型的训练并显著提高了它们的性能。我们的贡献如下：

- 使用基本参数，我们为模型及其相应的时间反转引起的前向过程建立了几个属性，从而提高了我们对模型类的理解。

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- We provide a remarkably simple expression of the Evidence Lower Bound (ELBO) for masked diffusion models, showing that it corresponds to a weighted integral over time of cross-entropy losses. Similarly to continuous space diffusions [33], this objective can be rewritten in terms of signal-to-noise ratio and exhibits invariance properties.

- We develop a unifying understanding of previously proposed continuous-time discrete diffusion models [29, 32, 35], revealing the changes they made to our ELBO objective and/or model parameterization. We show that these changes either lead to expensive model evaluations, or large variance in training, or breaking the consistency between forward and reverse processes.

- On GPT-2 scale text modeling and pixel-level image modeling tasks, masked diffusions trained using our simple ELBO objective outperform previous proposals, leading to the best likelihood and zero-shot transfer performance among discrete diffusion models.

- Finally, based on our simplified masked diffusion formulation, we propose a generalized masked diffusion model that allows state-dependent masking schedules. This generalized masked diffusion model further improves predictive performance measured by test likelihoods.

Concurrent work by Ou et al. [36] and Sahoo et al. [37] derives a similar simplified expression of the ELBO. Ou et al. [36]’s derivation relies on an observation similar to the one we made in Proposition 1.

2 Masked Diffusion

Consider a sentence where we progressively replace each word with a special mask token, transforming the sentence into a sequence of masks. Our goal is to train a generative model that reverses this process, effectively turning a sentence of masks back into meaningful text. More formally, assume our data consists of tokens from a finite discrete state space with m possible states, represented by integers $0, 1, \dots, m-1$ and their corresponding one-hot vectors $e_0, e_1, \dots, e_{m-1}$. To accommodate the masking process, we augment this space with an additional mask state, denoted by the index m . The masking process transitions each token to the mask state at a random time. This process, known as the forward process, is applied independently to each token (e.g., each word), progressively converting the data into a sequence of mask tokens. By learning to reverse this masking process, we create a generative model capable of producing coherent discrete data.

Discrete-time forward process. We start with the case of a single token and later expand to multiple dimensions. We define the forward process as a Markovian sequence of discrete random variables x_t indexed by time t , where t runs from 0 to 1. Throughout the work, we abuse the notation such that x_t can be either an integer or its corresponding one-hot vector, whenever it is clear from the context. We divide $[0, 1]$ into T intervals, and let $s(i) = (i-1)/T$, $t(i) = i/T$. Following Austin et al. [14], the state transition between $[s(i), t(i)]$ is determined by a transition matrix of size $(m+1) \times (m+1)$: $Q_i = (1 - \beta_i)I + \beta_i \mathbf{1} e_m^\top$, where $\mathbf{1}$ is an all-one vector of size $m+1$, e_m represents a one-hot vector where element at index m is 1. Each entry $[Q_i]_{jk}$ denotes the probability of transition from the state j to the state k :

$$[Q_i]_{jk} = q(x_{t(i)} = k | x_{s(i)} = j) = (1 - \beta_i)\delta_{jk} + \beta_i\delta_{km}.$$

This means that, with probability $1 - \beta_i$, $x_{t(i)} = x_{s(i)}$, otherwise it jumps to the mask state. Given the above transition matrix, the marginal distribution at time $t(i)$ given x_0 is

$$q(x_{t(i)} | x_0) = \text{Cat}(x_{t(i)}; \bar{Q}_i^\top x_0) = x_0^\top \bar{Q}_i x_{t(i)}.$$

Here, we use $\text{Cat}(x; p)$ to denote a Categorical distribution where p is the vector of probabilities of being in each category, and $\bar{Q}_i \triangleq \prod_{j=1}^i Q_j = \alpha_i I + (1 - \alpha_i) \mathbf{1} e_m^\top$ for $\alpha_i = \prod_{j=1}^i (1 - \beta_j)$. We expect α_T to become very small or zero for a sufficiently large T such that $q(x_1 | x_0)$ for any x_0 will become a delta mass at the mask state.

Continuous-time limit. We can define a continuous-time forward process by taking a limit of the above discrete-time process. We first specify a continuous function $\beta(t)$ such that $\beta_i = \beta(t(i))/T$. We then let $T \rightarrow \infty$ in the discrete-time process and compute the limit of $\bar{Q}_i$ (proved in Austin et al. 14, Appendix A.6, see also App. A) as

$$\bar{Q}(t) \triangleq \lim_{T \rightarrow \infty} \bar{Q}_i = \alpha_t I + (1 - \alpha_t) \mathbf{1} e_m^\top, \text{ where } \alpha_t \triangleq \exp\left(-\int_0^t \beta(s) ds\right), \quad (1)$$

- 我们为掩蔽扩散模型提供了一个非常简单的证据下界（ELBO）表达式，表明它对应于交叉熵损失随时间的加权积分。与连续空间扩散[33]类似，该目标可以用信噪比重写，并表现出不变性。

- 我们对之前提出的连续时间离散扩散模型 [29,32,35] 有了统一的理解，揭示了它们对我们的 ELBO 目标和/或模型参数化所做的改变。我们表明，这些变化要么导致昂贵的模型评估，要么导致训练出现巨大差异，或者破坏正向和反向过程之间的一致性。

- 在 GPT-2 规模文本建模和像素级图像建模任务中，使用我们简单的 ELBO 目标训练的掩模扩散优于之前的建议，从而在离散扩散模型中实现最佳似然和零样本传输性能。

- 最后，基于我们简化的掩蔽扩散公式，我们提出了一种广义掩蔽扩散模型，该模型允许状态相关的掩蔽时间表。这种广义掩蔽扩散模型进一步提高了通过测试可能性衡量的预测性能。

Ou 等人的同时工作。[36] 和 Sahoo 等人。[37] 推导了 ELBO 的类似简化表达式。欧等人。[36] 的推导依赖于类似于我们在命题 1 中所做的观察。

2 掩模扩散

考虑一个句子，我们逐步用特殊的掩码标记替换每个单词，将句子转换为一系列掩码。我们的目标是训练一个生成模型来逆转这个过程，有效地将面具句子转回有意义的文本。更正式地说，假设我们的数据由来自有限离散状态空间的标记组成，具有 m 可能状态，由整数 $0, 1, \dots, m-1$ 及其相应的 one-hot 向量 $e_0, e_1, \dots, e_{m-1}$ 表示。为了适应掩码过程，我们用一个额外的掩码状态来扩充这个空间，用索引 m 表示。屏蔽过程在随机时间将每个令牌转变为屏蔽状态。该过程称为前向过程，独立应用于每个标记（例如每个单词），逐步将数据转换为一系列掩码标记。通过学习逆转这种掩蔽过程，我们创建了一个能够生成连贯离散数据的生成模型。

离散时间前向过程。我们从单个 token 的情况开始，然后扩展到多个维度。我们将前向过程定义为由时间 t 索引的离散随机变量 x_t 的马尔可夫序列，其中 t 从 0 到 1。在整个工作中，我们滥用这种表示法，使得 x_t 可以是整数或其相应的单热向量，只要从上下文中可以清楚地看出。我们将 $[0, 1]$ 划分为 T 区间，并令 $s(i) = (i-1)/T$ 、 $t(i) = i/T$ 。继奥斯汀等人之后。[14]， $[s(i), t(i)]$ 之间的状态转换由大小为 $(m+1) \times (m+1)$ 的 $Q_i = (1 - \beta_i)I + \beta_i \mathbf{1}e_m^\top$ 的转移矩阵确定，其中 $\mathbf{1}$ 是大小为 $m+1$ 的全一向量， e_m 表示索引 m 处的元素为 1 的单热向量。每个条目 $[Q_i]_{jk}$ 表示从状态 j 转移到状态 k 的概率：

$$[Q_i]_{jk} = q(x_{t(i)} = k | x_{s(i)} = j) = (1 - \beta_i)\delta_{jk} + \beta_i\delta_{km}.$$

这意味着，以 $1 - \beta_i$ 的概率， $x_{t(i)} = x_{s(i)}$ ，否则跳转到掩码状态。给定上述转移矩阵，给定 x_0 时，时间 $t(i)$ 的边缘分布为

$$q(x_{t(i)} | x_0) = \text{Cat}(x_{t(i)}; \bar{Q}_i^\top x_0) = x_0^\top \bar{Q}_i x_{t(i)}.$$

在这里，我们使用 $\text{Cat}(x; p)$ 表示分类分布，其中 p 是每个类别中存在的概率向量，而 $\bar{Q}_i \triangleq \prod_{j=1}^i Q_j = \alpha_i I + (1 - \alpha_i) \mathbf{1}e_m^\top$ 表示 $\alpha_i = \prod_{j=1}^i (1 - \beta_j)$ 。我们期望对于足够大的 T ， α_T 变得非常小或为零，使得任何 x_0 的 $q(x_1 | x_0)$ 将成为掩模状态下的增量质量。

连续时间限制。我们可以通过对上述离散时间过程进行限制来定义连续时间前向过程。我们首先指定一个连续函数 $\beta(t)$ 使得 $\beta_i = \beta(t(i))/T$ 。然后我们让 $T \rightarrow \infty$ 进入离散时间过程并计算 Austin 等人证明的 $\bar{Q}_i$ 的极限。14，附录 A.6，另见附录。A) 作为

$$\bar{Q}(t) \triangleq \lim_{T \rightarrow \infty} \bar{Q}_i = \alpha_t I + (1 - \alpha_t) \mathbf{1}e_m^\top, \text{ where } \alpha_t \triangleq \exp\left(-\int_0^t \beta(s) ds\right), \quad (1)$$

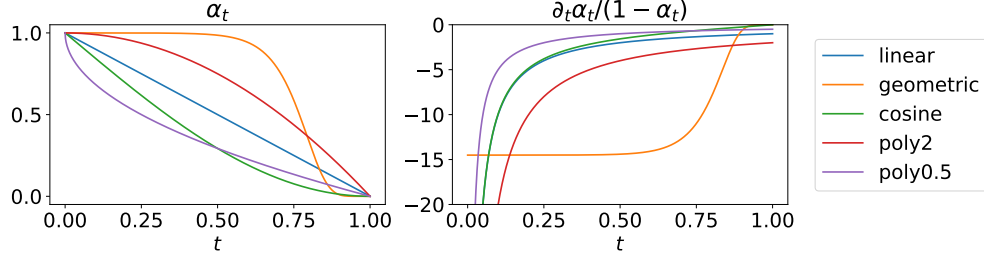


Figure 1: Masking schedules in the literature: (Left) α_t ; (Right) weight of the cross-entropy loss w.r.t. t ; Equations for these schedules are given in Tab. 4 in Appendix.

so that $q(x_t|x_0) = \text{Cat}(x_t; \bar{Q}(t)^\top x_0)$. For two arbitrary times, $0 \leq s < t \leq 1$, the transition distribution that is compatible with the above marginal (i.e., $q(x_t|x_0) = \sum_{x_s} q(x_t|x_s)q(x_s|x_0)$) is

$$q(x_t|x_s) = \text{Cat}(x_t; \bar{Q}(s,t)^\top x_s), \text{ where } \bar{Q}(s,t) \triangleq \bar{Q}(s)^{-1}\bar{Q}(t) = \frac{\alpha_t}{\alpha_s}I + \left(1 - \frac{\alpha_t}{\alpha_s}\right)\mathbf{1}e_m^\top.$$

Note that Austin et al. [14] did not derive this explicit form of transition matrix between two arbitrary time s and t , which appeared later in Zhao et al. [38] concurrently with our work.

Masking schedules. From the definition of α_t , we have that $\alpha_0 = 1$. And similar to the discrete-time formulation, we would like α_1 be zero or very close to zero. We provide a summary of masking schedules from literature that satisfy these properties in Fig. 1. The linear schedule was proposed in Sohl-Dickstein et al. [1] for binary variables and then re-derived by Austin et al. [14] from mutual information for discrete-time models. The geometric schedule α_t is plotted for $\beta_{\min} = 10^{-5}$ and $\beta_{\max} = 20$. It was first used for continuous diffusions [3] and then for discrete by Lou et al. [32]. The cosine schedule was originally proposed in MaskGIT [39], an iterative unmasking generative model inspired by diffusion. This schedule has the property of slowing down the unmasking process at the beginning of the reverse generation. Aligning with their observation, we find that this results in a lower chance of conflicting tokens being unmasked simultaneously at the start of generation, thereby enhancing the overall generation quality.

Time reversal of the forward process given x_0 . The analytic property of our forward process allows to compute many quantities of interest in closed form. One such quantity frequently used in diffusion models is the time reversal of the forward process given x_0 : $q(x_s|x_t, x_0)$ for $s \leq t$. We derive it in App. C as

$$q(x_s|x_t, x_0) = \text{Cat}(x_s; \bar{R}^{x_0}(t, s)^\top x_t), \text{ where } \bar{R}^{x_0}(t, s) = I + \frac{\alpha_s - \alpha_t}{1 - \alpha_t}e_m(x_0 - e_m)^\top.$$

From the transition matrix $\bar{R}^{x_0}(t, s) \in \mathbb{R}^{(m+1) \times (m+1)}$ we can see the reverse process conditioned on x_0 has a very simple logic—if x_t is a mask, with probability $\frac{\alpha_s - \alpha_t}{1 - \alpha_t}$, it will jump to the state x_0 at time s , otherwise it will stay masked. Once x_t is unmasked, it remains in the same state until the end.

3 Model and Objective

For a discrete-time masked diffusion process, we define our generative model by approximately reversing the forward transitions using a reverse model $p_\theta(x_s|x_t)$. One way to define this model is

$$p_\theta(x_s|x_t) \triangleq q(x_s|x_t, \mu_\theta(x_t, t)), \quad (2)$$

where $\mu_\theta(x_t, t) \in \Delta^{m+1}$ is a probability vector parametrized by a neural network f_θ with a softmax applied to the output logits (note the m -th output is forced to 0 since the clean data cannot be masks):

$$\mu_\theta(x_t, t) = \begin{cases} \text{softmax}(f_\theta(x_t, t)) & x_t = m, \\ x_t & x_t \neq m. \end{cases} \quad (3)$$

This is known as mean-parameterization since it leverages a prediction model for the mean of x_0 . A matrix-form depiction of $p_\theta(x_s|x_t)$ is shown in Fig. 7 (right). In fact, we can select a time-invariant parametrization $\mu_\theta(x_t, t) = \mu_\theta(x_t)$ as [36] showed that $p(x_0|x_t)$ given $x_t = x$ is identical for any t .

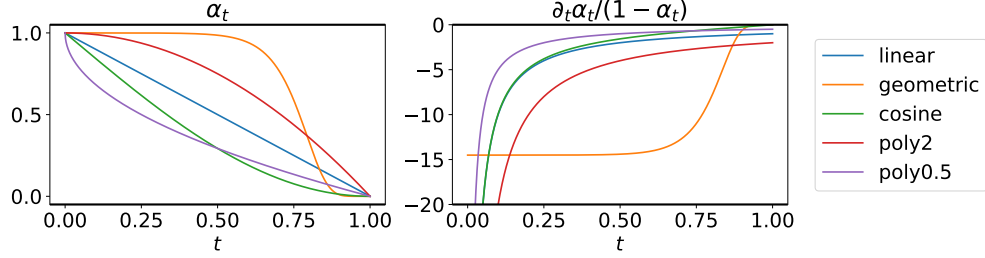


图 1: 文献中的掩蔽方案: (左) α_t ; (右) 交叉熵损失的权重 t ; 表中给出了这些时间表的方程式。4 见附录。

这样 $q(x_t|x_0) = \text{Cat}(x_t; \bar{Q}(t)^\top x_0)$ 。对于任意两个时间 $0 \leq s < t \leq 1$, 与上述边际 (即 $q(x_t|x_0) = \sum_{x_s} q(x_t|x_s)q(x_s|x_0)$) 兼容的转移分布为

$$q(x_t|x_s) = \text{Cat}(x_t; \bar{Q}(s,t)^\top x_s), \text{ where } \bar{Q}(s,t) \triangleq \bar{Q}(s)^{-1} \bar{Q}(t) = \frac{\alpha_t}{\alpha_s} I + \left(1 - \frac{\alpha_t}{\alpha_s}\right) \mathbf{1} e_m^\top.$$

请注意, 奥斯汀等人。[14]没有推导出两个任意时间 s 和 t 之间的这种明确形式的转移矩阵, 这后来出现在Zhao等人的论文中。[38]与我们的工作同时进行。

屏蔽时间表。根据 α_t 的定义, 我们得到 $\alpha_0 = 1$ 。与离散时间公式类似, 我们希望 α_1 为零或非常接近于零。我们在图 1 中总结了满足这些属性的文献中的掩蔽方案。Sohl-Dickstein 等人提出了线性方案。[1] 对于二元变量, 然后由 Austin 等人重新推导。[14]来自离散时间模型的互信息。几何调度 α_t 是针对 $\beta_{\min} = 10^{-5}$ 和 $\beta_{\max} = 20$ 绘制的。它首先用于连续扩散 [3], 然后由 Lou 等人用于离散扩散。[32]。余弦调度最初是在 MaskGIT [39] 中提出的, 这是一种受扩散启发的迭代揭蔽生成模型。该时间表具有在反向生成开始时减慢解锁过程的特性。与他们的观察一致, 我们发现这导致在生成开始时同时揭露冲突代币的可能性较低, 从而提高了整体生成质量。

给定 x_0 的前向过程的时间反转。我们的正向过程的分析属性允许以封闭形式计算许多感兴趣的量。扩散模型中经常使用的此类量之一是给定 x_0 : $q(x_s|x_t, x_0)$ for $s \leq t$ 的前向过程的时间反转。我们在App中导出它。C为

$$q(x_s|x_t, x_0) = \text{Cat}(x_s; \bar{R}^{x_0}(t,s)^\top x_t), \text{ where } \bar{R}^{x_0}(t,s) = I + \frac{\alpha_s - \alpha_t}{1 - \alpha_t} e_m (x_0 - e_m)^\top.$$

从转移矩阵 $\bar{R}^{x_0}(t,s) \in \mathbb{R}^{(m+1) \times (m+1)}$ 我们可以看到以 x_0 为条件的逆过程有一个非常简单的逻辑 - 如果 x_t 是一个掩码, 则概率为 $\frac{\alpha_s - \alpha_t}{1 - \alpha_t}$, 它将在 s 时刻跳转到状态 x_0 , 否则它将保持掩码状态。一旦 x_t 被解除屏蔽, 它就会保持相同的状态直到结束。

3 模型与目标

对于离散时间掩蔽扩散过程, 我们通过使用反向模型 $p_\theta(x_s|x_t)$ 近似反转前向转换来定义生成模型。定义该模型的一种方法是

$$p_\theta(x_s|x_t) \triangleq q(x_s|x_t, \mu_\theta(x_t, t)), \quad (2)$$

其中 $\mu_\theta(x_t, t) \in \Delta^{m+1}$ 是由神经网络 f_θ 参数化的概率向量, 并对输出 logits 应用了 softmax (请注意, 第 m 个输出被迫为 0, 因为干净的数据不能是掩码):

$$\mu_\theta(x_t, t) = \begin{cases} \text{softmax}(f_\theta(x_t, t)) & x_t = m, \\ x_t & x_t \neq m. \end{cases} \quad (3)$$

这称为均值参数化, 因为它利用 x_0 均值的预测模型。 $p_\theta(x_s|x_t)$ 的矩阵形式如图 7 (右) 所示。事实上, 我们可以选择一个时不变的参数化 $\mu_\theta(x_t, t) = \mu_\theta(x_t)$, 因为[36]表明, 给定 $x_t = x$ 的 $p(x_0|x_t)$ 对于任何 t 都是相同的。

Besides $p_\theta(x_s|x_t)$, we also need to specify $p(x_0|x_{t(1)})$ and the prior distribution $p(x_{t(T)}) = p(x_1)$. Following the practice in continuous diffusion models [33], we choose $p(x_0|x_{t(1)}) \propto q(x_{t(1)}|x_0)$. And since $q(x_1|x_0) \approx \delta_{x_1,m}$ for any x_0 as $\alpha_1 \approx 0$, we set $p(x_1) \approx \delta_{x_1,m}$, see App. E.

We then write out the discrete-time diffusion model objective [1, 2], which is a lower bound of the log marginal likelihood of data x_0 under the model p (known as the Evidence Lower Bound, or ELBO):

$$\log p(x_0) \geq \mathbb{E}_{q(x_{t(1)}|x_0)} [\log p(x_0|x_{t(1)})] - \text{KL}(q(x_1|x_0)||p(x_1)) - \mathcal{L}_T,$$

where $\mathcal{L}_T = \sum_{i=2}^T \mathbb{E}_{q(x_{t(i)}|x_0)} [\text{KL}(q(x_{s(i)}|x_{t(i)}, x_0)||p_\theta(x_{s(i)}|x_{t(i)}))]$. For the above choices of the prior distribution, the term $\text{KL}(q(x_1|x_0)||p(x_1))$ becomes zero. Under the reverse model (2), the KL divergence terms in $\mathcal{L}_T$ becomes (proof in App. D)

$$\text{KL}(q(x_s|x_t, x_0)||p_\theta(x_s|x_t)) = -\frac{\alpha_s - \alpha_t}{1 - \alpha_t} \delta_{x_t,m} \cdot x_0^\top \log \mu_\theta(x_t, t),$$

which is a simple cross-entropy loss between the predicted logits and the clean data. In App. D, we show that $\mathcal{L}_T$ is a Riemann sum and is lower bounded by the corresponding continuous integral:

$$\mathcal{L}_\infty \triangleq \lim_{T \rightarrow \infty} \mathcal{L}_T = \int_{t(1)}^1 \frac{\alpha'_t}{1 - \alpha_t} \mathbb{E}_{q(x_t|x_0)} [\delta_{x_t,m} \cdot x_0^\top \log \mu_\theta(x_t, t)] dt, \quad (4)$$

where α'_t denotes the derivative of α_t with respect to t . Therefore, we can obtain an ELBO that is tighter than that of any finite T by pushing $T \rightarrow \infty$. This ELBO can be further simplified by letting $t(1) \rightarrow 0$. As a result, $\mathbb{E}_{q(x_{t(1)}|x_0)} [\log p(x_0|x_{t(1)})]$ goes to 0 and the ELBO becomes $-\mathcal{L}_\infty$.

For continuous state-space diffusions, the ELBO depends on the signal-to-noise ratio (SNR) at its endpoints but is otherwise invariant to the noise schedule [33]. We establish here a similar result for discrete diffusions. Consider choosing $\alpha_t = \sigma(\lambda_t)$, where σ represents the sigmoid function $\sigma(x) = \frac{1}{1+e^{-x}}$. In this context, the log-SNR is defined by $\lambda_t = \log \frac{\alpha_t}{1-\alpha_t} = \log\text{-SNR}(t)$. By making a change of variables in (4) to make everything a function of the log-SNR, we obtain

$$\mathcal{L}_\infty = \int_{\lambda_{t(1)}}^{\lambda_1} \sigma(\lambda) \mathbb{E}_{\tilde{q}(x_\lambda|x_0)} [\delta_{x_\lambda,m} \cdot x_0^\top \log \tilde{\mu}_\theta(x_\lambda, \lambda)] d\lambda.$$

where $\tilde{\mu}_\theta(x, \lambda) := \mu_\theta(x, t)$ and $\tilde{q}(x_\lambda|x_0) := q(x_t|x_0)$ for $t = \log\text{-SNR}^{-1}(\lambda)$. This shows that the only effect α_t has on the loss is through the values of the SNR at the endpoints. Still, because we draw uniform samples of t to estimate the integral, the choice of masking schedule affects the variance.

Multidimensional data. In the previous sections, x_t was assumed to be a single discrete token. To extend the method to multidimensional data, let x_t be now a sequence $(x_t^{(1)}, x_t^{(2)}, \dots, x_t^{(N)})$, where each element $x_t^{(n)}$ represents a discrete token. We select a forward process which factorizes across all N tokens: $q(x_t|x_s) = \prod_{n=1}^N q(x_t^{(n)}|x_s^{(n)})$. As a result, the forward marginals $q(x_t|x_0)$ and reversal $q(x_s|x_t, x_0)$ also factorize. In this case, we define the reverse model as $p_\theta(x_s|x_t) \triangleq \prod_{n=1}^N q(x_s^{(n)}|x_t^{(n)}, \mu_\theta^{(n)}(x_t, t))$, where $\mu_\theta(x_t, t)$ is a neural network that takes the full N tokens as input and outputs N probability vectors.² The n -th output $\mu_\theta^{(n)}(x_t, t)$ is a prediction model for $\mathbb{E}[x_0^{(n)}|x_t]$, the mean value of the n -th token. Repeating above derivations gives

$$\mathcal{L}_\infty^{(N)} \triangleq \int_0^1 \frac{\alpha'_t}{1 - \alpha_t} \mathbb{E}_{q(x_t|x_0)} \left[\sum_{n: x_t^{(n)}=m} (x_0^{(n)})^\top \log \mu_\theta^{(n)}(x_t, t) \right] dt. \quad (5)$$

We term our simple masked diffusion model trained with loss (5) **MD4** (Masked Discrete Diffusion for Discrete Data). A single step of MD4 training algorithm is described in Alg. 1 in Appendix.

4 Sampling

We use ancestral sampling from our discrete-time reverse process for generation. We have found this yields slightly higher sample quality compared to other methods such as Euler discretization [29, 32]. For conditional generation tasks such as infilling, we find that the simple approach works best — we keep the conditioning tokens unmasked throughout the generation process. A complete description of the sampling algorithm can be found in Alg. 2 in Appendix.

²We intentionally choose the reverse model to factorize across dimensions because the true reverse transition $q(x_s|x_t)$ factorizes in the continuous-time limit (as s approaches t).

除了 $p_\theta(x_s|x_t)$ 之外，我们还需要指定 $p(x_0|x_{t(1)})$ 和先验分布 $p(x_{t(T)}) = p(x_1)$ 。遵循连续扩散模型[33]的实践，我们选择 $p(x_0|x_{t(1)}) \propto q(x_{t(1)}|x_0)$ 。由于任何 x_0 的 $q(x_1|x_0) \approx \delta_{x_1,m}$ 为 $\alpha_1 \approx 0$ ，因此我们设置 $p(x_1) \approx \delta_{x_1,m}$ ，请参阅 App. E。

然后，我们写出离散时间扩散模型目标 [1, 2]，它是模型 p (下数据 x_0 的对数边际似然的下界，称为证据下界，或 ELBO)：

$$\log p(x_0) \geq \mathbb{E}_{q(x_{t(1)}|x_0)} [\log p(x_0|x_{t(1)})] - \text{KL}(q(x_1|x_0)||p(x_1)) - \mathcal{L}_T,$$

其中 $\mathcal{L}_T = \sum_{i=2}^T \mathbb{E}_{q(x_{t(i)}|x_0)} [\text{KL}(q(x_{s(i)}|x_{t(i)}, x_0)||p_\theta(x_{s(i)}|x_{t(i)}))]$ 。对于先验分布的上述选择，项 $\text{KL}(q(x_1|x_0)||p(x_1))$ 变为零。在逆向模型(2)下， $\mathcal{L}_T$ 中的KL散度项变为 (App.D中的证明)

$$\text{KL}(q(x_s|x_t, x_0)||p_\theta(x_s|x_t)) = -\frac{\alpha_s - \alpha_t}{1 - \alpha_t} \delta_{x_t,m} \cdot x_0^\top \log \mu_\theta(x_t, t),$$

这是预测逻辑和干净数据之间的简单交叉熵损失。在应用程序中。D，我们证明 $\mathcal{L}_T$ 是黎曼和，并且下界为相应的连续积分：

$$\mathcal{L}_\infty \triangleq \lim_{T \rightarrow \infty} \mathcal{L}_T = \int_{t(1)}^1 \frac{\alpha'_t}{1 - \alpha_t} \mathbb{E}_{q(x_t|x_0)} [\delta_{x_t,m} \cdot x_0^\top \log \mu_\theta(x_t, t)] dt, \quad (4)$$

其中 α'_t 表示 α_t 相对于 t 的导数。因此，我们可以通过推 $T \rightarrow \infty$ 来获得比任何有限 T 更紧的ELBO。通过让 $t(1) \rightarrow 0$ 可以进一步简化此ELBO。因此， $\mathbb{E}_{q(x_{t(1)}|x_0)} [\log p(x_0|x_{t(1)})]$ 变为0，ELBO变为 $-\mathcal{L}_\infty$ 。

对于连续状态空间扩散，ELBO 取决于其端点处的信噪比 (SNR)，但在其他方面对噪声调度是不变的 [33]。我们在这里为离散扩散建立了类似的结果。考虑选择 $\alpha_t = \sigma(\lambda_t)$ ，其中 σ 表示 sigmoid 函数 $\sigma(x) = \frac{1}{1+e^{-x}}$ 。在这种情况下， $\log\text{-SNR}$ 由 $\lambda_t = \log \frac{\alpha_t}{1-\alpha_t} = \log\text{-SNR}(t)$ 定义。通过更改 (4) 中的变量以使所有内容成为对数 SNR 的函数，我们得到

$$\mathcal{L}_\infty = \int_{\lambda_{t(1)}}^{\lambda_1} \sigma(\lambda) \mathbb{E}_{\tilde{q}(x_\lambda|x_0)} [\delta_{x_\lambda,m} \cdot x_0^\top \log \tilde{\mu}_\theta(x_\lambda, \lambda)] d\lambda.$$

其中 $\tilde{\mu}_\theta(x, \lambda) := \mu_\theta(x, t)$ 和 $\tilde{q}(x_\lambda|x_0) := q(x_t|x_0)$ 表示 $t = \log\text{-SNR}^{-1}(\lambda)$ 。这表明 α_t 对损耗的唯一影响是通过端点处的 SNR 值。尽管如此，因为我们抽取 t 的统一样本来估计积分，所以掩蔽方案的选择会影响方差。

多维数据。在前面的部分中， x_t 被假定为单个离散标记。为了将该方法扩展到多维数据，让 x_t 现在是一个序列 $(x_t^{(1)}, x_t^{(2)}, \dots, x_t^{(N)})$ ，其中每个元素 $x_t^{(n)}$ 表示一个离散标记。我们选择一个对所有 N 标记进行因式分解的前向过程： $q(x_t|x_s) = \prod_{n=1}^N q(x_t^{(n)}|x_s^{(n)})$ 。因此，前向边缘 $q(x_t|x_0)$ 和反向边缘 $q(x_s|x_t, x_0)$ 也会因式分解。在这种情况下，我们将反向模型定义为 $p_\theta(x_s|x_t) \triangleq \prod_{n=1}^N q(x_s^{(n)}|x_t^{(n)}, \mu_\theta^{(n)}(x_t, t))$ ，其中 $\mu_\theta(x_t, t)$ 是一个神经网络，它将完整的 N 标记作为输入并输出 N 概率向量。² 第 n 个输出 $\mu_\theta^{(n)}(x_t, t)$ 是 $\mathbb{E}[x_0^{(n)}|x_t]$ (第 n 个标记的平均值) 的预测模型。重复以上推导可得

$$\mathcal{L}_\infty^{(N)} \triangleq \int_0^1 \frac{\alpha'_t}{1 - \alpha_t} \mathbb{E}_{q(x_t|x_0)} \left[\sum_{n: x_t^{(n)}=m} (x_0^{(n)})^\top \log \mu_\theta^{(n)}(x_t, t) \right] dt. \quad (5)$$

我们将使用损失 (5) MD4 (离散数据的掩模离散扩散) 训练的简单掩模扩散模型称为。Alg 中描述了 MD4 训练算法的单步。¹ 见附录。

4 取样

我们使用离散时间逆过程中的祖先采样来进行生成。我们发现，与欧拉离散化等其他方法相比，这种方法产生的样本质量略高[29, 32]。对于填充等条件生成任务，我们发现简单的方法效果最好——我们在整个生成过程中保持条件标记的暴露。采样算法的完整描述可以在 Alg 中找到。² 见附录。

²我们有意选择反向模型来跨维度进行因式分解，因为真正的反向转换 $q(x_s|x_t)$ 在连续时间限制中进行因式分解 (当 s 接近 t 时)。

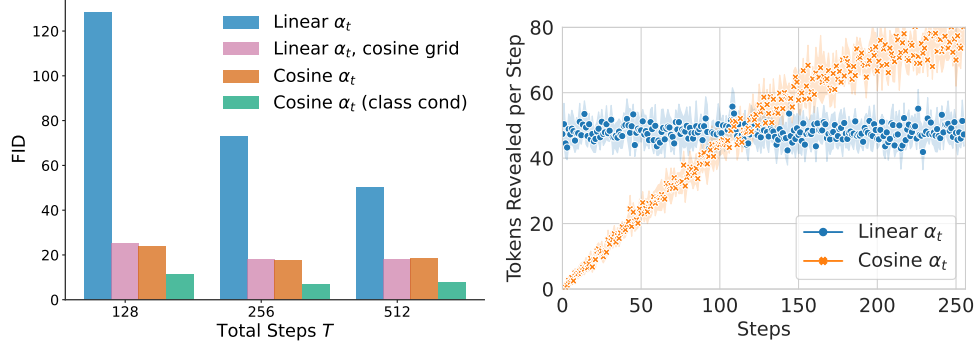


Figure 2: Left: FID evaluation for 50k samples randomly generated from MD4 on pixel-level modeling of ImageNet 64×64 (numbers in Tab. 6). Right: Number of tokens revealed per generation step ($T = 256$). Each image consists of $64 \times 64 \times 3 = 12288$ tokens.

Impact of schedules and discretization. For comparing different sampling configurations, we primarily use the FID score [40] on image datasets as our evaluation metric. We favor it over text generative perplexity³ used in prior work [32], as the latter can be misleadingly reduced by lowering sample diversity [41]. We initially trained our model using the linear schedule, which achieves the best final ELBO overall; however, we found that sampling did not perform well with a standard uniform discretization grid $t(i) = \frac{i}{T}$. We hypothesize that time discretization can lead to conflicts by generating multiple tokens in a single step. We then switched to the cosine schedule (Tab. 4) that slows down unmasking at the beginning of reverse process. This drastically improves the FID on ImageNet 64×64 from 70 to 17 for $T = 256$ steps (Fig. 2, left). Building on this observation, we suggest using a “cosine” discretization grid for sampling in models trained with a linear schedule:

$$t(i) = \cos\left(\frac{\pi}{2}\left(1 - \frac{i}{T}\right)\right). \quad (6)$$

This induces the same discretization in α_t as the cosine schedule with a uniform grid, leading to comparable sample quality, as shown in Fig. 2 (left). In Fig. 2 (right), we plot the number of tokens unmasked per step for linear and cosine schedules with a uniform grid. We believe the cosine schedule performs better because it leverages information redundancy: with more tokens revealed, the remaining tokens become more predictable, reducing conflicts when unmasking them in a single step.

Although these findings were originally developed on images, we find them translate well to text (see Fig. 10). we expect other techniques such as top- p sampling [41], classifier-free guidance [42, 43], and predictor-correctors [29, 44] to further improve sample quality of our models. While we reserve these for future work, we note that the JAX [45] implementation of categorical sampling implicitly truncates small probabilities, creating a similar effect to top- p sampling. See App. G for details.

5 Relation to Existing Work

We discuss how to unify several existing masked diffusion models using our framework.

Continuous-Time Markov Chains (CTMC). To show the connection with the CTMC view presented in Austin et al. [14], Campbell et al. [29], we can write out the forward and reverse masked diffusion using CTMC machinery. To see this, for a short time Δt , given x_0 , the Taylor expansions of our forward and reverse transition matrices at t are

$$\bar{Q}(t, t + \Delta t) = I + Q(t)\Delta t + o(\Delta t) \quad \text{for} \quad Q(t) \triangleq \beta(t)(\mathbf{1}e_m^\top - I), \quad (7)$$

$$\bar{R}^{x_0}(t, t - \Delta t) = I + R^{x_0}(t)\Delta t + o(\Delta t) \quad \text{for} \quad R^{x_0}(t) \triangleq -\frac{\alpha'_t}{1 - \alpha_t}e_m(x_0 - e_m)^\top, \quad (8)$$

where $Q(t)$ and $R^{x_0}(t)$ are known as the *transition rate* matrices. Austin et al. [14] derived the same $Q(t)$ in App. A.6 of their paper. However, they did not explore the reverse process or a

³Perplexity of generated samples scored by a large language model such as GPT-2.

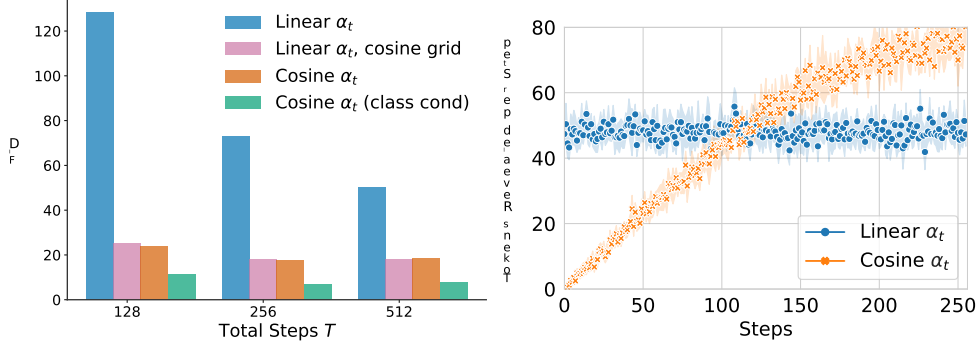


图 2: 左图: 在 ImageNet 64×64 (表 6 中的数字) 的像素级建模上对从 MD4 随机生成的 50 k 样本进行 FID 评估。右: 每个生成步骤显示的代币数量 ($T = 256$)。每个图像由 $64 \times 64 \times 3 = 12288$ 个标记组成。

时间表和离散化的影响。为了比较不同的采样配置，我们主要使用图像数据集上的 FID 分数 [40] 作为我们的评估指标。我们更喜欢它而不是先前工作中使用的文本生成困惑度³ [32]，因为后者可以通过降低样本多样性来误导性地减少[41]。我们最初使用线性计划训练我们的模型，这实现了总体上最好的最终 ELBO；然而，我们发现使用标准均匀离散网格 $t(i) = \frac{i}{T}$ 进行采样效果不佳。我们假设时间离散化可能会通过在一个步骤中生成多个令牌而导致冲突。然后，我们切换到余弦时间表（表 4），这会减慢反向过程开始时的揭露速度。这将 ImageNet 64×64 上的 FID 从 70 大幅提高到 17， $T = 256$ 个步骤（图 2，左）。基于这一观察，我们建议使用“余弦”离散网格在使用线性计划训练的模型中进行采样：

$$t(i) = \cos\left(\frac{\pi}{2}\left(1 - \frac{i}{T}\right)\right). \quad (6)$$

这会在 α_t 中产生与具有均匀网格的余弦计划相同的离散化，从而产生可比较的样本质量，如图 2（左）所示。在图 2（右）中，我们使用统一网格绘制了线性和余弦计划每一步未屏蔽的标记数量。我们相信余弦计划表现更好，因为它利用了信息冗余：随着更多令牌的显示，剩余的令牌变得更加可预测，从而减少了在一步中揭开它们时的冲突。

尽管这些发现最初是在图像上开发的，但我们发现它们可以很好地转化为文本（见图 10）。我们期望其他技术，例如 top- p 采样 [41]、无分类器指导 [42, 43] 和预测校正器 [29, 44] 来进一步提高模型的样本质量。虽然我们为未来的工作保留这些，但我们注意到分类采样的 J AX [45] 实现隐式截断了小概率，从而产生了与 top- p 采样类似的效果。参见应用程序。G 了解详情。

5 与现有工作的关系

我们讨论如何使用我们的框架统一几个现有的掩模扩散模型。

连续时间马尔可夫链 (CTMC)。为了显示与 Austin 等人提出的 CTMC 观点的联系。[14]，坎贝尔等人。[29]，我们可以使用 CTMC 机制写出前向和反向掩模扩散。为了看到这一点，对于短时间 Δt ，给定 x_0 ，我们在 t 处的正向和反向转移矩阵的泰勒展开式为

$$\bar{Q}(t, t + \Delta t) = I + Q(t)\Delta t + o(\Delta t) \quad \text{for} \quad Q(t) \triangleq \beta(t)(\mathbf{1}e_m^\top - I), \quad (7)$$

$$\bar{R}^{x_0}(t, t - \Delta t) = I + R^{x_0}(t)\Delta t + o(\Delta t) \quad \text{for} \quad R^{x_0}(t) \triangleq -\frac{\alpha'_t}{1 - \alpha_t}e_m(x_0 - e_m)^\top, \quad (8)$$

其中 $Q(t)$ 和 $R^{x_0}(t)$ 称为 *transition rate* 矩阵。奥斯汀等人。[14]在App中导出了相同的 $Q(t)$ 。他们论文的 A.6。然而，他们并没有探索相反的过程或

³ 通过 GPT-2 等大型语言模型对生成样本进行评分的困惑度。

continuous-time objective. Campbell et al. [29] derived an alternative ELBO expression using rate matrices, which Kitouni et al. [46] further simplified for absorbing diffusion. In App. H.1, we show how to recover their expression by separating out a constant from our ELBO expression (4) and applying a discrete “integration-by-part”. A key limitation of their expression is that it needs N evaluations of the prediction model $\mu_\theta(\cdot, t)$ to compute an inner summation. To circumvent this computational burden, they used a doubly stochastic estimate. However, this leads to significantly higher variance compared to the analytic cross-entropy (4) which only requires one pass of $\mu_\theta(\cdot, t)$. Please refer to App. H.2 for more details.

Score parameterization. While so far we used a prediction model $\mu_\theta(x_t, t)$ for the mean of clean data given x_t (i.e., mean parameterization), one can choose other ways of parameterizing the reverse model. Lou et al. [32], Benton et al. [35] proposed to parameterize the discrete “score” $s(x_t, t)_j \triangleq \frac{q_t(j)}{q_t(x_t)}$ and introduced a score-based loss for discrete diffusions. In App. H.3, we provide an alternative derivation of their loss which is simpler. We show the link between score and mean parameterizations through the following proposition.

Proposition 1 (Score Parameterization vs. Mean Parameterization). *Let q_t be the marginal distribution of the masked diffusion defined in Sec. 2 at time t . The discrete score $s(x_t, t)_j = \frac{q_t(j)}{q_t(x_t)}$ for a mask state $x_t = m$ and $j \neq m$ can be expressed as*

$$s(m, t)_j = \frac{\alpha_t}{1 - \alpha_t} \mathbb{E}[x_0 | x_t = m]^\top e_j, \text{ which satisfies } \sum_{j \neq m} s(m, t)_j = \frac{\alpha_t}{1 - \alpha_t}. \quad (9)$$

Proposition 1 (proved in App. H.3) implies that a reasonable score model for a mask state is

$$s_\theta(m, t)_j = \frac{\alpha_t}{1 - \alpha_t} \mu_\theta(m, t)_j. \quad (10)$$

Indeed, substituting (10) into the score-based loss of Lou et al. [32], Benton et al. [35] recovers our objective (4). In Lou et al. [32], the score is parameterized as a neural network without enforcing the constraint in (9). This means the learned reverse model can be incompatible with the forward process. We find that our parameterization, which enforces the constraint, leads to more stable training and better results.

Any-order autoregressive models. The continuous-time reverse process of our masked diffusion model can be viewed as an any-order autoregressive model (AO-ARM) [47]. To see this, we reorder the tokens according to the timing of their unmasking events in the reverse process. For all tokens, the cumulative distribution functions (CDFs) of unmasking times $\{\tau_n\}_{n=1}^N$ are identical and satisfy $P(\tau_n \leq t) = P(x_t^{(n)} = m) = 1 - \alpha_t$. As a result, the ordering is uniformly random across all possible arrangements, and the token prediction during each unmasking event represents a prediction step in AO-ARMs. This connection was initially pointed out in Hooigeboom et al. [48, App. C]. The relation between our simplified ELBO (5) and the AO-ARM objective is independently clarified by Ou et al. [36]. Despite this equivalence, our work demonstrates that the masking schedule α_t introduces a new degree of freedom in the design of such models. Variations in α_t can lead to different distributions of unmasking times, significantly impacting performance in diffusion-style parallel sampling under time discretization, as shown in Fig. 2.

Other related work. Due to space constraint, we defer the discussion on other related work, including MaskGIT [39], discrete flow matching [49], SDDM [30], Blackout diffusion [50] and SUNDAE [51], to App. H.4.

6 Generalization to State-dependent Masking Schedules

Consider a scenario where some tokens hold more significance than others and we would like to unmask them earlier in the process. To achieve this, we introduce state-dependent masking schedules, where the probability of unmasking a token depends not only on time, but also on the token’s value.

We first define the forward process for a single token x_t . Let α_t be a $m + 1$ dimensional vector function, i.e., there is a different function $\alpha_{t,i}$ for each possible value i of the token x_t . Also, by

连续时间目标。坎贝尔等人。[29] 使用速率矩阵导出了另一种 ELBO 表达式，Kitouni 等人。[46] 进一步简化了吸收扩散。在应用程序中。H.1，我们展示了如何通过从 ELBO 表达式 (4) 中分离出一个常数并应用离散的“按部分积分”来恢复它们的表达式。它们的表达的一个关键限制是它需要预测模型 $\mu_\theta(\cdot, t)$ 的 N 评估来计算内部求和。为了规避这种计算负担，他们使用了双重随机估计。然而，与仅需要一次 $\mu_\theta(\cdot, t)$ 的分析交叉熵 (4) 相比，这会导致显着更高的方差。请参阅应用程序。H.2 了解更多详情。

分数参数化。虽然到目前为止，我们使用预测模型 $\mu_\theta(x_t, t)$ 来表示给定 x_t (的干净数据的平均值，即平均参数化)，但我们可以选择其他方法来参数化反向模型。卢等人。[32]，本顿等人。[35] 提出参数化离散“分数” $s(x_t, t)_j \triangleq \frac{q_t(j)}{q_t(x_t)}$ ，并为离散扩散引入基于分数的损失。在应用程序中。H.3，我们提供了一种更简单的损失的替代推导。我们通过以下命题展示分数和平均参数化之间的联系。

命题 1 (分数参数化与平均参数化) 。 *Let q_t be the marginal distribution of the masked diffusion defined in Sec. 2 at time t . The discrete score $s(x_t, t)_j = \frac{q_t(j)}{q_t(x_t)}$ for a mask state $x_t = m$ and $j \neq m$ can be expressed as*

$$s(m, t)_j = \frac{\alpha_t}{1 - \alpha_t} \mathbb{E}[x_0 | x_t = m]^\top e_j, \text{ which satisfies } \sum_{j \neq m} s(m, t)_j = \frac{\alpha_t}{1 - \alpha_t}. \quad (9)$$

命题 1 (在 App.H.3 中证明) 意味着掩模状态的合理评分模型是

$$s_\theta(m, t)_j = \frac{\alpha_t}{1 - \alpha_t} \mu_\theta(m, t)_j. \quad (10)$$

事实上，将 (10) 代入 Lou 等人的基于分数的损失中。[32]，本顿等人。[35] 恢复了我们的目标 (4)。在卢等人。[32]，分数被参数化为神经网络，而不强制执行 (9) 中的约束。这意味着学习到的反向模型可能与正向过程不兼容。我们发现，强制约束的参数化可以带来更稳定的训练和更好的结果。

任意阶自回归模型。我们的掩模扩散模型的连续时间逆过程可以被视为任意阶自回归模型 (AO-ARM) [47]。为了看到这一点，我们根据相反过程中的解锁事件的时间对令牌进行重新排序。对于所有令牌，解锁时间 $\{\tau_n\}_{n=1}^N$ 的累积分布函数 (CDF) 是相同的并且满足 $P(\tau_n \leq t) = P(x_t^{(n)} = m) = 1 - \alpha_t$ 。因此，所有可能的排列中的排序都是均匀随机的，并且每个揭开事件期间的令牌预测代表 AO-ARM 中的预测步骤。Hoogeboom 等人最初指出了这种联系。[48，应用程序。C]。我们的简化 ELBO (5) 和 AO-ARM 目标之间的关系由 O u 等人独立阐明。[36]。尽管存在这种等价性，但我们的工作表明，掩蔽方案 α_t 在此类模型的设计中引入了新的自由度。 α_t 的变化可能导致揭露时间的不同分布，从而显着影响时间离散化下扩散式并行采样的性能，如图 2 所示。

其他相关工作。由于空间限制，我们推迟了对其他相关工作的讨论，包括 MaskGIT [39]、离散流匹配[49]、SDDM [30]、Blackout Diffusion [50] 和 SUNDAE [51]，到 App。H.4。

6 状态相关屏蔽时间表的推广

考虑这样一种情况，其中某些代币比其他代币更重要，我们希望在此过程中尽早揭开它们的面纱。为了实现这一目标，我们引入了与状态相关的屏蔽时间表，其中解密令牌的概率不仅取决于时间，还取决于令牌的价值。

我们首先定义单个令牌 x_t 的转发过程。令 α_t 为 $m + 1$ 维向量函数，即，对于标记 x_t 的每个可能值 i 都有一个不同的函数 $\alpha_{t,i}$ 。另外，通过

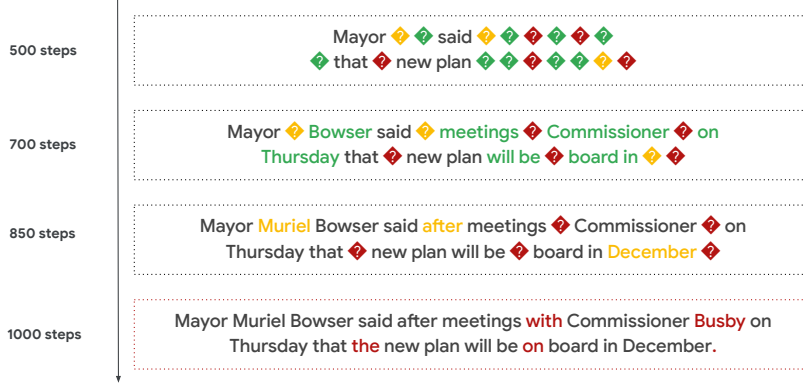


Figure 3: Iterative unmasking process for an unconditionally generated sample by MD4. This visualization only includes a subsequence from a generated sequence of 1024 tokens. "?" represents masks. Masked tokens are revealed sequentially: green (steps 500-700), yellow (700-850), and red (850-1000). Additional unconditional generation from MD4 can be found in App. K.5.

vector $\frac{\alpha_t}{\alpha_s}$ we denote the element-wise division of the two vectors. We define the forward transition as $q(x_t|x_s) = \text{Cat}(x_t; \bar{Q}(s, t)^\top x_s)$ where

$$\bar{Q}(s, t) = \text{diag}\left(\frac{\alpha_t}{\alpha_s}\right) + \left(I - \text{diag}\left(\frac{\alpha_t}{\alpha_s}\right)\right) \mathbf{1} e_m^\top$$

and $\text{diag}\left(\frac{\alpha_t}{\alpha_s}\right)$ is a diagonal matrix with the vector $\frac{\alpha_t}{\alpha_s}$ in its diagonal. The probability of moving from current state x_s to a future state x_t (either the same as x_s or mask) is determined by a state-dependent rate $\left(\frac{\alpha_t}{\alpha_s}\right)^\top x_s$, while the marginal at time s given x_0 is

$$q(x_s|x_0) = \text{Cat}(x_s; \bar{Q}(s)^\top x_0) \quad \text{for} \quad \bar{Q}(s) = \text{diag}(\alpha_s) + (I - \text{diag}(\alpha_s)) \mathbf{1} e_m^\top.$$

Further, for any time $0 \leq s < t \leq 1$ it holds that $q(x_t|x_0) = \sum_{x_s} q(x_t|x_s)q(x_s|x_0)$ so the above is a valid continuous-time Markov chain.

Given the forward conditionals and marginals, we can now compute the time reversal conditioned on x_0 . The full form of $q(x_s|x_t, x_0)$ is derived in App. I.1. For $x_t = m$, we have

$$q(x_s|x_t = m, x_0) = q(x_s|x_t = m, x_0, x_0^\top) = \left(\frac{1-\alpha_s}{1-\alpha_t}\right)^\top x_0 e_m^\top x_s + \left(\frac{\alpha_s-\alpha_t}{1-\alpha_t}\right)^\top x_0 x_0^\top x_s. \quad (11)$$

This suggests that the reverse model given $x_t = m$ can be chosen as $p_\theta(x_s|x_t = m) \triangleq q(x_s|x_t = m, \mu_\theta(x_t, t), \text{diag}(\mu_\theta(x_t, t)))$ where $\mu_\theta(x_t, t)$ is a neural network that approximates $\mathbb{E}[x_0|x_t]$ while $\text{diag}(\mu_\theta(x_t, t))$ approximates $\mathbb{E}[x_0 x_0^\top | x_t] = \text{diag}(\mathbb{E}[x_0 | x_t])$. We show in App. I.1 that the negative continuous-time ELBO for the state-dependent rate case is

$$\mathcal{L}_\infty = \int_0^1 \left(\frac{\alpha'_t}{1-\alpha_t}\right)^\top \mathbb{E}_{q(x_t|x_0)} [\delta_{x_t, m} \cdot (x_0 - \mu_\theta(x_t, t) + x_0 x_0^\top \log \mu_\theta(x_t, t))] dt. \quad (12)$$

Here, α'_t is the elementwise derivative of α_t . This generalizes the MD4 loss (4), which is recovered when α_t is a scalar schedule times a vector of ones. For N tokens, the model further generalize similarly to Sec. 3 and the loss is given in (32). We call this generalized model **GenMD4**.

To learn the token dependent masking schedule using ELBO optimization, we parametrize the $m+1$ dimensional function α_t using the polynomial schedule (see Fig. 1) as $\alpha_{t,i} = 1 - t^{w_i}$ and optimize each parameter $w_i > 0$.⁴ The value of w_i , through the masking probability $1 - \alpha_{t,i}$, determines how fast the token with value i jumps to the mask state. Since in the loss (12) the distribution $q(x_t|x_0)$ depends on α_t and thus the vector w , optimizing w poses a discrete gradient estimation problem [see, e.g., 52]. Naive autodiff leads to biased gradients and pushes w towards zero because the gradients cannot propagate through the (discrete) samples drawn from $q(x_t|x_0)$. To fix this, we used the REINFORCE leave-one-out estimator [53, 54] to compute low-variance unbiased gradients for optimizing w . Details are given in App. I.2.

⁴We only need m learnable parameters w_i , for $i = 0, \dots, m-1$, since x_0 can never be the mask token. For the final mask dimension we can choose an arbitrary fixed value such as $w_m = 0$.

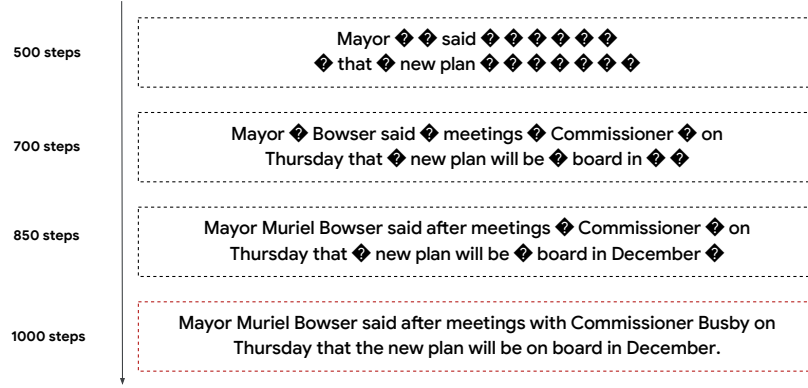


图 3: MD4 无条件生成样本的迭代解密过程。此可视化仅包含生成的 1024 个标记序列中的一个子序列。“?”代表面具。被遮盖的标记按顺序显示: 绿色 (步骤 500-700)、黄色 (700-850) 和红色 (850-1000)。MD4 的附加无条件生成可以在 App 中找到。K.5。

向量 $\frac{\alpha_t}{\alpha_s}$ 我们表示两个向量的逐元素除法。我们将前向转换定义为 $q(x_t|x_s) = \text{Cat}(x_t; \bar{Q}(s, t)^\top x_s)$, 其中

$$\bar{Q}(s, t) = \text{diag}\left(\frac{\alpha_t}{\alpha_s}\right) + \left(I - \text{diag}\left(\frac{\alpha_t}{\alpha_s}\right)\right) \mathbf{1} e_m^\top$$

$\text{diag}\left(\frac{\alpha_t}{\alpha_s}\right)$ 是对角矩阵, 其对角线为向量 $\frac{\alpha_t}{\alpha_s}$ 。从当前状态 x_s 移动到与 x_s 或掩模) 相同的未来状态 x_t 的概率由状态相关速率 $\left(\frac{\alpha_t}{\alpha_s}\right)^\top x_s$ 确定, 而给定 x_0 时 s 时刻的边为

$$q(x_s|x_0) = \text{Cat}(x_s; \bar{Q}(s)^\top x_0) \quad \text{for} \quad \bar{Q}(s) = \text{diag}(\alpha_s) + (I - \text{diag}(\alpha_s)) \mathbf{1} e_m^\top.$$

此外, 对于任何时间 $0 \leq s < t \leq 1$ 都保持 $q(x_t|x_0) = \sum_{x_s} q(x_t|x_s)q(x_s|x_0)$, 因此上面是有效的连续时间马尔可夫链。

给定前向条件和边, 我们现在可以计算以 x_0 为条件的时间反转。 $q(x_s|x_t, x_0)$ 的完整形式是在 App 中导出的。一.1. 对于 $x_t = m$, 我们有

$$q(x_s|x_t = m, x_0) = q(x_s|x_t = m, x_0, x_0 x_0^\top) = \left(\frac{1-\alpha_s}{1-\alpha_t}\right)^\top x_0 e_m^\top x_s + \left(\frac{\alpha_s - \alpha_t}{1-\alpha_t}\right)^\top x_0 x_0^\top x_s. \quad (11)$$

这表明给定 $x_t = m$ 的反向模型可以选择为 $p_\theta(x_s|x_t = m) \triangleq q(x_s|x_t = m, \mu_\theta(x_t, t), \text{diag}(\mu_\theta(x_t, t)))$, 其中 $\mu_\theta(x_t, t)$ 是近似 $\mathbb{E}[x_0|x_t]$ 的神经网络, 而 $\text{diag}(\mu_\theta(x_t, t))$ 近似 $\mathbb{E}[x_0 x_0^\top | x_t] = \text{diag}(\mathbb{E}[x_0 | x_t])$ 。我们在应用程序中显示。I.1 状态相关速率情况下的负连续时间 ELBO 为

$$\mathcal{L}_\infty = \int_0^1 \left(\frac{\alpha'_t}{1-\alpha_t}\right)^\top \mathbb{E}_{q(x_t|x_0)} [\delta_{x_t, m} \cdot (x_0 - \mu_\theta(x_t, t) + x_0 x_0^\top \log \mu_\theta(x_t, t))] dt. \quad (12)$$

这里, α'_t 是 α_t 的元素导数。这概括了 MD4 损失 (4), 当 α_t 是标量调度乘以 1 向量时, MD 4 损失即可恢复。对于 N 令牌, 该模型进一步推广, 类似于第 2 节。3 and the loss is given in (32). We call this generalized model GenMD4.

为了使用 ELBO 优化学习依赖于令牌的掩码调度, 我们使用多项式调度 (见图 1) 将 $m+1$ 维函数 α_t 参数化为 $\alpha_{t,i} = 1 - t^{w_i}$ 并优化每个参数 $w_i > 0$ 。⁴ w_i 的值通过掩码概率 $1 - \alpha_{t,i}$ 确定值为 i 的令牌跳转到掩码的速度状态。由于在损失 (12) 中, 分布 $q(x_t|x_0)$ 取决于 α_t , 因此取决于向量 w , 因此优化 w 会带来离散梯度估计问题 [参见, 例如, 52]。朴素的自动微分会导致有偏差的梯度并将 w 推向零, 因为梯度无法通过从 $q(x_t|x_0)$ 提取的 (离散) 样本传播。为了解决这个问题, 我们使用 REINFORCE 留一法估计器 [53, 54] 来计算低方差无偏梯度以优化 w 。详细信息在应用程序中给出。I.2.

⁴对于 $i = 0, \dots, m-1$, 我们只需要 m 可学习参数 w_i , 因为 x_0 永远不能成为掩码标记。对于最终的掩模尺寸, 我们可以选择任意固定值, 例如 $w_m = 0$ 。

Table 1: Zero-shot unconditional perplexity on five benchmark datasets from Radford et al. [57]. The numbers for other methods are from Lou et al. [32] except our reimplementation of SEDD Absorb. Our MD4 model achieves the best result on all benchmarks except LAMBADA where it is the second best. *The GPT-2 numbers are reported for the GPT-2 checkpoint pretrained on WebText instead of OWT thus is not a direct comparison.

Size	Method	LAMBADA	WikiText2	PTB	WikiText103	IBW
Small	GPT-2 (WebText)*	45.04	42.43	138.43	41.60	75.20
	D3PM	≤ 93.47	≤ 77.28	≤ 200.82	≤ 75.16	≤ 138.92
	Plaid	≤ 57.28	≤ 51.80	≤ 142.60	≤ 50.86	≤ 91.12
	SEDD Absorb	≤ 50.92	≤ 41.84	≤ 114.24	≤ 40.62	≤ 79.29
	SEDD Absorb (reimpl.)	≤ 49.73	≤ 38.94	≤ 107.54	≤ 39.15	≤ 72.96
	MD4 (Ours)	≤ 48.43	$\leq \mathbf{34.94}$	$\leq \mathbf{102.26}$	$\leq \mathbf{35.90}$	$\leq \mathbf{68.10}$
Medium	GPT-2 (WebText)*	35.66	31.80	123.14	31.39	55.72
	SEDD Absorb	≤ 42.77	≤ 31.04	≤ 87.12	≤ 29.98	≤ 61.19
	MD4 (Ours)	≤ 44.12	$\leq \mathbf{25.84}$	$\leq \mathbf{66.07}$	$\leq \mathbf{25.84}$	$\leq \mathbf{51.45}$

7 Experiments

7.1 Text

Text is natural discrete data with rich structures. For comparison with prior work, we evaluate likelihood on two datasets: **text8** [55], a character-level text modeling benchmark, and **OpenWebText** [56], an open clone of the unreleased WebText dataset used to train GPT-2 [57]. We also assess our model’s performance on downstream tasks by training on **FineWeb-Edu** [58], a high-quality dataset of fine educational text commonly used by the open-source community for comparing LLMs. Unless otherwise specified, a linear schedule and a cosine sampling grid are employed.

OpenWebText. We train MD4 of GPT-2 small (S) and GPT-2 medium (M) sizes on OpenWebText and evaluate zero-shot perplexity on five benchmark datasets used in Radford et al. [57]. We keep our evaluation setup the same as SEDD [32]. To ensure fair comparison, we reimplemented SEDD in our codebase. Our implementation led to slightly better results than those reported in their paper.

As seen in Tab. 1, our small model outperforms previous best discrete diffusion models on all five tasks. We are also better than GPT-2 on all tasks except LAMBADA where we are the second best method. When scaling up to medium size, MD4 similarly beats SEDD and GPT-2 on 4 out of 5 tasks.

To confirm that the strong zero-shot performance stems from improved training, we plot perplexity on 2% OpenWebText validation set in Fig. 4. Our models converge faster and have better final likelihoods than prior methods. We also observed that SEDD [32] has training instabilities, likely due to score parameterization breaking consistency between forward and reverse processes (Sec. 5). Although GenMD4 achieves lower perplexity than MD4, we observed that the learned ws can overfit to dataset statistics, making it less effective on zero-shot transfer tasks.

We also assess our models’ generation quality. Fig. 3 shows a randomly selected, notably coherent sample from MD4-medium and its denoising process. Fig. 10 demonstrates MD4’s text infilling ability and highlights a substantial quality gain when transitioning from uniform to cosine discretization (see Sec. 4). Despite MD4’s strong performance on quantitative metrics like generative perplexity, we have placed these results in Appendix Fig. 8 due to the metric’s inherent unreliability, as noted in Sec. 4. We emphasize the more reliable FID-based assessments found in our image experiments.

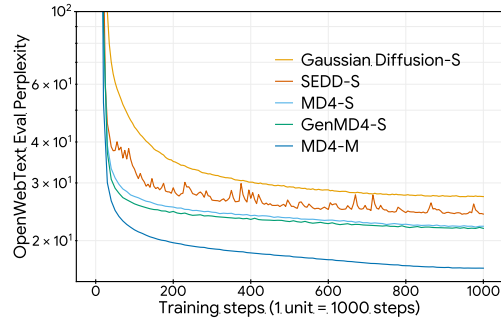


Figure 4: Perplexity on OpenWebText (OWT) validation set during training. The final numbers are reported in Tab. 5 in Appendix.

表 1: Radford 等人在五个基准数据集上的零样本无条件困惑度。[57]。其他方法的数字来自 Lou 等人。[32] 除了我们重新实现 SEDD Absorb 之外。我们的 MD4 模型在所有基准测试中都取得了最好的结果，除了 LAMBADA 之外，它是第二好的。*报告的 GPT-2 数字是在 WebText（而不是 OWT）上预训练的 GPT-2 检查点，因此不是直接比较。

Size	Method	LAMBADA	WikiText2	PTB	WikiText103	IBW
Small	GPT-2 (WebText)*	45.04	42.43	138.43	41.60	75.20
	D3PM	≤ 93.47	≤ 77.28	≤ 200.82	≤ 75.16	≤ 138.92
	Plaid	≤ 57.28	≤ 51.80	≤ 142.60	≤ 50.86	≤ 91.12
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	MD4 (Ours)	≤ 48.43	$\leq$ 34.94	$\leq$ 102.26	$\leq$ 35.90	$\leq$ 68.10
Medium	GPT-2 (WebText)*	35.66	31.80	123.14	31.39	55.72
	SEDD Absorb	≤ 42.77	≤ 31.04	≤ 87.12	≤ 29.98	≤ 61.19
	MD4 (Ours)	≤ 44.12	$\leq$ 25.84	$\leq$ 66.07	$\leq$ 25.84	$\leq$ 51.45

7 实验

7.1 正文

文本是具有丰富结构的自然离散数据。为了与之前的工作进行比较，我们评估了两个数据集的可能性：text8 [55]（字符级文本建模基准）和 OpenWebText [56]（用于训练 GPT-2 [57] 的未发布 WebText 数据集的开放克隆）。我们还通过在 FineWeb-Edu [58] 上进行训练来评估我们的模型在下游任务上的性能，FineWeb-Edu 是开源社区通常用于比较法学硕士的高质量优质教育文本数据集。除非另有说明，否则采用线性计划和余弦采样网格。

打开网络文本。我们在 OpenWeb-Text 上训练 GPT-2 小 (S) 和 GPT-2 中 (M) 大小的 MD 4，并在 Radford 等人使用的五个基准数据集上评估零样本困惑度。[57]。我们的评估设置与 SEDD [32] 相同。为了确保公平比较，我们在代码库中重新实现了 SEDD。我们的实施产生了比他们论文中报告的结果稍好的结果。

如表所示。如图 1 所示，我们的小模型在所有五项任务上都优于之前最好的离散扩散模型。我们在所有任务上都优于 GPT-2，除了 LAMBADA 之外，我们是第二好的方法。当扩展到中等规模时，MD4 在 5 个任务中的 4 个上同样击败了 SEDD 和 GPT-2。

为了确认强大的零样本性能源于改进的训练，我们在图 4 中绘制了 2% OpenWebText 验证集的困惑度。我们的模型收敛速度更快，并且比以前的方法具有更好的最终可能性。我们还观察到 SEDD [32] 存在训练不稳定，可能是由于分数参数化破坏了正向和反向过程之间的一致性（第 5 节）。尽管 GenMD4 实现了比 MD4 更低的复杂度，但我们观察到学习到的 w 可能会过度拟合数据集统计数据，从而使其在零样本传输任务上效果较差。

我们还评估模型的生成质量。图 3 显示了从 MD4 介质中随机选择的、特别一致的样本及其去噪过程。图 10 展示了 MD4 的文本填充能力，并强调了从均匀离散化过渡到余弦离散化时的显着质量增益（参见第 4 节）。尽管 MD4 在生成困惑度等定量指标上表现强劲，但由于该指标固有的不可靠性，我们将这些结果放在附录图 8 中，如第 2 节所述。4. 我们强调在我们的图像实验中发现的更可靠的基于 FID 的评估。

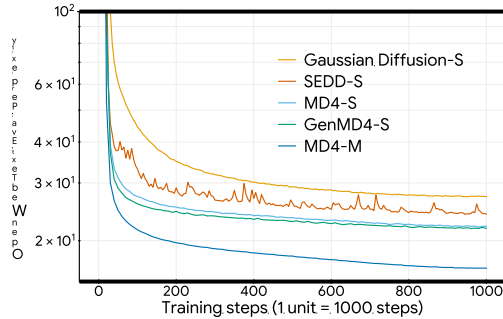


图 4: 训练期间 OpenWebText (OWT) 验证集的困惑。最终数字报告在表中。5 见附录。

Table 2: Bits Per Character (BPC) on Text8 test set. All models use standard 12-layer transformers similar to GPT-2 small [57] except Discrete Flow which uses 8×3 layers.

Method	BPC ($\downarrow$)
<i>Continuous Diffusion</i>	
Plaid [22] (Our impl.)	≤ 1.48
BFN [26]	≤ 1.41
<i>Any-order Autoregressive</i>	
ARDM [48]	≤ 1.43
MAC [61]	≤ 1.40
<i>Autoregressive</i>	
IAF/SCF [62]	1.88
AR Argmax Flow [15]	1.39
Discrete Flow [59]	1.23
Transformer AR [14]	1.23
<i>Discrete Diffusion</i>	
Mult. Diffusion [15]	≤ 1.72
D3PM Uniform [14]	≤ 1.61
D3PM Absorb [14]	≤ 1.45
SEDD Absorb [32]	≤ 1.39
MD4 (Ours)	≤ 1.37
GenMD4 (Ours)	≤ 1.34

Table 3: Bits Per Dimension (BPD) on CIFAR-10 test set and Downsampled ImageNet 64×64 [63] validation set. All models in the table are trained without data augmentation.

	Method	#Params	BPD ($\downarrow$)
CIFAR-10	<i>Autoregressive</i>		
	PixelRNN [63]		3.00
	Gated PixelCNN [64]		3.03
	PixelCNN++ [65]	53M	2.92
	PixelSNAIL [66]	46M	2.85
	Image Transformer [67]		2.90
	Sparse Transformer [68]	59M	2.80
	<i>Discrete Diffusion</i>		
	D3PM Absorb [14]	37M	≤ 4.40
	D3PM Gauss [14]	36M	≤ 3.44
ImageNet 64×64	Campbell et al. [29]	36M	≤ 3.59
	Campbell et al. [29] Absorb	28M	≤ 3.52
	MD4 (Ours)	28M	≤ 2.75
	<i>Autoregressive</i>		
	PixelRNN [63]		3.63
	Gated PixelCNN [64]		3.57
	Sparse Transformer [68]	152M	3.44
	Routing Transformer [69]		3.43
	Perceiver AR [68]	770M	3.40
	<i>Discrete Diffusion</i>		
	MD4 (Ours)	198M	≤ 3.40

Text8. Following prior work [14, 32], we trained masked diffusion models on text8 and evaluate the bits-per-character on the test set (details in App. J.1). As seen in Tab. 2, our models outperform previous discrete and continuous diffusion models, as well as state-of-the-art AO-ARMs which are closely related to discrete diffusion [48]. Our model is only beaten by an autoregressive (AR) transformer and the AR-backbone Discrete Flow [59]. We believe this is because AR models only require learning a fixed generation order thus better utilize model capacity. Text8’s small vocabulary (26 letters and a space) led us to expect limited flexibility from our state-dependent formulation. However, using the generalized objective in (12), GenMD4 achieved significantly better BPC than MD4, demonstrating the potential of state-dependent diffusion for discrete data.

FineWeb-Edu. We train MD4 on FineWeb-Edu and evaluate its zero-shot accuracy on the Hellaswag dataset [60], a popular common sense inference benchmark for LLMs. We directly compared MD4 to its AR counterparts – transformers with identical configurations (except for causal masking) trained on the same data. Results are summarized in Fig. 5.

MD4 demonstrates steady performance growth with increasing scale. While outperformed by AR models of the same size, the performance gap does not widen as model size increases. For example, AR-small reaches 30% accuracy in 50k steps, while MD4-small takes 200k steps (4x data efficiency difference). At the medium scale, AR achieves 37% in 270k steps, compared to MD4’s 1 million steps.

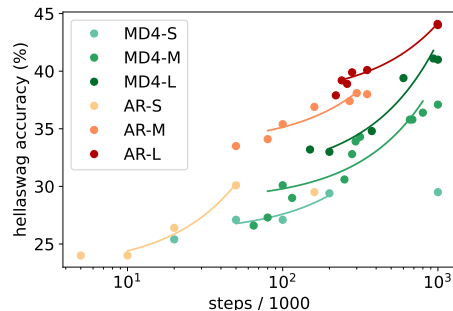


Figure 5: Hellaswag accuracy vs. training steps for MD4 and AR models at GPT-2 small, medium, and large scales.

7.2 Pixel-level image modeling

Unlike continuous diffusion which struggles with discrete data, we show that MD4, a discrete diffusion model, performs well on inherently continuous data, suggesting its potential for unifying

表 2: Text8 测试集上的每个字符位数 (BPC)。所有模型都使用类似于 GPT-2 小 [57] 的标准 12 层变压器, 除了使用 8×3 层的离散流之外。

Method	BPC ($\downarrow$)
<i>Continuous Diffusion</i>	
Plaid [22] (Our impl.)	≤ 1.48
BFN [26]	≤ 1.41
<i>Any-order Autoregressive</i>	
ARDM [48]	≤ 1.43
MAC [61]	≤ 1.40
<i>Autoregressive</i>	
IAF/SCF [62]	1.88
AR Argmax Flow [15]	1.39
Discrete Flow [59]	1.23
Transformer AR [14]	1.23
<i>Discrete Diffusion</i>	
Mult. Diffusion [15]	≤ 1.72
D3PM Uniform [14]	≤ 1.61
D3PM Absorb [14]	≤ 1.45
SEDD Absorb [32]	≤ 1.39
MD4 (Ours)	≤ 1.37
GenMD4 (Ours)	≤ 1.34

表 3: CIFAR-10 测试集和下采样 ImageNet 64×64 [63] 验证集的每维位数 (BPD)。表中的所有模型都是在没有数据增强的情况下进行训练的。

	Method	#Params	BPD ($\downarrow$)
O I R A F C	<i>Autoregressive</i>		
	PixelRNN [63]		3.00
	Gated PixelCNN [64]		3.03
	PixelCNN++ [65]	53M	2.92
	PixelSNAIL [66]	46M	2.85
	Image Transformer [67]		2.90
	Sparse Transformer [68]	59M	2.80
	<i>Discrete Diffusion</i>		
	D3PM Absorb [14]	37M	≤ 4.40
	D3PM Gauss [14]	36M	≤ 3.44
4 6 X 4 6 i c e N e g a m	Campbell et al. [29]	36M	≤ 3.59
	Campbell et al. [29] Absorb	28M	≤ 3.52
	MD4 (Ours)	28M	≤ 2.75
	<i>Autoregressive</i>		
	PixelRNN [63]		3.63
	Gated PixelCNN [64]		3.57
	Sparse Transformer [68]	152M	3.44
	Routing Transformer [69]		3.43
	Perceiver AR [68]	770M	3.40
	<i>Discrete Diffusion</i>		
	MD4 (Ours)	198M	≤ 3.40

文本8.继之前的工作 [14, 32] 之后, 我们在 text8 上训练了掩蔽扩散模型, 并评估了测试集上的每个字符的位数 (详细信息参见 App. J.1)。如表所示。如图2所示, 我们的模型优于以前的离散和连续扩散模型, 以及与离散扩散密切相关的最先进的AO-ARM [48]。我们的模型仅被自回归 (AR) 变压器和 AR 骨干离散流击败 [59]。我们认为这是因为 AR 模型只需要学习固定的生成顺序, 从而更好地利用模型容量。Text8 的词汇量较小 (26 个字母和一个空格), 导致我们对依赖于状态的表述的灵活性预期有限。然而, 使用 (12) 中的广义目标, GenMD4 实现了比 MD4 显著更好的 BPC, 证明了离散数据的状态相关扩散的潜力。

FineWeb-Edu. 我们在 FineWeb-Edu 上训练 MD4, 并在 Hellaswag 数据集 [60] 上评估其零样本精度, Hellaswag 数据集是法学硕士的流行常识推理基准。我们直接将 MD4 与其 AR 对应物进行比较——在相同数据上训练的具有相同配置 (因果屏蔽除外) 的变压器。结果总结于图 5 中。

随着规模的扩大, MD4 表现出稳定的性能增长。虽然性能优于相同尺寸的 AR 模型, 但性能差距并不会随着模型尺寸的增加而扩大。例如, AR-small 在 50k 步内达到 30% 的准确率, 而 MD4-small 需要 200k 步 (4 倍数据效率差异)。在中等规模下, AR 在 27 万步中实现了 37%, 而 MD4 为 100 万步。

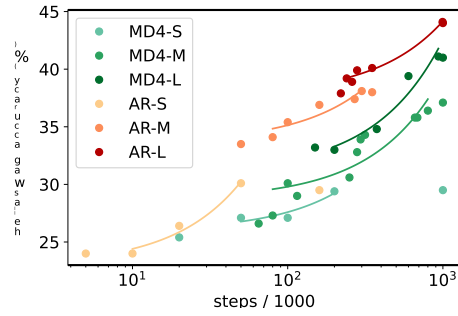


图 5: GPT-2 小、中、大尺度下 MD4 和 AR 模型의 Hellaswag 准确度与训练步骤。

7.2 像素级图像建模

与处理离散数据的连续扩散不同, 我们证明 MD4 (一种离散扩散模型) 在本质上连续的数据上表现良好, 这表明它具有统一数据的潜力。

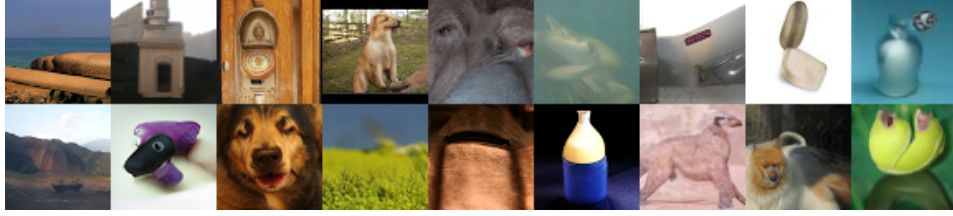


Figure 6: Non cherry-picked unconditional samples from MD4 trained on ImageNet 64x64, treating pixels as discrete tokens. More samples can be found in Fig. 9 in Appendix. The model is optimized for likelihood instead of visual quality—see e.g., Kingma et al. [33] for samples from a continuous diffusion model optimized similarly for likelihood.

modalities. We follow Austin et al. [14] and train MD4 on order-agnostic image data from CIFAR-10 and downsampled ImageNet 64×64 [63]. Each image is treated as a set of 256-valued discrete tokens, making the model agnostic to pixel proximity. We compare to other discrete diffusion and AR models with reported likelihood results on these datasets, although to our knowledge there are no published result on discrete diffusion for ImageNet 64×64 that directly model raw pixel space.

Tab. 3 summarizes our results. We establish a new state-of-the-art for discrete diffusion models, outperforming previous work [14, 29] by a significant margin. Our CIFAR-10 result surpasses the best reported AR result. On ImageNet 64×64 , our results are competitive with Transformer AR models that are $4 \times$ larger, as well as a strong continuous diffusion model VDM [33]. Notably, despite lacking knowledge of the ordinal structure of pixel values, MD4 outperforms models trained with this inductive bias, including D3PM Gauss and Campbell et al. [29] where the noising distribution is a discrete Gaussian that assigns larger probabilities to near pixel values. To isolate the differences caused by training objectives, we also implemented the Campbell et al. [29] objective with the absorbing process, showing its high variance hinders learning even with our architecture.

We provide a random sample from our ImageNet 64×64 model in Fig. 6. More results can be found in App. K. In Fig. 2, we plot the FID values of samples generated under different choices of schedules and discretization grids. We can see that the model with the linear schedule plus a cosine grid achieves an FID close to the model with cosine schedule, both significantly outperform the linear schedule with a uniform grid. We further trained a class-conditional model on ImageNet 64×64 that boosts the FID to around 7. Although these are not state-of-the-art FIDs on ImageNet 64×64 , we emphasize our models are optimized for likelihood instead of sample quality.

8 Conclusion

In this work, we revisit masked diffusion models, focusing on a flexible continuous-time formulation. Existing works in this area are not easily accessible to non-specialists and present ELBOs that are difficult to optimize, often resulting in performance that is not competitive with continuous diffusions and AR models. The framework we propose provides a very simple expression of the ELBO as a weighted integral of cross-entropy losses. Additionally, we propose a generalized masked diffusion formulation (GenMD4), where the masking schedule depends on the current state of the process, and derive its corresponding ELBO. On text data, our MD4 models outperform existing discrete and continuous diffusion models. For pixel-level image modeling, we significantly improve discrete diffusion results, outperforming similar-sized AR models and achieving comparable likelihoods to continuous diffusion models such as VDM. GenMD4 provides further improvements in terms of likelihoods over the state-independent case.

Although we have improved masked diffusion models, they still suffer from limitations. First, in some tasks such as text8, masked diffusions are not yet competitive with AR models. We conjecture that this is because AR models can better leverage model capacity since they only require learning one order. It would be interesting to develop better architectures for discrete diffusions. Moreover, GenMD4 is promising, but it can easily overfit to the dataset, making it less effective for zero-shot transfer compared to simpler versions. Additionally, inference with a state-dependent schedule is more challenging.

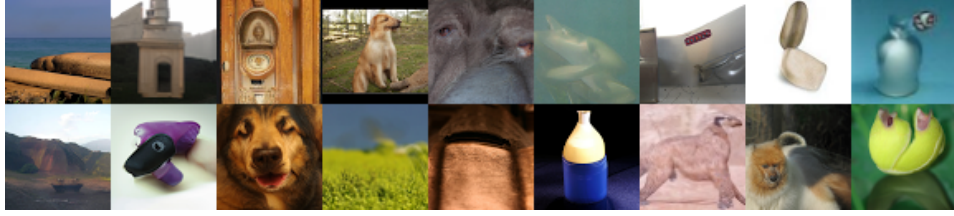


图 6: 来自在 ImageNet 64x64 上训练的 MD4 的非精选无条件样本, 将像素视为离散标记。更多示例请参见附录图 9。该模型针对可能性而不是视觉质量进行了优化, 例如, 参见 Kingma 等人。[33]对于来自连续扩散模型的样本, 对似然性进行了类似的优化。

方式。我们关注奥斯汀等人。[14] 并使用来自 CIFAR-10 和下采样 ImageNet 64×64 [63] 的顺序无关图像数据训练 MD4。每个图像都被视为一组 256 值的离散标记, 使模型与像素邻近度无关。我们将这些数据集上报告的似然结果与其他离散扩散和 AR 模型进行比较, 尽管据我们所知, 还没有直接对原始像素空间进行建模的 ImageNet 64×64 的离散扩散结果。

标签。3 总结了我们的结果。我们建立了一种新的最先进的离散扩散模型, 其性能显著优于之前的工作 [14, 29]。我们的 CIFAR-10 结果超过了报告的最佳 AR 结果。在 ImageNet 64×64 上, 我们的结果与大 $4 \times$ 的 Transformer AR 模型以及强大的连续扩散模型 VDM [33] 具有竞争力。值得注意的是, 尽管缺乏像素值顺序结构的知识, MD4 的性能优于使用这种归纳偏差训练的模型, 包括 D3PM Gauss 和 Campbell 等人。[29]其中噪声分布是离散高斯分布, 它将更大的概率分配给附近的像素值。为了隔离训练目标造成的差异, 我们还实施了 Campbell 等人的方法。[29] 吸收过程的目标表明, 即使在我们的架构中, 其高方差也会阻碍学习。

我们在图 6 中提供了来自 ImageNet 64×64 模型的随机样本。更多结果可以在 App 中找到。K. 在图 2 中, 我们绘制了在不同的计划和离散化网格选择下生成的样本的 FID 值。我们可以看到, 使用线性计划加余弦网格的模型实现的 FID 接近于使用余弦计划的模型, 两者都显著优于使用均匀网格的线性计划。我们在 ImageNet 64×64 上进一步训练了类条件模型, 将 FID 提高到 7 左右。尽管这些不是 ImageNet 64×64 上最先进的 FID, 但我们强调我们的模型针对可能性而不是样本质量进行了优化。

8 结论

在这项工作中, 我们重新审视了掩蔽扩散模型, 重点关注灵活的连续时间公式。该领域的现有工作对于非专业人士来说不容易访问, 并且存在难以优化的 ELBO, 通常导致性能与连续扩散和 AR 模型不具有竞争力。我们提出的框架提供了一种非常简单的 ELBO 表达式, 即交叉熵损失的加权积分。此外, 我们提出了一种广义掩蔽扩散公式 (GenMD4), 其中掩蔽时间表取决于过程的当前状态, 并推导其相应的 ELBO。在文本数据上, 我们的 MD4 模型优于现有的离散和连续扩散模型。对于像素级图像建模, 我们显著改进了离散扩散结果, 优于类似大小的 AR 模型, 并实现了与 VDM 等连续扩散模型相当的可能性。GenMD4 在与状态无关的情况的可能性方面提供了进一步的改进。

尽管我们改进了掩蔽扩散模型, 但它们仍然存在局限性。首先, 在某些任务 (例如 text8) 中, 掩模扩散尚无法与 AR 模型竞争。我们推测这是因为 AR 模型可以更好地利用模型能力, 因为它们只需要学习一个顺序。为离散扩散开发更好的架构将会很有趣。此外, GenMD4 很有前途, 但它很容易过度拟合数据集, 与更简单的版本相比, 它对于零样本传输的效果较差。此外, 使用依赖于状态的时间表进行推理更具挑战性。

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Table 4: Masking schedule formulas.

Masking schedules	α_t	Cross-entropy loss weight $\frac{\alpha'_t}{1-\alpha_t}$
Linear	$1 - t$	$-\frac{1}{t}$
Polynomial	$1 - t^w$	$-\frac{w}{t}$
Geometric	$\exp(-\bar{\beta}_{\min}^{1-t} \bar{\beta}_{\max}^t)$	$-\frac{\exp(-\bar{\beta}_{\min}^{1-t} \bar{\beta}_{\max}^t)}{1 - \exp(-\bar{\beta}_{\min}^{1-t} \bar{\beta}_{\max}^t)} \bar{\beta}_{\min}^{1-t} \bar{\beta}_{\max}^t \log \frac{\sigma_{\min}}{\sigma_{\max}}$
Cosine	$1 - \cos(\frac{\pi}{2}(1 - t))$	$-\frac{\pi}{2} \tan(\frac{\pi}{2}(1 - t))$

A Discrete-time derivation

We divide time from 0 to 1 into T intervals, and let $s(i) = (i - 1)/T$, $t(i) = i/T$. The forward transition matrix $Q_i \in \mathbb{R}^{(m+1) \times (m+1)}$ (m is vocabulary size) at time $t(i)$ is

$$[Q_i]_{jk} = \begin{cases} 1 & j = k = m \\ 1 - \beta_i & j = k \neq m \\ \beta_i & k = m, j \neq m \\ 0 & \text{otherwise} \end{cases}$$

or more compactly written as

$$Q_i = (1 - \beta_i)I + \beta_i \mathbf{1} e_m^\top,$$

where $\mathbf{1}$ denotes an all-one vector of size $m + 1$, and e_m is an one-hot vector of size $m + 1$ with the m -th element (recall that counting starts from 0) being one. We use an one-hot vector x_t of length $m + 1$ to denote the discrete state. The forward conditionals are defined as

$$q(x_{t(i)} | x_{s(i)}) = \text{Cat}(x_{t(i)}; Q_i^\top x_{s(i)}) = x_{s(i)}^\top Q_i x_{t(i)}, \quad (13)$$

where $Q_i^\top x_{s(i)}$ is the probabilities for each of the $m + 1$ categories that $x_{t(i)}$ can take. The marginal forward distribution at time $t(i)$ given x_0 is

$$q(x_{t(i)} | x_0) = \text{Cat}(x_{t(i)}; \bar{Q}_i^\top x_0) = x_0^\top \bar{Q}_i x_{t(i)},$$

where $\bar{Q}_i = \prod_{j=1}^i Q_j = \prod_{j=1}^i (1 - \beta_j)I + (1 - \prod_{j=1}^i (1 - \beta_j)) \mathbf{1} e_m^\top$. To see what this leads to in continuous time, we let $\beta_i = \frac{\beta(t(i))}{T}$ and $T \rightarrow \infty$:

$$\begin{aligned} \prod_{j=1}^i (1 - \beta_j) &= \exp\left(\sum_{j=1}^i \log(1 - \beta_j)\right) \\ &= \exp\left(\sum_{j=1}^i -\frac{\beta(t(j))}{T} + o(1/T)\right) \\ &\xrightarrow{T \rightarrow \infty} \exp\left(-\int_0^{t(i)} \beta(s) ds\right). \end{aligned}$$

We let $\bar{Q}(t)$ denote the limit of $\bar{Q}_i$ in this case:

$$\begin{aligned} \bar{Q}(t) &= \exp\left(-\int_0^t \beta(s) ds\right) I + \left(1 - \exp\left(-\int_0^t \beta(s) ds\right)\right) \mathbf{1} e_m^\top \\ &\triangleq \alpha_t I + (1 - \alpha_t) \mathbf{1} e_m^\top. \end{aligned}$$

Here we define $\alpha_t \triangleq \exp(-\int_0^t \beta(s) ds)$. And the marginal forward transition is

$$q(x_t | x_0) = \text{Cat}(x_t; \bar{Q}(t)^\top x_0) = x_0^\top \bar{Q}(t) x_t = \alpha_t x_0^\top x_t + (1 - \alpha_t) e_m^\top x_t. \quad (14)$$

表 4: 掩蔽方案公式。

Masking schedules	α_t	Cross-entropy loss weight $\frac{\alpha'_t}{1-\alpha_t}$
Linear	$1 - t$	$-\frac{1}{t}$
Polynomial	$1 - t^w$	$-\frac{w}{t}$
Geometric	$\exp(-\bar{\beta}_{\min}^{1-t} \bar{\beta}_{\max}^t)$	$-\frac{\exp(-\bar{\beta}_{\min}^{1-t} \bar{\beta}_{\max}^t)}{1 - \exp(-\bar{\beta}_{\min}^{1-t} \bar{\beta}_{\max}^t)} \bar{\beta}_{\min}^{1-t} \bar{\beta}_{\max}^t \log \frac{\sigma_{\min}}{\sigma_{\max}}$
Cosine	$1 - \cos(\frac{\pi}{2}(1 - t))$	$-\frac{\pi}{2} \tan(\frac{\pi}{2}(1 - t))$

离散时间推导

我们将从 0 到 1 的时间划分为 T 个区间, 并令 $s(i) = (i - 1)/T, t(i) = i/T$ 。前向转移矩阵 $Q_i \in \mathbb{R}^{(m+1) \times (m+1)}$ (m 是时间 $t(i)$ 时的词汇量大小)

$$[Q_i]_{jk} = \begin{cases} 1 & j = k = m \\ 1 - \beta_i & j = k \neq m \\ \beta_i & k = m, j \neq m \\ 0 & \text{otherwise} \end{cases}$$

或者更紧凑地写为

$$Q_i = (1 - \beta_i)I + \beta_i \mathbf{1} e_m^\top,$$

其中 $\mathbf{1}$ 表示大小为 $m + 1$ 的全一向量, 而 e_m 是大小为 $m + 1$ 的独热向量, 其中第 m 个元素 (回想一下, 从 0 开始计数) 为 1。我们使用长度为 $m + 1$ 的 one-hot 向量 x_t 来表示离散状态。前向条件定义为

$$q(x_{t(i)} | x_{s(i)}) = \text{Cat}(x_{t(i)}; Q_i^\top x_{s(i)}) = x_{s(i)}^\top Q_i x_{t(i)}, \quad (13)$$

其中 $Q_i^\top x_{s(i)}$ 是 $x_{t(i)}$ 可以采用的每个 $m + 1$ 类别的概率。给定 x_0 时, 时间 $t(i)$ 的边际前向分布为

$$q(x_{t(i)} | x_0) = \text{Cat}(x_{t(i)}; \bar{Q}_i^\top x_0) = x_0^\top \bar{Q}_i x_{t(i)},$$

其中 $\bar{Q}_i = \prod_{j=1}^i Q_j = \prod_{j=1}^i (1 - \beta_j)I + (1 - \prod_{j=1}^i (1 - \beta_j)) \mathbf{1} e_m^\top$ 。为了看看这在连续时间内会导致什么, 我们让 $\beta_i = \frac{\beta(t(i))}{T}$ 和 $T \rightarrow \infty$:

$$\begin{aligned} \prod_{j=1}^i (1 - \beta_j) &= \exp\left(\sum_{j=1}^i \log(1 - \beta_j)\right) \\ &= \exp\left(\sum_{j=1}^i -\frac{\beta(t(j))}{T} + o(1/T)\right) \\ &\xrightarrow{T \rightarrow \infty} \exp\left(-\int_0^{t(i)} \beta(s) ds\right). \end{aligned}$$

在这种情况下, 我们让 $\bar{Q}(t)$ 表示 $\bar{Q}_i$ 的极限:

$$\begin{aligned} \bar{Q}(t) &= \exp\left(-\int_0^t \beta(s) ds\right) I + \left(1 - \exp\left(-\int_0^t \beta(s) ds\right)\right) \mathbf{1} e_m^\top \\ &\triangleq \alpha_t I + (1 - \alpha_t) \mathbf{1} e_m^\top. \end{aligned}$$

这里我们定义 $\alpha_t \triangleq \exp(-\int_0^t \beta(s) ds)$ 。边际前向过渡是

$$q(x_t | x_0) = \text{Cat}(x_t; \bar{Q}(t)^\top x_0) = x_0^\top \bar{Q}(t) x_t = \alpha_t x_0^\top x_t + (1 - \alpha_t) e_m^\top x_t. \quad (14)$$

B Continuous-time derivation

We consider a continuous-time Markov chain with transition rates

$$Q(t) = (Q_i - I)/(1/T) = \beta(t)(\mathbf{1}e_m^\top - I). \quad (15)$$

For simplicity, we let $Q = \mathbf{1}e_m^\top - I$. The marginal forward distribution at time t given x_0 is $q(x_t|x_0) = \text{Cat}(x_t; \bar{Q}(t)^\top x_0)$, where

$$\bar{Q}(t) = \exp\left(\int_0^t Q(s)ds\right) = \exp\left(Q \int_0^t \beta(s)ds\right) = \exp(\bar{\beta}(t)Q).$$

Here we define $\bar{\beta}(t) \triangleq \int_0^t \beta(s)ds$. The matrix exponential can be computed via eigendecomposition:

$$\bar{\beta}(t)Q = U\Lambda U^{-1},$$

where

$$\begin{aligned} U &= I - e_me_m^\top + \frac{1}{\sqrt{n+1}}\mathbf{1}e_m^\top, \\ U^{-1} &= I + \sqrt{n+1}e_me_m^\top - \mathbf{1}e_m^\top, \\ \Lambda &= \bar{\beta}(t)(e_me_m^\top - I), \end{aligned}$$

and thus $\exp(\Lambda) = \alpha_t I + (1 - \alpha_t)e_me_m^\top$,

$$\bar{Q}(t) = U \exp(\Lambda) U^{-1} = \alpha_t I + (1 - \alpha_t)\mathbf{1}e_m^\top.$$

A simpler derivation uses the following property:

$$Q^2 = -Q.$$

Therefore,

$$\begin{aligned} \bar{Q}(t) &= \exp(\bar{\beta}(t)Q) \\ &= I + \bar{\beta}(t)Q + \frac{1}{2}\bar{\beta}(t)^2 Q^2 + \frac{1}{3}\bar{\beta}(t)^3 Q^3 + \dots \\ &= I + Q - (1 - \bar{\beta}(t) + \frac{1}{2}\bar{\beta}(t)^2 - \frac{1}{3}\bar{\beta}(t)^3 + \dots)Q \\ &= I + Q - \exp(-\bar{\beta}(t))Q \\ &= \alpha_t I + (1 - \alpha_t)\mathbf{1}e_m^\top. \end{aligned}$$

This marginal forward transition matrix at time t coincides with the result (1) we get by taking the limit of discrete-time derivation.

Arbitrary discretization of the continuous-time forward process. For the discrete-time process we have defined the per-step transition in (13). For the continuous-time process, we can derive the transition matrix $\bar{Q}(s, t)_{ij} \triangleq q(x_t = j | x_s = i)$ between two arbitrary time s and t as the solution to the following differential equation (known as Kolmogorov forward equation)

$$\frac{d}{dt}\bar{Q}(s, t) = \bar{Q}(s, t)Q(t) \text{ where } Q(t) = \beta(t)Q$$

with initial condition $\bar{Q}(s, s) = I$. The solution is given by

$$\bar{Q}(s, t) = \exp((\bar{\beta}(t) - \bar{\beta}(s))Q) = \bar{Q}(s)^{-1}\bar{Q}(t).$$

Routine work (using the Woodbury matrix inversion lemma) shows that

$$\bar{Q}(t)^{-1} = \alpha_t^{-1}I + (1 - \alpha_t^{-1})\mathbf{1}e_m^\top.$$

Plugging the result back, we get the forward transition distribution from s to t :

$$\begin{aligned} q(x_t|x_s) &= \text{Cat}(x_t; \bar{Q}(s, t)^\top x_s) = x_s^\top \bar{Q}(s, t)x_t, \\ \text{where } \bar{Q}(s, t) &\triangleq \bar{Q}(s)^{-1}\bar{Q}(t) = \frac{\alpha_t}{\alpha_s}I + \left(1 - \frac{\alpha_t}{\alpha_s}\right)\mathbf{1}e_m^\top. \end{aligned} \quad (16)$$

B 连续时间求导

我们考虑具有转移率的连续时间马尔可夫链

$$Q(t) = (Q_i - I)/(1/T) = \beta(t)(\mathbf{1}e_m^\top - I). \quad (15)$$

为了简单起见, 我们让 $Q = \mathbf{1}e_m^\top - I$ 。给定 x_0 时, 时间 t 的边际前向分布为 $q(x_t|x_0) = \text{Cat}(x_t; \bar{Q}(t)^\top x_0)$, 其中

$$\bar{Q}(t) = \exp\left(\int_0^t Q(s)ds\right) = \exp\left(Q \int_0^t \beta(s)ds\right) = \exp(\bar{\beta}(t)Q).$$

这里我们定义 $\bar{\beta}(t) \triangleq \int_0^t \beta(s)ds$ 。矩阵指数可以通过特征分解计算:

$$\bar{\beta}(t)Q = U\Lambda U^{-1},$$

在哪里

$$\begin{aligned} U &= I - e_me_m^\top + \frac{1}{\sqrt{n+1}}\mathbf{1}e_m^\top, \\ U^{-1} &= I + \sqrt{n+1}e_me_m^\top - \mathbf{1}e_m^\top, \\ \Lambda &= \bar{\beta}(t)(e_me_m^\top - I), \end{aligned}$$

因此 $\exp(\Lambda) = \alpha_t I + (1 - \alpha_t)e_me_m^\top$,

$$\bar{Q}(t) = U \exp(\Lambda) U^{-1} = \alpha_t I + (1 - \alpha_t)\mathbf{1}e_m^\top.$$

更简单的推导使用以下属性:

$$Q^2 = -Q.$$

所以,

$$\begin{aligned} \bar{Q}(t) &= \exp(\bar{\beta}(t)Q) \\ &= I + \bar{\beta}(t)Q + \frac{1}{2}\bar{\beta}(t)^2Q^2 + \frac{1}{3}\bar{\beta}(t)^3Q^3 + \dots \\ &= I + Q - (1 - \bar{\beta}(t) + \frac{1}{2}\bar{\beta}(t)^2 - \frac{1}{3}\bar{\beta}(t)^3 + \dots)Q \\ &= I + Q - \exp(-\bar{\beta}(t))Q \\ &= \alpha_t I + (1 - \alpha_t)\mathbf{1}e_m^\top. \end{aligned}$$

这个在时间 t 的边际前向转移矩阵与我们通过离散时间求导极限得到的结果 (1) 一致。

连续时间前向过程的任意离散化。对于离散时间过程, 我们在 (13) 中定义了每步转换。

对于连续时间过程, 我们可以推导出任意两个时间 s 和 t 之间的转移矩阵

$\bar{Q}(s, t)_{ij} \triangleq q(x_t = j | x_s = i)$ 作为以下微分方程 (称为柯尔莫哥洛夫前向方程) 的解

$$\frac{d}{dt}\bar{Q}(s, t) = \bar{Q}(s, t)Q(t) \text{ where } Q(t) = \beta(t)Q$$

初始条件为 $\bar{Q}(s, s) = I$ 。解由下式给出

$$\bar{Q}(s, t) = \exp((\bar{\beta}(t) - \bar{\beta}(s))Q) = \bar{Q}(s)^{-1}\bar{Q}(t).$$

罗日常工作 (使用伍德伯里矩阵求逆引理) 显示 是这样的

$$\bar{Q}(t)^{-1} = \alpha_t^{-1}I + (1 - \alpha_t^{-1})\mathbf{1}e_m^\top.$$

将结果插回, 我们得到从 s 到 t 的前向转移分布 :

$$\begin{aligned} q(x_t|x_s) &= \text{Cat}(x_t; \bar{Q}(s, t)^\top x_s) = x_s^\top \bar{Q}(s, t)x_t, \\ \text{where } \bar{Q}(s, t) &\triangleq \bar{Q}(s)^{-1}\bar{Q}(t) = \frac{\alpha_t}{\alpha_s}I + \left(1 - \frac{\alpha_t}{\alpha_s}\right)\mathbf{1}e_m^\top. \end{aligned} \quad (16)$$

$$\begin{array}{ccc}
& & x_0 & \text{MASK} \\
\begin{array}{c} \text{MASK} \end{array} & \begin{bmatrix} 1 & & & \\ & 1 & & \\ & & 1 & \\ & & & 1 \\ & & & & 1 \\ & & & & & 1 \end{bmatrix} & \begin{bmatrix} 1 & & & \\ & 1 & & \\ & & 1 & \\ & & & 1 \\ & & & & 1 \\ & & & & & 1 \end{bmatrix} \\
& & \begin{array}{cc} 0 & \frac{\alpha_s - \alpha_t}{1 - \alpha_t} \end{array} & \begin{array}{cc} 0 & \frac{1 - \alpha_s}{1 - \alpha_t} \end{array} \\
& & q(x_s = \cdot | x_t = \cdot, x_0) & q(x_s = \cdot | x_t = \cdot, \mu_\theta(x_t, t))
\end{array}$$

Figure 7: The reverse transition probability and our generative model. Left: $q(x_s = \cdot | x_t = \cdot, x_0)$ in matrix form where first index is x_t and second index is x_s . Right: $p_\theta(x_s = \cdot | x_t = \cdot) \triangleq q(x_s = \cdot | x_t = \cdot, \mu_\theta(x_t, t))$ also in matrix form.

C Time reversal of the forward process given x_0

The analytic property of our forward process allows to compute many quantities of interest in closed form. One such quantity frequently used in diffusion models is the time reversal of the forward process given x_0 : $q(x_s | x_t, x_0)$. We can compute it using (14) and (16) as

$$\begin{aligned}
q(x_s | x_t, x_0) &= \frac{q(x_t | x_s) q(x_s | x_0)}{q(x_t | x_0)} \\
&= \begin{cases} \frac{\alpha_s - \alpha_t}{1 - \alpha_t} x_s^\top x_0 & x_s \neq m, x_t = m \\ \frac{1 - \alpha_s}{1 - \alpha_t} & x_s = m, x_t = m \\ x_s^\top x_t & x_t \neq m. \end{cases} \quad (17)
\end{aligned}$$

Visually, eqn (17) is a $\mathbb{R}^{(m+1) \times (m+1)}$ matrix (Fig. 7, left) whose first index is x_t and the second is x_s . The matrix is almost an identity matrix except the last row corresponding to x_t is the mask token. The last row means with probability of $\frac{\alpha_s - \alpha_t}{1 - \alpha_t}$ the mask token gets unmasked to become x_0 , and with probability of $\frac{1 - \alpha_s}{1 - \alpha_t}$ it remains masked.

Alternatively, we can rewrite the above using reverse transition matrix $\bar{R}^{x_0}(t, s) \in \mathbb{R}^{(m+1) \times (m+1)}$ as

$$q(x_s | x_t, x_0) = \text{Cat}(x_s; \bar{R}^{x_0}(t, s)^\top x_t), \text{ where } \bar{R}^{x_0}(t, s) = I + \frac{\alpha_s - \alpha_t}{1 - \alpha_t} e_m (x_0 - e_m)^\top.$$

We are also interested in what would happen in the infinitesimal time limit, i.e., when $s = t - \Delta t$ and $\Delta t \rightarrow 0$. Note that

$$\alpha_{t-\Delta t} - \alpha_t = -\alpha'_t \Delta t + o(\Delta t).$$

Plugging it into the original formula, we get

$$\bar{R}^{x_0}(t, t - \Delta t) = I - \frac{\alpha'_t}{1 - \alpha_t} e_m (x_0 - e_m)^\top \Delta t + o(\Delta t).$$

Comparing the above with the transition rate matrix $R^{x_0}(t)$ definition

$$\bar{R}^{x_0}(t, t - \Delta t) = I + R^{x_0}(t) \Delta t + o(\Delta t),$$

we have determined the transition rate matrix for the reverse process conditioned on x_0 :

$$R^{x_0}(t) = -\frac{\alpha'_t}{1 - \alpha_t} e_m (x_0 - e_m)^\top. \quad (18)$$

$$\begin{array}{ccc}
& & x_0 & \text{MASK} & & & \text{MASK} \\
\begin{array}{c} \text{MASK} \end{array} & \left[\begin{array}{cccc} 1 & & & \\ & 1 & & \\ & & 1 & \\ & & & 1 \\ & & & & 1 \\ & & & & & 1 \\ 0 & \frac{\alpha_s - \alpha_t}{1 - \alpha_t} & 0 & \frac{1 - \alpha_s}{1 - \alpha_t} \end{array} \right] & & \left[\begin{array}{cccc} 1 & & & \\ & 1 & & \\ & & 1 & \\ & & & 1 \\ & & & & 1 \\ & & & & & 1 \\ \frac{\alpha_s - \alpha_t}{1 - \alpha_t} \mu_\theta(x_t, t) & & & \frac{1 - \alpha_s}{1 - \alpha_t} \end{array} \right] \\
& q(x_s = \cdot | x_t = \cdot, x_0) & & q(x_s = \cdot | x_t = \cdot, \mu_\theta(x_t, t))
\end{array}$$

图 7: 反向转移概率和我们的生成模型。左: 矩阵形式的 $q(x_s = \cdot | x_t = \cdot, x_0)$, 其中第一个索引是 x_t , 第二个索引是 x_s 。右: $p_\theta(x_s = \cdot | x_t = \cdot) \triangleq q(x_s = \cdot | x_t = \cdot, \mu_\theta(x_t, t))$ 也是矩阵形式。

C 给定 x_0 的前向过程的时间反转

我们的正向过程的分析属性允许以封闭形式计算许多感兴趣的量。扩散模型中经常使用的此类量之一是给定 x_0 : $q(x_s | x_t, x_0)$ 的前向过程的时间反转。我们可以使用 (14) 和 (16) 计算它:

$$\begin{aligned}
q(x_s | x_t, x_0) &= \frac{q(x_t | x_s) q(x_s | x_0)}{q(x_t | x_0)} \\
&= \begin{cases} \frac{\alpha_s - \alpha_t}{1 - \alpha_t} x_s^\top x_0 & x_s \neq m, x_t = m \\ \frac{1 - \alpha_s}{1 - \alpha_t} & x_s = m, x_t = m \\ x_s^\top x_t & x_t \neq m. \end{cases} \quad (17)
\end{aligned}$$

从视觉上看, eqn (17) 是一个 $\mathbb{R}^{(m+1) \times (m+1)}$ 矩阵 (图 7, 左), 其第一个索引是 x_t , 第二个索引是 x_s 。该矩阵几乎是单位矩阵, 除了对应于 x_t 的最后一行是掩码标记。最后一行意味着掩码令牌以 $\frac{\alpha_s - \alpha_t}{1 - \alpha_t}$ 的概率被取消掩码变为 x_0 , 并且以 $\frac{1 - \alpha_s}{1 - \alpha_t}$ 的概率保持掩码。

或者, 我们可以使用反向转移矩阵 $\bar{R}^{x_0}(t, s) \in \mathbb{R}^{(m+1) \times (m+1)}$ 将上面的重写为

$$q(x_s | x_t, x_0) = \text{Cat}(x_s; \bar{R}^{x_0}(t, s)^\top x_t), \text{ where } \bar{R}^{x_0}(t, s) = I + \frac{\alpha_s - \alpha_t}{1 - \alpha_t} e_m (x_0 - e_m)^\top.$$

我们还对无穷小的时间限制内会发生什么感兴趣, 即当 $s = t - \Delta t$ 和 $\Delta t \rightarrow 0$ 时。请注意

$$\alpha_{t-\Delta t} - \alpha_t = -\alpha'_t \Delta t + o(\Delta t).$$

将其代入原公式, 我们得到

$$\bar{R}^{x_0}(t, t - \Delta t) = I - \frac{\alpha'_t}{1 - \alpha_t} e_m (x_0 - e_m)^\top \Delta t + o(\Delta t).$$

将上述与转移率矩阵 $R^{x_0}(t)$ 定义进行比较

$$\bar{R}^{x_0}(t, t - \Delta t) = I + R^{x_0}(t) \Delta t + o(\Delta t),$$

我们已经确定 编辑逆向过程的转移率矩阵

在 x_0 上:

$$R^{x_0}(t) = -\frac{\alpha'_t}{1 - \alpha_t} e_m (x_0 - e_m)^\top. \quad (18)$$

D Details of the ELBO

Using (17) and (3), we compute the KL divergences between forward and reverse transitions

$$\begin{aligned}
\text{KL}(q(x_s|x_t, x_0)||p_\theta(x_s|x_t)) &= \text{KL}(q(x_s|x_t, x_0)||q(x_s|x_t, \mu_\theta(x_t, t))) \\
&= \begin{cases} \sum_{x_s=0}^m q(x_s|x_t, x_0) \log \frac{q(x_s|x_t, x_0)}{q(x_s|x_t, \mu_\theta(x_t, t))} & x_t = m \\ 0 & x_t \neq m \end{cases} \\
&= \delta_{x_t=m} \sum_{k \neq m} \frac{\alpha_s - \alpha_t}{1 - \alpha_t} x_0^\top e_k \log \frac{x_0^\top e_k}{\mu_\theta(x_t, t)^\top e_k} \\
&= -\delta_{x_t=m} \frac{\alpha_s - \alpha_t}{1 - \alpha_t} x_0^\top \log \mu_\theta(x_t, t).
\end{aligned} \tag{19}$$

Note that $0 \log 0 = 0$. Alternatively, this result can be easily obtained from the visual depictions of $q(x_s|x_t, x_0)$ and $p_\theta(x_s|x_t)$ shown in Fig. 7. In this case, the reconstruction term becomes

$$\begin{aligned}
\mathbb{E}_{q(x_{t(1)}|x_0)}[\log p(x_0|x_{t(1)})] &= \sum_{k=0}^m q_{t(1)|0}(k|x_0) \log \frac{q_{t(1)|0}(k|x_0)}{\sum_{j \neq m} q_{t(1)|0}(k|j)} \\
&= \alpha_{t(1)} \cdot \log \frac{\alpha_{t(1)}}{\alpha_{t(1)}} + (1 - \alpha_{t(1)}) \log \frac{1}{m} \\
&= -(1 - \alpha_{t(1)}) \log m.
\end{aligned}$$

The prior KL term can be computed as

$$\text{KL}(q(x_1|x_0)||p(x_1)) = \text{KL}(\delta_{x_1,m}||\delta_{x_1,m}) = 0.$$

As usual, we take the continuous-time limit by letting $T \rightarrow \infty$:

$$\begin{aligned}
\mathcal{L}_\infty &\triangleq \lim_{T \rightarrow \infty} \mathcal{L}_T \\
&= \lim_{T \rightarrow \infty} \sum_{i=2}^T -\frac{\alpha_{s(i)} - \alpha_{t(i)}}{s(i) - t(i)} \frac{s(i) - t(i)}{1 - \alpha_{t(i)}} x_0^\top \mathbb{E}_{q(x_{t(i)}|x_0)} [\delta_{x_{t(i)},m} \log \mu_\theta(x_{t(i)}, t(i))] \\
&= \int_{t(1)}^1 \frac{\alpha'_t}{1 - \alpha_t} x_0^\top \mathbb{E}_{q(x_t|x_0)} [\delta_{x_t,m} \log \mu_\theta(x_t, t)] dt.
\end{aligned}$$

E Avoiding undefined KL divergence

When defining the forward process, we often do not want α_1 to be exactly 0, or equivalently, λ_1 to be ∞ for numerical stability reasons. Instead, we set λ_1 to be a finite value, and thereby α_1 has a small positive value. This has a problem that the support of $q(x_1|x_0)$ is no longer $\{m\}$ and instead becomes $\{m, x_0\}$. As a result, the KL divergence between $q(x_1|x_0)$ and $p(x_1)$ is undefined because $q(x_1|x_0)$ is not absolutely continuous with respect to $p(x_1) = \delta_{x_1,m}$. To resolve the issue, we modify the prior distribution $p(x_1)$ such that it has support over all $m+1$ values. One such choice is letting

$$p(x_1) = \frac{\alpha_1}{m} \sum_{j \neq m} \delta_{x_1,j} + (1 - \alpha_1) \delta_{x_1,m}.$$

Then, the prior KL divergence term becomes

$$\begin{aligned}
\text{KL}(q(x_1|x_0)||p(x_1)) &= \sum_{x_1=0}^m q(x_1|x_0) \log \frac{q(x_1|x_0)}{p(x_1)} \\
&= \sum_{x_1=0}^m (\alpha_1 \delta_{x_1,x_0} + (1 - \alpha_1) \delta_{x_1,m}) \log \frac{\alpha_1 \delta_{x_1,x_0} + (1 - \alpha_1) \delta_{x_1,m}}{p(x_1)} \\
&= \alpha_1 \log \frac{\alpha_1}{\alpha_1/m} + (1 - \alpha_1) \log \frac{1 - \alpha_1}{1 - \alpha_1} \\
&= \alpha_1 \log m.
\end{aligned}$$

D ELBO 的详细信息

使用 (17) 和 (3)，我们计算正向和反向转换之间的 KL 散度

$$\begin{aligned}
\text{KL}(q(x_s|x_t, x_0)||p_\theta(x_s|x_t)) &= \text{KL}(q(x_s|x_t, x_0)||q(x_s|x_t, \mu_\theta(x_t, t))) \\
&= \begin{cases} \sum_{x_s=0}^m q(x_s|x_t, x_0) \log \frac{q(x_s|x_t, x_0)}{q(x_s|x_t, \mu_\theta(x_t, t))} & x_t = m \\ 0 & x_t \neq m \end{cases} \\
&= \delta_{x_t=m} \sum_{k \neq m} \frac{\alpha_s - \alpha_t}{1 - \alpha_t} x_0^\top e_k \log \frac{x_0^\top e_k}{\mu_\theta(x_t, t)^\top e_k} \\
&= -\delta_{x_t=m} \frac{\alpha_s - \alpha_t}{1 - \alpha_t} x_0^\top \log \mu_\theta(x_t, t).
\end{aligned} \tag{19}$$

请注意， $0 \log 0 = 0$ 。或者，可以从图 7 所示的 $q(x_s|x_t, x_0)$ 和 $p_\theta(x_s|x_t)$ 的直观描述中轻松获得此结果。在这种情况下，重建项变为

$$\begin{aligned}
\mathbb{E}_{q(x_{t(1)}|x_0)}[\log p(x_0|x_{t(1)})] &= \sum_{k=0}^m q_{t(1)|0}(k|x_0) \log \frac{q_{t(1)|0}(k|x_0)}{\sum_{j \neq m} q_{t(1)|0}(k|j)} \\
&= \alpha_{t(1)} \cdot \log \frac{\alpha_{t(1)}}{\alpha_{t(1)}} + (1 - \alpha_{t(1)}) \log \frac{1}{m} \\
&= -(1 - \alpha_{t(1)}) \log m.
\end{aligned}$$

先前的 KL 项可以计算为

$$\text{KL}(q(x_1|x_0)||p(x_1)) = \text{KL}(\delta_{x_1,m}||\delta_{x_1,m}) = 0.$$

像往常一样，我们通过令 $T \rightarrow \infty$ 来获取连续时间限制：

$$\begin{aligned}
\mathcal{L}_\infty &\triangleq \lim_{T \rightarrow \infty} \mathcal{L}_T \\
&= \lim_{T \rightarrow \infty} \sum_{i=2}^T -\frac{\alpha_{s(i)} - \alpha_{t(i)}}{s(i) - t(i)} \frac{s(i) - t(i)}{1 - \alpha_{t(i)}} x_0^\top \mathbb{E}_{q(x_{t(i)}|x_0)} [\delta_{x_{t(i)},m} \log \mu_\theta(x_{t(i)}, t(i))] \\
&= \int_{t(1)}^1 \frac{\alpha'_t}{1 - \alpha_t} x_0^\top \mathbb{E}_{q(x_t|x_0)} [\delta_{x_t,m} \log \mu_\theta(x_t, t)] dt.
\end{aligned}$$

E 避免未定义的 KL 散度

在定义前向过程时，出于数值稳定性的原因，我们通常不希望 α_1 恰好为 0，或者等效地， λ_1 为 ∞ 。相反，我们将 λ_1 设置为有限值，因此 α_1 具有较小的正值。这样就有一个问题，就是 $q(x_1|x_0)$ 的支持度不再是 $\{m\}$ ，而是变成了 $\{m, x_0\}$ 。因此， $q(x_1|x_0)$ 和 $p(x_1)$ 之间的 KL 散度未定义，因为 $q(x_1|x_0)$ 相对于 $p(x_1) = \delta_{x_1,m}$ 不是绝对连续的。为了解决这个问题，我们修改了先验分布 $p(x_1)$ ，使其支持所有 $m+1$ 值。其中一个选择是让

$$p(x_1) = \frac{\alpha_1}{m} \sum_{j \neq m} \delta_{x_1,j} + (1 - \alpha_1) \delta_{x_1,m}.$$

那么，先验 KL 散度项就变为

$$\begin{aligned}
\text{KL}(q(x_1|x_0)||p(x_1)) &= \sum_{x_1=0}^m q(x_1|x_0) \log \frac{q(x_1|x_0)}{p(x_1)} \\
&= \sum_{x_1=0}^m (\alpha_1 \delta_{x_1,x_0} + (1 - \alpha_1) \delta_{x_1,m}) \log \frac{\alpha_1 \delta_{x_1,x_0} + (1 - \alpha_1) \delta_{x_1=m}}{p(x_1)} \\
&= \alpha_1 \log \frac{\alpha_1}{\alpha_1/m} + (1 - \alpha_1) \log \frac{1 - \alpha_1}{1 - \alpha_1} \\
&= \alpha_1 \log m.
\end{aligned}$$

F Details of Training and Sampling with MD4

F.1 Training

Algorithm 1 A single step of training with MD4.

Input: data minibatch $\{x_t^i\}_{i=1}^B$, network $\mu_\theta(\cdot, t)$, masking schedule α_t
for $i = 1, \dots, B$ **do** (in parallel):
 $t_i \leftarrow \text{mod}(u + i/B, 1)$, $u \sim U[0, 1]$
for $n \in [N]$, mask out each token $x_0^{i,(n)}$ independently with probability $1 - \alpha_{t_i}$ to obtain $x_{t_i}^i$
for $n \in [N]$, if $x_{t_i}^{(n)} = m$, compute weighted cross entropy loss $\frac{\alpha'_{t_i}}{1 - \alpha_{t_i}} (x_0^{i,(n)})^\top \log \mu_\theta^{(n)}(x_{t_i}^i, t_i)$
Sum over all weighted cross entropy losses for mask positions and optimize via autodiff

F.2 Sampling

Algorithm 2 Unconditional and conditional generation (e.g., infilling) with MD4.

Input: Context sequence x^c of length N , with masks indicating the target areas for generation
Init: $\{t(i)\}_{i=0}^T \leftarrow \text{discretize}([0, 1])$, $x_{t(T)} \leftarrow x^c$
for $i = T, T-1, \dots, 1$ **do**
 $t \leftarrow t(i)$, $s \leftarrow t(i-1)$
for $n \in [N]$, if $x_t^{(n)} = m$, draw $x_s^{(n)} \sim \text{Cat}(\frac{\alpha_s - \alpha_t}{1 - \alpha_t} \mu_\theta^{(n)}(x_t, t) + \frac{1 - \alpha_s}{1 - \alpha_t} e_m)$ else $x_s^{(n)} \leftarrow x_t^{(n)}$
return x_0 .

G JAX Categorical Sampling and Implicit Top- p

We noticed that the following equivalent implementation of Alg. 2 leads to significantly worse sample quality in JAX:

Algorithm 3 Variant of Alg. 2 that yields lower sample quality when implemented in JAX.

Input: Token sequence x^c of length N , with masks indicating the target areas for generation
Init: $\{t(i)\}_{i=0}^T \leftarrow \text{discretize}([0, 1])$, $x_{t(T)} \leftarrow x^c$
for $i = T, T-1, \dots, 1$ **do**
 $t \leftarrow t(i)$, $s \leftarrow t(i-1)$
for $n \in [N]$ **do** (in parallel)
draw $u \sim U[0, 1]$
if $x_t^{(n)} = m$ **and** $u < \frac{\alpha_s - \alpha_t}{1 - \alpha_t}$ **then**
draw $x_s^{(n)} \sim \text{Cat}(\mu_\theta^{(n)}(x_t, t))$
else
 $x_s^{(n)} \leftarrow x_t^{(n)}$
return x_0 .

However, mathematically it is equivalent to Alg. 2 and should produce identical results. Our investigation revealed that the issue arises because Alg. 2 scales the output probabilities of μ_θ by a small factor $\frac{\alpha_s - \alpha_t}{1 - \alpha_t}$ as s is close to t , causing some categories to have very low probabilities. JAX, however, implements categorical sampling using Gumbel argmax, which is less numerically stable than methods like binary search. As a result, categories with low probabilities are rarely sampled, even when their cumulative probability is significant. In our experiment, we found that categories with probabilities below $1e-8$ are rarely sampled out of a total of 50K categories. Thus, Alg. 2 implicitly performs top- p sampling (with a dynamic p) under JAX’s categorical sampling, yielding better sample quality than Alg. 3 where μ_θ is not scaled by a small factor and has fewer categories truncated.

F MD4 训练和采样细节

F.1 培训

Algorithm 1 A single step of training with MD4.

Input: data minibatch $\{x_t^i\}_{i=1}^B$, network $\mu_\theta(\cdot, t)$, masking schedule α_t
for $i = 1, \dots, B$ **do** (in parallel):
 $t_i \leftarrow \text{mod}(u + i/B, 1)$, $u \sim U[0, 1]$
for $n \in [N]$, mask out each token $x_0^{i,(n)}$ independently with probability $1 - \alpha_{t_i}$ to obtain $x_{t_i}^i$
for $n \in [N]$, if $x_{t_i}^{(n)} = m$, compute weighted cross entropy loss $\frac{\alpha'_{t_i}}{1 - \alpha_{t_i}} (x_0^{i,(n)})^\top \log \mu_\theta^{(n)}(x_{t_i}^i, t_i)$
Sum over all weighted cross entropy losses for mask positions and optimize via autodiff

F.2 采样

Algorithm 2 Unconditional and conditional generation (e.g., infilling) with MD4.

Input: Context sequence x^c of length N , with masks indicating the target areas for generation
Init: $\{t(i)\}_{i=0}^T \leftarrow \text{discretize}([0, 1])$, $x_{t(T)} \leftarrow x^c$
for $i = T, T-1, \dots, 1$ **do**
 $t \leftarrow t(i)$, $s \leftarrow t(i-1)$
for $n \in [N]$, if $x_t^{(n)} = m$, draw $x_s^{(n)} \sim \text{Cat}(\frac{\alpha_s - \alpha_t}{1 - \alpha_t} \mu_\theta^{(n)}(x_t, t) + \frac{1 - \alpha_s}{1 - \alpha_t} e_m)$ else $x_s^{(n)} \leftarrow x_t^{(n)}$
return x_0 .

G JAX 分类采样和隐式置顶- p

我们注意到 Alg 的以下等效实现。2 导致 JAX 中的样本质量明显变差：

Algorithm 3 Variant of Alg. 2 that yields lower sample quality when implemented in JAX.

Input: Token sequence x^c of length N , with masks indicating the target areas for generation
Init: $\{t(i)\}_{i=0}^T \leftarrow \text{discretize}([0, 1])$, $x_{t(T)} \leftarrow x^c$
for $i = T, T-1, \dots, 1$ **do**
 $t \leftarrow t(i)$, $s \leftarrow t(i-1)$
for $n \in [N]$ **do** (in parallel)
draw $u \sim U[0, 1]$
if $x_t^{(n)} = m$ **and** $u < \frac{\alpha_s - \alpha_t}{1 - \alpha_t}$ **then**
draw $x_s^{(n)} \sim \text{Cat}(\mu_\theta^{(n)}(x_t, t))$
else
 $x_s^{(n)} \leftarrow x_t^{(n)}$
return x_0 .

然而，从数学上讲，它相当于 Alg. 2 并且应该产生相同的结果。我们的调查显示，问题的出现是因为 Alg. 2 将 μ_θ 的输出概率缩放一个小因子 $\frac{\alpha_s - \alpha_t}{1 - \alpha_t}$ ，因为 s 接近 t ，导致某些类别的概率非常低。然而，JAX 使用 Gumbel argmax 实现分类采样，这在数值上不如二分搜索等方法稳定。因此，即使概率较低的类别的累积概率很大，也很少对它们进行采样。在我们的实验中，我们发现概率低于 $1e-8$ 的类别很少在总共 50K 个类别中被采样。因此，阿尔格。2 在 JAX 的分类采样下隐式执行 top- p 采样（使用动态 p ），产生比 Alg 更好的样本质量。3，其中 μ_θ 未按小因子缩放，并且被截断的类别较少。

H Unifying Existing Masked Diffusion Models

H.1 The CTMC point of view

We first prove a lemma that connects the forward and reverse transition rate matrices. This follows from the results in [29] but we give a proof for completeness.

Lemma 2. *The forward transition rate matrix $Q(t)$ and the reverse transition rate matrix (given x_0) $R^{x_0}(t)$ satisfy:*

$$R^{x_0}(t)_{kj} = Q(t)_{jk} \frac{q_{t|0}(j|x_0)}{q_{t|0}(k|x_0)} \text{ for } j \neq k. \quad (20)$$

Proof Consider the reverse transition from time $t + \tau$ to t . For $j \neq k$, Bayes' rule yields

$$\begin{aligned} q(x_t = j | x_{t+\tau} = k, x_0) &= \frac{q(x_t = j | x_0) q(x_{t+\tau} = k | x_t = j)}{q(x_{t+\tau} = k | x_0)} \\ &= \frac{q(x_t = j | x_0) (\delta_{jk} + Q(t)_{jk} \tau + o(\tau))}{q(x_{t+\tau} = k | x_0)} \\ &\stackrel{\tau \rightarrow 0}{=} \delta_{kj} + \frac{q(x_t = j | x_0)}{q(x_t = k | x_0)} Q(t)_{jk} \tau + o(\tau). \end{aligned}$$

Then, it follows from the definition of the transition rate matrix that $R^{x_0}(t)_{kj} = Q(t)_{jk} \frac{q_{t|0}(j|x_0)}{q_{t|0}(k|x_0)}$. $\square$

Proposition 3. *We use the shorthand $R_\theta(t)_{kj}$ to denote the approximate reverse transition rate from the state k to j obtained by substituting our prediction model $\mu_\theta(k)$ for x_0 in $R^{x_0}(t)_{kj}$. Then, the continuous-time objective (4) can be equivalently expressed as*

$$\mathcal{L}_\infty = - \int_{t(1)}^1 \mathbb{E}_{q_{t|0}(k|x_0)} \left[R_\theta(t)_{kk} + \sum_{j \neq k} Q(t)_{kj} \log R_\theta(t)_{jk} \right] dt + C, \quad (21)$$

where C is a constant independent of θ .

Proof To rewrite our objective $\mathcal{L}_\infty$ with the transition rate matrices, we first go back to (19). There, instead of plugging in the explicit form of $R^{x_0}(t, s)$, we substitute it with (8) which leverages the transition rate $R^{x_0}(t)$. To simplify the notation, we assume $x_t = k$ and use the shorthand $R_\theta(t)_{kj} \triangleq R^{\mu_\theta(k)}(t)_{kj}$. We then have

$$\begin{aligned} &\text{KL}(q(x_{t-\Delta t} | x_t, x_0) \| p_\theta(x_{t-\Delta t} | x_t)) \\ &= \text{KL}(\text{Cat}(x_s; \bar{R}^{x_0}(t, t - \Delta t)^\top e_k) \| \text{Cat}(x_s; \bar{R}^{\mu_\theta(k)}(t, t - \Delta t)^\top e_k)) \\ &= \sum_{j=0}^m e_k^\top (I + R^{x_0}(t) \Delta t + o(\Delta t)) e_j \log \frac{e_k^\top (I + R^{x_0}(t) \Delta t + o(\Delta t)) e_j}{e_k^\top (I + R_\theta(t) \Delta t + o(\Delta t)) e_j} \\ &= (1 + R^{x_0}(t)_{kk} \Delta t) \log \frac{1 + R^{x_0}(t)_{kk} \Delta t + o(\Delta t)}{1 + R_\theta(t)_{kk} \Delta t + o(\Delta t)} \\ &\quad + \sum_{j \neq k} (R^{x_0}(t)_{kj} \Delta t) \log \frac{R^{x_0}(t)_{kj} \Delta t + o(\Delta t)}{R_\theta(t)_{kj} \Delta t + o(\Delta t)} + o(\Delta t) \\ &= (R^{x_0}(t)_{kk} - R_\theta(t)_{kk}) \Delta t + \sum_{j \neq k} (R^{x_0}(t)_{kj} \Delta t) \log \frac{R^{x_0}(t)_{kj} \Delta t + o(\Delta t)}{R_\theta(t)_{kj} \Delta t + o(\Delta t)} + o(\Delta t). \end{aligned}$$

H 统一现有的掩模扩散模型

H.1 CTMC 的观点

我们首先证明连接正向和反向转移率矩阵的引理。这是根据[29]中的结果得出的，但我们给出了完整性证明。

引理 2. *The forward transition rate matrix $Q(t)$ and the reverse transition rate matrix (given x_0) $R^{x_0}(t)$ satisfy:*

$$R^{x_0}(t)_{kj} = Q(t)_{jk} \frac{q_{t|0}(j|x_0)}{q_{t|0}(k|x_0)} \text{ for } j \neq k. \quad (20)$$

证明考虑从时间 $t + \tau$ 到 t 的反向转换。对于 $j \neq k$ ，贝叶斯规则得出

$$\begin{aligned} q(x_t = j | x_{t+\tau} = k, x_0) &= \frac{q(x_t = j | x_0) q(x_{t+\tau} = k | x_t = j)}{q(x_{t+\tau} = k | x_0)} \\ &= \frac{q(x_t = j | x_0) (\delta_{jk} + Q(t)_{jk} \tau + o(\tau))}{q(x_{t+\tau} = k | x_0)} \\ &\stackrel{\tau \rightarrow 0}{=} \delta_{kj} + \frac{q(x_t = j | x_0)}{q(x_t = k | x_0)} Q(t)_{jk} \tau + o(\tau). \end{aligned}$$

然后，根据转移率矩阵的定义可以得出 $R^{x_0}(t)_{kj} = Q(t)_{jk} \frac{q_{t|0}(j|x_0)}{q_{t|0}(k|x_0)}$ $\square$

命题 3. *We use the shorthand $R_\theta(t)_{kj}$ to denote the approximate reverse transition rate from the state k to j obtained by substituting our prediction model $\mu_\theta(k)$ for x_0 in $R^{x_0}(t)_{kj}$. Then, the continuous-time objective (4) can be equivalently expressed as*

$$\mathcal{L}_\infty = - \int_{t(1)}^1 \mathbb{E}_{q_{t|0}(k|x_0)} \left[R_\theta(t)_{kk} + \sum_{j \neq k} Q(t)_{kj} \log R_\theta(t)_{jk} \right] dt + C, \quad (21)$$

where C is a constant independent of θ .

证明为了用转移率矩阵重写我们的目标 $\mathcal{L}_\infty$ ，我们首先回到(19)。在那里，我们没有插入 $\bar{R}^{x_0}(t, s)$ 的显式形式，而是用利用转换率 $R^{x_0}(t)$ 的 (8) 代替它。为了简化表示法，我们假设 $x_t = k$ 并使用简写 $R_\theta(t)_{kj} \triangleq R^{\mu_\theta(k)}(t)_{kj}$ 。然后我们有

$$\begin{aligned} &\text{KL}(q(x_{t-\Delta t} | x_t, x_0) \| p_\theta(x_{t-\Delta t} | x_t)) \\ &= \text{KL}(\text{Cat}(x_s; \bar{R}^{x_0}(t, t - \Delta t)^\top e_k) \| \text{Cat}(x_s; \bar{R}^{\mu_\theta(k)}(t, t - \Delta t)^\top e_k)) \\ &= \sum_{j=0}^m e_k^\top (I + R^{x_0}(t) \Delta t + o(\Delta t)) e_j \log \frac{e_k^\top (I + R^{x_0}(t) \Delta t + o(\Delta t)) e_j}{e_k^\top (I + R_\theta(t) \Delta t + o(\Delta t)) e_j} \\ &= (1 + R^{x_0}(t)_{kk} \Delta t) \log \frac{1 + R^{x_0}(t)_{kk} \Delta t + o(\Delta t)}{1 + R_\theta(t)_{kk} \Delta t + o(\Delta t)} \\ &\quad + \sum_{j \neq k} (R^{x_0}(t)_{kj} \Delta t) \log \frac{R^{x_0}(t)_{kj} \Delta t + o(\Delta t)}{R_\theta(t)_{kj} \Delta t + o(\Delta t)} + o(\Delta t) \\ &= (R^{x_0}(t)_{kk} - R_\theta(t)_{kk}) \Delta t + \sum_{j \neq k} (R^{x_0}(t)_{kj} \Delta t) \log \frac{R^{x_0}(t)_{kj} \Delta t + o(\Delta t)}{R_\theta(t)_{kj} \Delta t + o(\Delta t)} + o(\Delta t). \end{aligned}$$

For the last identity, we have used the fact that $\log(1+x) = x + o(x)$. To obtain $\mathcal{L}_\infty$, we take the limit of $\mathcal{L}_T$ as $T \rightarrow \infty$, which is equivalent to letting $\Delta t = 1/T \rightarrow 0$. We obtain

$$\begin{aligned}\mathcal{L}_\infty &= \lim_{T \rightarrow \infty} \sum_{i=2}^T \mathbb{E}_{q(x_{t(i)}|x_0)} [\text{KL}(q(x_{s(i)}|x_{t(i)}, x_0) \| p_\theta(x_{s(i)}|x_{t(i)}))] \\ &= \lim_{T \rightarrow \infty} \sum_{i=2}^T \mathbb{E}_{q(x_{t(i)}|x_0)} \left[\left(R^{x_0}(t(i))_{kk} - R_\theta(t(i))_{kk} \right. \right. \\ &\quad \left. \left. + \sum_{j \neq k} R^{x_0}(t(i))_{kj} \log \frac{R^{x_0}(t(i))_{kj} \Delta t + o(\Delta t)}{R_\theta(t(i))_{kj} \Delta t + o(\Delta t)} \right) \Delta t + o(\Delta t) \right] \\ &= \int_{t(1)}^1 \mathbb{E}_{q_{t|0}(k|x_0)} \left[R^{x_0}(t)_{kk} - R_\theta(t)_{kk} + \sum_{j \neq k} R^{x_0}(t)_{kj} \log \frac{R^{x_0}(t)_{kj}}{R_\theta(t)_{kj}} \right] dt.\end{aligned}$$

Note that $R^{x_0}(t)$ is a constant matrix independent of θ . Absorbing all constant terms into C , we have

$$\mathcal{L}_\infty = - \int_{t(1)}^1 \mathbb{E}_{q_{t|0}(k|x_0)} \left[R_\theta(t)_{kk} + \sum_{j \neq k} R^{x_0}(t)_{kj} \log R_\theta(t)_{kj} \right] dt + C.$$

Next, we substitute $R^{x_0}(t)$ with the forward transition rate using Lemma 2:

$$\begin{aligned}\mathcal{L}_\infty &= - \int_{t(1)}^1 \mathbb{E}_{q_{t|0}(k|x_0)} \left[R_\theta(t)_{kk} + \sum_{j \neq k} Q(t)_{jk} \frac{q_{t|0}(j|x_0)}{q_{t|0}(k|x_0)} \log R_\theta(t)_{kj} \right] dt + C \\ &= - \int_{t(1)}^1 \left[\sum_{k=0}^m q_{t|0}(k|x_0) R_\theta(t)_{kk} + \sum_{k=0}^m \sum_{j \neq k} Q(t)_{jk} q_{t|0}(j|x_0) \log R_\theta(t)_{kj} \right] dt + C \\ &= - \int_{t(1)}^1 \left[\sum_{k=0}^m q_{t|0}(k|x_0) R_\theta(t)_{kk} + \sum_{k=0}^m \sum_{j \neq k} Q(t)_{kj} q_{t|0}(k|x_0) \log R_\theta(t)_{jk} \right] dt + C,\end{aligned}$$

where the last identity used the discrete analog to integration-by-part (or summation-by-part): $\sum_{k=0}^m \sum_{j \neq k} f(j, k) = \sum_{k=0}^m \sum_{j \neq k} f(k, j)$. Rearranging the terms then gives (21). $\square$

H.2 Differences from Campbell et al. [29]

Campbell et al. [29] used the first term of (21) as the training loss. A key limitation of this loss function is from the inner summation term

$$\sum_{j \neq k} Q(t)_{kj} \log R_\theta(t)_{jk}.$$

For single dimension case, the sum is analytically computable due to the sparse structure of $R_\theta(t)$ —if $x_t = k$ is mask, the second term disappears; otherwise the only possible neighbor j is a mask. However, for multidimensional data, j will represent all $N - 1$ neighbors in the forward process, i.e., the states we get from mask out a single unmasked dimension of $x_t = k$. Recall that $R_\theta(t)_{jk}$ is computed as substituting our neural network prediction model $\mu_\theta(j)$ for x_0 in $R^{x_0}(t)_{jk}$. Therefore, the summation together with $R_\theta(t)_{kk}$ requires N evaluations of $\mu_\theta(\cdot)$. This is prohibitive since the neural network model is usually expensive. To resolve this issue, Campbell et al. [29] proposed to rewrite the sum as

$$\mathbb{E}_{j \sim \tilde{q}(\cdot|k)} [Z_k \log R_\theta(t)_{jk}] \quad \text{where} \quad \tilde{q}(j|k) = \frac{Q(t)_{kj}}{Z_k}, Z_k \triangleq \sum_{j' \neq k} Q(t)_{kj'}$$

and estimate it through Monte Carlo. Taking into account the outer expectation under $q_{t|0}(k|x_0)$, the computation of the loss then becomes a doubly stochastic estimate (using $k \sim q_{t|0}(k|x_0)$ and $j \sim \tilde{q}(j|k)$) which suffers from large variance. In contrast, the form of our loss (4) only requires evaluating μ_θ once for a single stochastic estimation of the expectation w.r.t. $q(x_t|x_0)$.

对于最后一个恒等式，我们使用了 $\log(1+x) = x + o(x)$ 这一事实。为了得到 $\mathcal{L}_\infty$ ，我们将 $\mathcal{L}_T$ 的极限取为 $T \rightarrow \infty$ ，相当于让 $\Delta t = 1/T \rightarrow 0$ 。我们得到

$$\begin{aligned}\mathcal{L}_\infty &= \lim_{T \rightarrow \infty} \sum_{i=2}^T \mathbb{E}_{q(x_{t(i)}|x_0)} [\text{KL}(q(x_{s(i)}|x_{t(i)}, x_0) \| p_\theta(x_{s(i)}|x_{t(i)}))] \\ &= \lim_{T \rightarrow \infty} \sum_{i=2}^T \mathbb{E}_{q(x_{t(i)}|x_0)} \left[\left(R^{x_0}(t(i))_{kk} - R_\theta(t(i))_{kk} \right. \right. \\ &\quad \left. \left. + \sum_{j \neq k} R^{x_0}(t(i))_{kj} \log \frac{R^{x_0}(t(i))_{kj} \Delta t + o(\Delta t)}{R_\theta(t(i))_{kj} \Delta t + o(\Delta t)} \right) \Delta t + o(\Delta t) \right] \\ &= \int_{t(1)}^1 \mathbb{E}_{q_{t|0}(k|x_0)} \left[R^{x_0}(t)_{kk} - R_\theta(t)_{kk} + \sum_{j \neq k} R^{x_0}(t)_{kj} \log \frac{R^{x_0}(t)_{kj}}{R_\theta(t)_{kj}} \right] dt.\end{aligned}$$

请注意， $R^{x_0}(t)$ 是独立于 θ 的常量矩阵。将所有常数项吸收到 C 中，我们有

$$\mathcal{L}_\infty = - \int_{t(1)}^1 \mathbb{E}_{q_{t|0}(k|x_0)} \left[R_\theta(t)_{kk} + \sum_{j \neq k} R^{x_0}(t)_{kj} \log R_\theta(t)_{kj} \right] dt + C.$$

下一个我们使用 Lem 将 $R^{x_0}(t)$ 替换为前向转移率

马2:

$$\begin{aligned}\mathcal{L}_\infty &= - \int_{t(1)}^1 \mathbb{E}_{q_{t|0}(k|x_0)} \left[R_\theta(t)_{kk} + \sum_{j \neq k} Q(t)_{jk} \frac{q_{t|0}(j|x_0)}{q_{t|0}(k|x_0)} \log R_\theta(t)_{kj} \right] dt + C \\ &= - \int_{t(1)}^1 \left[\sum_{k=0}^m q_{t|0}(k|x_0) R_\theta(t)_{kk} + \sum_{k=0}^m \sum_{j \neq k} Q(t)_{jk} q_{t|0}(j|x_0) \log R_\theta(t)_{kj} \right] dt + C \\ &= - \int_{t(1)}^1 \left[\sum_{k=0}^m q_{t|0}(k|x_0) R_\theta(t)_{kk} + \sum_{k=0}^m \sum_{j \neq k} Q(t)_{kj} q_{t|0}(k|x_0) \log R_\theta(t)_{jk} \right] dt + C,\end{aligned}$$

其中最后一个恒等式使用离散模拟来进行按部分积分（或按部分求和）：

$\sum_{k=0} \sum_{j \neq k} f(j, k) = \sum_{k=0} \sum_{j \neq k} f(k, j)$ 。重新排列各项即可得出 (21)。

□

H.2 与 Campbell 等人的差异[29]

坎贝尔等人。[29]使用 (21) 的第一项作为训练损失。该损失函数的一个关键限制来自内部求和项

$$\sum_{j \neq k} Q(t)_{kj} \log R_\theta(t)_{jk}.$$

对于单维情况，由于 $R_\theta(t)$ 的稀疏结构，总和是可解析计算的 - 如果 $x_t = k$ 是掩码，则第二项消失；否则唯一可能的邻居 j 是一个掩码。然而，对于多维数据， j 将表示前向过程中的所有 $N-1$ 邻居，即我们从屏蔽 $x_t = k$ 的单个未屏蔽维度获得的状态。回想一下， $R_\theta(t)_{jk}$ 的计算方法是将我们的神经网络预测模型 $\mu_\theta(j)$ 替换为 $R^{x_0}(t)_{jk}$ 中的 x_0 。因此，与 $R_\theta(t)_{kk}$ 一起求和需要对 $\mu_\theta(\cdot)$ 进行 N 评估。这是令人望而却步的，因为神经网络模型通常很昂贵。为了解决这个问题，坎贝尔等人。[29]建议将总和重写为

$$\mathbb{E}_{j \sim \tilde{q}(\cdot|k)} [Z_k \log R_\theta(t)_{jk}] \quad \text{where} \quad \tilde{q}(j|k) = \frac{Q(t)_{kj}}{Z_k}, Z_k \triangleq \sum_{j' \neq k} Q(t)_{kj'}$$

并通过蒙特卡洛进行估计。考虑到 $q_{t|0}(k|x_0)$ 下的外部期望，损失的计算就变成了双随机估计（使用 $k \sim q_{t|0}(k|x_0)$ 和 $j \sim \tilde{q}(j|k)$ ），其方差很大。相比之下，我们的损失 (4) 的形式只需要评估 μ_θ 一次即可对期望进行一次随机估计。 $q(x_t|x_0)$ 。

H.3 Score parameterization

We provide a simpler derivation of the score-based loss [32, 35] below. We start from the form of the ELBO in (21) and rewrite it as

$$\mathcal{L}_\infty = \int_{t(1)}^1 \mathbb{E}_{q_{t|0}(k|x_0)} \left[\sum_{j \neq k} \left(R^{\mu_\theta}(t)_{kj} - R^{x_0}(t)_{kj} + R^{x_0}(t)_{kj} \log \frac{R^{x_0}(t)_{kj}}{R^{\mu_\theta}(t)_{kj}} \right) \right] dt. \quad (22)$$

For the last identity we used the zero-row-sum property of transition rate matrix:

$$R^{x_0}(t)_{kk} = - \sum_{j \neq k} R^{x_0}(t)_{kj}.$$

If we plug (20) into (22) and reparameterize with a score model

$$s_\theta(x_t)_j \triangleq \frac{q_{t|0}(j|\mu_\theta(x_t))}{q(x_t|\mu_\theta(x_t))}, \quad (23)$$

we recover the score entropy loss function from Lou et al. [32], Benton et al. [35]:

$$\mathcal{L}_\infty = \int_{t(1)}^1 \mathbb{E}_{q_{t|0}(k|x_0)} \left[\sum_{j \neq k} Q(t)_{jk} \left(s_\theta(k)_j - \frac{q_{t|0}(j|x_0)}{q_{t|0}(k|x_0)} \log s_\theta(k)_j + \psi \left(\frac{q_{t|0}(j|x_0)}{q_{t|0}(k|x_0)} \right) \right) \right] dt, \quad (24)$$

where $\psi(y) \triangleq y \log y - y$. Note that our derivation above is different and simpler than that of Campbell et al. [29] (which Lou et al. [32] is based on) since we leverage the conditional reverse transition rate given x_0 instead of the transition rate matrix of the reverse process. We can further simplify the loss with the following relationship between the conditional score and x_0 :

$$\frac{q_{t|0}(j|x_0)}{q_{t|0}(k|x_0)} = \frac{x_0^\top \bar{Q}(t) e_j}{x_0^\top \bar{Q}(t) e_k} = \frac{\alpha_t}{1 - \alpha_t} x_0^\top e_j \text{ for } k = m, j \neq k. \quad (25)$$

Note that only the result under the case $k = m$ is needed. This is because when x_t is unmasked, at any time between 0 and t , the state must stay unchanged and remain x_0 . As a result, $\text{KL}(q(x_{t-\Delta t}|x_t, x_0) \| p_\theta(x_{t-\Delta t}|x_t)) = 0$ for $x_t \neq m$. From (15), we know $Q(t)_{jk} = \beta(t)(\delta_{mk} - \delta_{jk})$. Combining (25) and (24), we get

$$\mathcal{L}_\infty = \int_{t(1)}^1 \beta(t) \left(\mathbb{E}_{q_{t|0}(k|x_0)} \left[\delta_{mk} \left(\sum_{j \neq k} s_\theta(k)_j - \frac{\alpha_t}{1 - \alpha_t} x_0^\top \log s_\theta(k) \right) \right] + \psi \left(\frac{\alpha_t}{1 - \alpha_t} \right) \right) dt. \quad (26)$$

Further, we can show the connection between (26) and (4) by reverting the score parameterization to a mean parameterization using (23), or equivalently $s_\theta(x_t)_j = \frac{\alpha_t}{1 - \alpha_t} \mu_\theta(x_t)^\top e_j$. By doing so, we obtain

$$\mathcal{L}_\infty = \int_{t(1)}^1 \beta(t) \left(\mathbb{E}_{q_{t|0}(k|x_0)} \left[\delta_{mk} \left(\sum_{j \neq k} s_\theta(k)_j - \frac{\alpha_t}{1 - \alpha_t} x_0^\top \log \mu_\theta(k) \right) \right] + \frac{\alpha_t}{1 - \alpha_t} \right) dt.$$

Observing that

$$\sum_{j \neq m} s_\theta(m)_j = \frac{\alpha_t}{1 - \alpha_t}, \quad (27)$$

we conclude that this recovers the objective in (4). Interestingly, in Lou et al. [32] the score parameterization is not constrained to satisfy (27). That means the learned reverse model might be incompatible with the forward process.

Below, we prove Proposition 1 using the result from Eq. (25).

Proof of Proposition 1

$$\begin{aligned} \frac{q_t(j)}{q_t(m)} &= \frac{\sum_{x_0} q_{t|0}(j|x_0) q(x_0)}{q_t(m)} = \frac{\sum_{x_0} q_{t|0}(j|x_0) q_{0|t}(x_0|m)}{q_{t|0}(m|x_0)} = \mathbb{E}_{x_0|x_t=m} \left[\frac{q_{t|0}(j|x_0)}{q_{t|0}(m|x_0)} \right] \\ &= \mathbb{E}_{x_0|x_t=m} \left[\frac{\alpha_t}{1 - \alpha_t} x_0^\top e_j \right] = \frac{\alpha_t}{1 - \alpha_t} \mathbb{E}[x_0|x_t = m]^\top e_j. \end{aligned}$$

□

H.3 Score parameterization

我们在下面提供了基于分数的损失的更简单的推导 [32, 35]。我们从(21)中的ELBO形式开始，将其重写为

$$\mathcal{L}_\infty = \int_{t(1)}^1 \mathbb{E}_{q_{t|0}(k|x_0)} \left[\sum_{j \neq k} \left(R^{\mu_\theta}(t)_{kj} - R^{x_0}(t)_{kj} + R^{x_0}(t)_{kj} \log \frac{R^{x_0}(t)_{kj}}{R^{\mu_\theta}(t)_{kj}} \right) \right] dt. \quad (22)$$

对于最后一个恒等式，我们使用转移率矩阵的零行和性质：

$$R^{x_0}(t)_{kk} = - \sum_{j \neq k} R^{x_0}(t)_{kj}.$$

如果我们将 (20) 代入 (22) 并使用评分模型重新参数化

$$s_\theta(x_t)_j \triangleq \frac{q_{t|0}(j|\mu_\theta(x_t))}{q(x_t|\mu_\theta(x_t))}, \quad (23)$$

我们从 Lou 等人那里恢复了得分熵损失函数。[32]，本顿等人。[35]：

$$\mathcal{L}_\infty = \int_{t(1)}^1 \mathbb{E}_{q_{t|0}(k|x_0)} \left[\sum_{j \neq k} Q(t)_{jk} \left(s_\theta(k)_j - \frac{q_{t|0}(j|x_0)}{q_{t|0}(k|x_0)} \log s_\theta(k)_j + \psi \left(\frac{q_{t|0}(j|x_0)}{q_{t|0}(k|x_0)} \right) \right) \right] dt, \quad (24)$$

其中 $\psi(y) \triangleq y \log y - y$ 。请注意，我们上面的推导与 Campbell 等人的推导不同且更简单。[29] (Lou 等人[32] 的基础)，因为我们利用给定 x_0 的条件反向转移率，而不是反向过程的转移率矩阵。我们可以通过条件分数和 x_0 之间的关系进一步简化损失：

$$\frac{q_{t|0}(j|x_0)}{q_{t|0}(k|x_0)} = \frac{x_0^\top \bar{Q}(t) e_j}{x_0^\top \bar{Q}(t) e_k} = \frac{\alpha_t}{1 - \alpha_t} x_0^\top e_j \text{ for } k = m, j \neq k. \quad (25)$$

请注意，仅需要案例 $k = m$ 下的结果。这是因为当 x_t 未被屏蔽时，在 0 到 t 之间的任何时间，状态必须保持不变并保持 x_0 。结果，对于 $x_t \neq m$ ， $\text{KL}(q(x_{t-\Delta t}|x_t, x_0) \| p_\theta(x_{t-\Delta t}|x_t)) = 0$ 。从(15)，我们知道 $Q(t)_{jk} = \beta(t)(\delta_{mk} - \delta_{jk})$ 。结合(25)和(24)，我们得到

$$\mathcal{L}_\infty = \int_{t(1)}^1 \beta(t) \left(\mathbb{E}_{q_{t|0}(k|x_0)} \left[\delta_{mk} \left(\sum_{j \neq k} s_\theta(k)_j - \frac{\alpha_t}{1 - \alpha_t} x_0^\top \log s_\theta(k) \right) \right] + \psi \left(\frac{\alpha_t}{1 - \alpha_t} \right) \right) dt. \quad (26)$$

此外，我们可以通过使用 (23) 或等效的 $s_\theta(x_t)_j = \frac{\alpha_t}{1 - \alpha_t} \mu_\theta(x_t)^\top e_j$ 将分数参数化恢复为平均参数化来显示 (26) 和 (4) 之间的联系。通过这样做，我们得到

$$\mathcal{L}_\infty = \int_{t(1)}^1 \beta(t) \left(\mathbb{E}_{q_{t|0}(k|x_0)} \left[\delta_{mk} \left(\sum_{j \neq k} s_\theta(k)_j - \frac{\alpha_t}{1 - \alpha_t} x_0^\top \log \mu_\theta(k) \right) \right] + \frac{\alpha_t}{1 - \alpha_t} \right) dt.$$

观察到

$$\sum_{j \neq m} s_\theta(m)_j = \frac{\alpha_t}{1 - \alpha_t}, \quad (27)$$

我们得出结论，这恢复了 (4) 中的目标。有趣的是，在 Lou 等人中。[32] 分数参数化不限于满足 (27)。这意味着学习到的反向模型可能与正向过程不兼容。

下面，我们使用式 (1) 的结果证明命题 1。(25)。

命题1的证明

$$\begin{aligned} \frac{q_t(j)}{q_t(m)} &= \frac{\sum_{x_0} q_{t|0}(j|x_0)q(x_0)}{q_t(m)} = \frac{\sum_{x_0} q_{t|0}(j|x_0)q_0(x_0|m)}{q_{t|0}(m|x_0)} = \mathbb{E}_{x_0|x_t=m} \left[\frac{q_{t|0}(j|x_0)}{q_{t|0}(m|x_0)} \right] \\ &= \mathbb{E}_{x_0|x_t=m} \left[\frac{\alpha_t}{1 - \alpha_t} x_0^\top e_j \right] = \frac{\alpha_t}{1 - \alpha_t} \mathbb{E}[x_0|x_t = m]^\top e_j. \end{aligned}$$

□

H.4 Other related work.

MaskGIT [39]. MaskGIT is a diffusion-inspired iterative denoising model for discrete image tokens obtained through models such as VQ-VAE [70]. Training of MaskGIT follows the steps: (a) Sample $t \in [0, 1]$. (b) Given a mask scheduling function $\gamma(t)$, sample $\gamma(t)N$ tokens to place masks. (c) For data x_0 of size $(m + 1) \times N$ and the partially masked state x_t , minimize the negative log-likelihood

$$\mathcal{L}_{\text{MaskGIT}} = - \int_0^1 \mathbb{E}_{x_t} \left[\sum_{n: x_t^{(n)} = m} (x_0^{(n)})^\top \log \mu_\theta^{(n)}(x_t, t) \right] dt. \quad (28)$$

Our forward process satisfies $q_{t|0}(m|x_0) = 1 - \alpha_t$. Therefore, when we set the mask scheduling function as $\gamma(t) = 1 - \alpha_t$ we obtain a loss similar to (5) without the $\frac{\alpha'_t}{1-\alpha_t}$ weighting. Note that there remains a difference in the sampling distribution of x_t : in the masked diffusion forward process, tokens are sampled independently and do not necessarily yield exactly $(1 - \alpha_t)N$ mask tokens at time t , though the expected number is $(1 - \alpha_t)N$. One might be interested in whether the uniform weighting can be recovered by selecting an appropriate schedule α_t . However, solving α_t such that $\alpha'_t = \alpha_t - 1$ yields $\alpha_t = ce^t + 1$ and there is no c that satisfies both $\alpha_0 = 1$ and $\alpha_1 = 0$. This shows that training with the MaskGIT loss (28) may not be faithfully optimizing the model likelihood.

Discrete flow matching [49]. For the linear schedule $\alpha_t = 1 - t$, our reverse transition rate matrix (8) conditioned on x_0 is:

$$R^{x_0}(t) = -\frac{\alpha'_t}{1 - \alpha_t} e_m(x_0 - e_m)^\top = \frac{1}{t} e_m(x_0 - e_m)^\top.$$

This is the same as the conditional reverse transition rate used in Campbell et al. [49, Eq. (22)]—note that their time t is reversed, and the rate matrix was therefore in the form $R^{x_0}(t) = \frac{1}{1-t} e_m(x_0 - e_m)^\top$.

SDDM [30]. Sun et al. [30] proposed a pseudo-likelihood-like objective for training discrete diffusion models that can also be applied to masked diffusion. However, their objective encounters the same challenge as Campbell et al. [29] — requiring N passes of the mask prediction model. To mitigate this, they introduced a new transformer architecture, which unfortunately leads to some performance degradation.

Blackout diffusion [50]. Santos et al. [50] proposed a “blackout” diffusion process that gradually diffuses images to a black state. While this approach is similar to masked diffusion on binary data, key differences emerge when dealing with larger state spaces. In their method, image pixel intensities gradually fade out, whereas ours directly transition to a mask state. Our method offers more flexibility, being applicable to general discrete state spaces without requiring predefined structural relationships. It also demonstrates competitive performance in image generation, achieving this without knowing pixel value proximity.

SUNDAE [51, 71]. Unlike masked diffusion, SUNDAE uniformly corrupts data with random tokens in the vocab (known as uniform discrete diffusion [14]). Additionally, it uses a second loss term from cross entropy between clean data and 1-step unrolled model prediction. Similar ideas have been proposed in [72].

I Details for state-dependent rates

I.1 Derivations and time continuous limit

All derivations in this section assume that x_t is a single token, while for N tokens the masked diffusion with state-dependent rates factorises across the N tokens. Learning from data of N tokens using variational inference is discussed in App. I.2.

H.4 其他相关工作。

MaskGIT [39]。MaskGIT 是一种受扩散启发的迭代去噪模型，用于通过 VQ-VAE [70] 等模型获得的离散图像标记。MaskGIT 的训练遵循以下步骤：(a) 样本 $t \in [0, 1]$ 。(b) 给定掩码调度函数 $\gamma(t)$ ，采样 $\gamma(t)N$ 标记来放置掩码。(c) 对于大小为 $(m+1) \times N$ 的数据 x_0 和部分屏蔽状态 x_t ，最小化负对数似然

$$\mathcal{L}_{\text{MaskGIT}} = - \int_0^1 \mathbb{E}_{x_t} \left[\sum_{n: x_t^{(n)} = m} (x_0^{(n)})^\top \log \mu_\theta^{(n)}(x_t, t) \right] dt. \quad (28)$$

我们的前向过程满足 $q_{t|0}(m|x_0) = 1 - \alpha_t$ 。因此，当我们将掩码调度函数设置为 $\gamma(t) = 1 - \alpha_t$ 时，我们会得到类似于 (5) 的损失，而无需 $\frac{\alpha'_t}{1-\alpha_t}$ 权重。请注意， x_t 的采样分布仍然存在差异：在掩码扩散前向过程中，令牌是独立采样的，并且不一定在时间 t 准确产生 $(1 - \alpha_t)N$ 掩码令牌，尽管预期数量是 $(1 - \alpha_t)N$ 。人们可能会对是否可以通过选择适当的时间表 α_t 来恢复统一权重感兴趣。然而，求解 α_t 使得 $\alpha'_t = \alpha_t - 1$ 产生 $\alpha_t = ce^t + 1$ ，并且不存在同时满足 $\alpha_0 = 1$ 和 $\alpha_1 = 0$ 的 c 。这表明使用 MaskGIT 损失 (28) 进行训练可能无法忠实地优化模型似然。

离散流匹配[49]。对于线性调度 $\alpha_t = 1 - t$ ，我们以 x_0 为条件的反向转移率矩阵 (8) 为：

$$R^{x_0}(t) = -\frac{\alpha'_t}{1-\alpha_t} e_m(x_0 - e_m)^\top = \frac{1}{t} e_m(x_0 - e_m)^\top.$$

这与 Campbell 等人使用的条件反向转换率相同。[49, 等式。(22)]—请注意，它们的时间 t 是相反的，因此速率矩阵的形式为 $R^{x_0}(t) = \frac{1}{1-t} e_m(x_0 - e_m)^\top$ 。

SDDM [30]。孙等人。[30]提出了一种类似伪似然的目标来训练离散扩散模型，该模型也可以应用于掩模扩散。然而，他们的目标遇到了与坎贝尔等人相同的挑战。[29] - 需要掩模预测模型的 N 遍。为了解决这个问题，他们引入了一种新的变压器架构，不幸的是，这导致了一些性能下降。

停电扩散[50]。桑托斯等人。[50]提出了一种“黑化”扩散过程，逐渐将图像扩散到黑色状态。虽然这种方法类似于二进制数据上的掩码扩散，但在处理更大的状态空间时出现了关键差异。在他们的方法中，图像像素强度逐渐淡出，而我们的方法直接过渡到掩模状态。我们的方法提供了更大的灵活性，适用于一般的离散状态空间，而不需要预定义的结构关系。它还展示了图像生成方面的竞争性能，在不知道像素值接近度的情况下实现这一点。

圣代 [51, 71]。与掩码扩散不同，SUNDAE 使用词汇中的随机标记统一破坏数据（称为均匀离散扩散 [14]）。此外，它还使用来自干净数据和一步展开模型预测之间的交叉熵的第二个损失项。[72]中也提出了类似的想法。

I 各州费率详情

I.1 导数和时间连续极限

本节中的所有推导均假设 x_t 是单个令牌，而对于 N 令牌，具有状态相关速率的掩码扩散在 N 令牌中进行因式分解。App 中讨论了使用变分推理从 N 标记的数据中学习。I.2.

Given the forward transition $q(x_t|x_s)$ and marginal $q(x_s|x_0)$ derived in main text (Sec. 6) The reversal given x_0 is $q(x_s|x_t, x_0) = \text{Cat}(x_s; \bar{R}^{x_0}(t, s)^\top x_t)$ for

$$\bar{R}^{x_0}(t, s)_{jk} = \begin{cases} \left(\frac{\alpha_s - \alpha_t}{1 - \alpha_t} \right)^\top x_0 x_0^\top e_k & j = m, k \neq m \\ \left(\frac{1 - \alpha_s}{1 - \alpha_t} \right)^\top x_0 & j = m, k = m \\ \delta_{jk} & j \neq m. \end{cases}$$

or alternatively can be written as

$$\begin{aligned} q(x_s|x_t, x_0) &= \frac{q(x_t|x_s)q(x_s|x_0)}{q(x_t|x_0)} \\ &= \frac{\left[\frac{\alpha_t^\top x_s}{\alpha_s^\top x_s} x_s^\top x_t + \left(1 - \frac{\alpha_t^\top x_s}{\alpha_s^\top x_s} \right) e_m^\top x_t \right] [\alpha_s^\top x_0 x_0^\top x_s + (1 - \alpha_s^\top x_0) e_m^\top x_s]}{[\alpha_t^\top x_0 x_0^\top x_t + (1 - \alpha_t^\top x_0) e_m^\top x_t]}. \end{aligned} \quad (29)$$

To simplify this expression we consider the two cases: either $x_t = m$ (i.e. x_t is mask) or $x_t \neq m$ where in the second case $x_t = x_0$. For the case $x_t = m$, the denominator in (29) simplifies as

$$q(x_t = m|x_0) = 1 - \alpha_t^\top x_0$$

due to $x_0^\top x_t = 0$ since $x_0 \neq m$, i.e. the observed token x_0 cannot be a mask. Then given that $x_t = m$ the probability that $x_s = x_t = m$ is

$$\frac{1 - \alpha_s^\top x_0}{1 - \alpha_t^\top x_0} = \frac{(1 - \alpha_s)^\top x_0}{(1 - \alpha_t)^\top x_0} = \left(\frac{1 - \alpha_s}{1 - \alpha_t} \right)^\top x_0 \quad (30)$$

while the remaining probability for $x_s = x_0 \neq m$ is

$$\frac{(\alpha_s - \alpha_t)^\top x_0}{1 - \alpha_t^\top x_0} = \frac{(\alpha_s - \alpha_t)^\top x_0}{(1 - \alpha_t)^\top x_0} = \left(\frac{\alpha_s - \alpha_t}{1 - \alpha_t} \right)^\top x_0. \quad (31)$$

Then, combining (30) and (31) to write $q(x_s|x_t = m, x_0)$ in an unified way yields the expression (11) in the main Sec. 6. In the second case, when $x_t = x_0 \neq m$, $q(x_s|x_t \neq m, x_0)$ from (29) simplifies dramatically and it becomes $q(x_s|x_t \neq m, x_0) = x_t^\top x_s$ which is a point mass that sets $x_s = x_t$.

Derivation of the continuous-time limit of the loss in (12). To simplify the notation, we let $\xi_{s,t} \triangleq \frac{\alpha_s - \alpha_t}{1 - \alpha_t}$. We first compute the KL divergence terms in the discrete-time ELBO as

$$\begin{aligned} &\text{KL}(q(x_s|x_t, x_0) \| p_\theta(x_s|x_t)) \\ &= \begin{cases} \sum_{x_s=0}^m q(x_s|x_t, x_0) \log \frac{q(x_s|x_t, x_0)}{p_\theta(x_s|x_t)} & x_t = m \\ 0 & x_t \neq m \end{cases} \\ &= \delta_{x_t, m} \left[\sum_{k \neq m} \xi_{s,t}^\top x_0 x_0^\top e_k \log \frac{\xi_{s,t}^\top x_0 x_0^\top e_k}{\xi_{s,t}^\top \text{diag}(\mu_\theta(x_t, t)) e_k} + (1 - \xi_{s,t})^\top x_0 \log \frac{(1 - \xi_{s,t})^\top x_0}{(1 - \xi_{s,t})^\top \mu_\theta(x_t, t)} \right] \\ &= \delta_{x_t, m} \left[-\xi_{s,t}^\top x_0 x_0^\top \log \mu_\theta(x_t, t) + (1 - \xi_{s,t})^\top x_0 \log \frac{(1 - \xi_{s,t})^\top x_0}{(1 - \xi_{s,t})^\top \mu_\theta(x_t, t)} \right]. \end{aligned}$$

Let $\Delta_t \triangleq \frac{1}{T} = t(i) - s(i)$ for all i . Plugging $\alpha_{t-\Delta_t} = \alpha_t - \alpha'_t \Delta_t + o(\Delta_t)$ into the above formula and letting $\gamma_t = \frac{\alpha'_t}{1 - \alpha_t}$, we get

$$\begin{aligned} &\text{KL}(q(x_s|x_t, x_0) \| p_\theta(x_s|x_t)) \\ &= \delta_{x_t, m} \left[\gamma_t^\top x_0 x_0^\top \log \mu_\theta(x_t, t) \Delta_t + (1 + \gamma_t^\top x_0 \Delta_t) \cdot \log \frac{1 + \gamma_t^\top x_0 \Delta_t + o(\Delta_t)}{1 + \gamma_t^\top \mu_\theta(x_t, t) \Delta_t + o(\Delta_t)} + o(\Delta_t) \right] \\ &= \delta_{x_t, m} \left[\gamma_t^\top x_0 x_0^\top \log \mu_\theta(x_t, t) \Delta_t + (1 + \gamma_t^\top x_0 \Delta_t) (\gamma_t^\top x_0 \Delta_t - \gamma_t^\top \mu_\theta(x_t, t) \Delta_t + o(\Delta_t)) + o(\Delta_t) \right] \\ &= \delta_{x_t, m} \left[\gamma_t^\top x_0 x_0^\top \log \mu_\theta(x_t, t) \Delta_t + \gamma_t^\top x_0 \Delta_t - \gamma_t^\top \mu_\theta(x_t, t) \Delta_t + o(\Delta_t) \right] \\ &= \delta_{x_t, m} \cdot \gamma_t^\top (x_0 x_0^\top \log \mu_\theta(x_t, t) + x_0 - \mu_\theta(x_t, t)) \Delta_t + o(\Delta_t). \end{aligned}$$

给定正文中导出的正向转移 $q(x_t|x_s)$ 和边际 $q(x_s|x_0)$ (第6节), 给定 x_0 的反转为 $q(x_s|x_t, x_0) = \text{Cat}(x_s; \bar{R}^{x_0}(t, s)^\top x_t)$

$$\bar{R}^{x_0}(t, s)_{jk} = \begin{cases} \left(\frac{\alpha_s - \alpha_t}{1 - \alpha_t}\right)^\top x_0 x_0^\top e_k & j = m, k \neq m \\ \left(\frac{1 - \alpha_s}{1 - \alpha_t}\right)^\top x_0 & j = m, k = m \\ \delta_{jk} & j \neq m. \end{cases}$$

或者也可以写成

$$\begin{aligned} q(x_s|x_t, x_0) &= \frac{q(x_t|x_s)q(x_s|x_0)}{q(x_t|x_0)} \\ &= \frac{\left[\frac{\alpha_t^\top x_s}{\alpha_s^\top x_s} x_s^\top x_t + (1 - \frac{\alpha_t^\top x_s}{\alpha_s^\top x_s}) e_m^\top x_t\right] [\alpha_s^\top x_0 x_0^\top x_s + (1 - \alpha_s^\top x_0) e_m^\top x_s]}{[\alpha_t^\top x_0 x_0^\top x_t + (1 - \alpha_t^\top x_0) e_m^\top x_t]}. \end{aligned} \quad (29)$$

为了简化这个表达式, 我们考虑两种情况: 要么 $x_t = m$ (即 x_t 是掩码) 或 $x_t \neq m$, 其中第二种情况是 $x_t = x_0$. 对于情况 $x_t = m$, (29) 中的分母简化为

$$q(x_t = m|x_0) = 1 - \alpha_t^\top x_0$$

由于自 $x_0 \neq m$ 以来 $x_0^\top x_t = 0$, 即观察到的标记 x_0 不能是掩码。然后给定 $x_t = m$ 的概率是 $x_s = x_t = m$

$$\frac{1 - \alpha_s^\top x_0}{1 - \alpha_t^\top x_0} = \frac{(1 - \alpha_s)^\top x_0}{(1 - \alpha_t)^\top x_0} = \left(\frac{1 - \alpha_s}{1 - \alpha_t}\right)^\top x_0 \quad (30)$$

而 $x_s = x_0 \neq m$ 的剩余概率是

$$\frac{(\alpha_s - \alpha_t)^\top x_0}{1 - \alpha_t^\top x_0} = \frac{(\alpha_s - \alpha_t)^\top x_0}{(1 - \alpha_t)^\top x_0} = \left(\frac{\alpha_s - \alpha_t}{1 - \alpha_t}\right)^\top x_0. \quad (31)$$

然后, 将(30)和(31)组合起来统一写成 $q(x_s|x_t = m, x_0)$, 得到主Sec中的表达式(11)。6. 在第二种情况下, 当 $x_t = x_0 \neq m$ 时, (29) 中的 $q(x_s|x_t \neq m, x_0)$ 急剧简化, 变成 $q(x_s|x_t \neq m, x_0) = x_t^\top x_s$, 它是设置 $x_s = x_t$ 的点质量。

(12) 中损失的连续时间极限的推导。为了简化符号, 我们让 $\xi_{s,t} \triangleq \frac{\alpha_s - \alpha_t}{1 - \alpha_t}$ 。我们首先计算离散时间 ELBO 中的 KL 散度项为

$$\begin{aligned} &\text{KL}(q(x_s|x_t, x_0) \| p_\theta(x_s|x_t)) \\ &= \begin{cases} \sum_{x_s=0}^m q(x_s|x_t, x_0) \log \frac{q(x_s|x_t, x_0)}{p_\theta(x_s|x_t)} & x_t = m \\ 0 & x_t \neq m \end{cases} \\ &= \delta_{x_t, m} \left[\sum_{k \neq m} \xi_{s,t}^\top x_0 x_0^\top e_k \log \frac{\xi_{s,t}^\top x_0 x_0^\top e_k}{\xi_{s,t}^\top \text{diag}(\mu_\theta(x_t, t)) e_k} + (1 - \xi_{s,t})^\top x_0 \log \frac{(1 - \xi_{s,t})^\top x_0}{(1 - \xi_{s,t})^\top \mu_\theta(x_t, t)} \right] \\ &= \delta_{x_t, m} \left[-\xi_{s,t}^\top x_0 x_0^\top \log \mu_\theta(x_t, t) + (1 - \xi_{s,t})^\top x_0 \log \frac{(1 - \xi_{s,t})^\top x_0}{(1 - \xi_{s,t})^\top \mu_\theta(x_t, t)} \right]. \end{aligned}$$

让 $\Delta_t \triangleq \frac{1}{T} = t(i) - s(i)$ 代表所有 i 。将 $\alpha_{t-\Delta_t} = \alpha_t - \alpha'_t \Delta_t + o(\Delta_t)$ 代入上面的公式并让 $\gamma_t = \frac{\alpha'_t}{1 - \alpha_t}$, 我们得到

$$\begin{aligned} &\text{KL}(q(x_s|x_t, x_0) \| p_\theta(x_s|x_t)) \\ &= \delta_{x_t, m} \left[\gamma_t^\top x_0 x_0^\top \log \mu_\theta(x_t, t) \Delta_t + (1 + \gamma_t^\top x_0 \Delta_t) \cdot \log \frac{1 + \gamma_t^\top x_0 \Delta_t + o(\Delta_t)}{1 + \gamma_t^\top \mu_\theta(x_t, t) \Delta_t + o(\Delta_t)} + o(\Delta_t) \right] \\ &= \delta_{x_t, m} \left[\gamma_t^\top x_0 x_0^\top \log \mu_\theta(x_t, t) \Delta_t + (1 + \gamma_t^\top x_0 \Delta_t) (\gamma_t^\top x_0 \Delta_t - \gamma_t^\top \mu_\theta(x_t, t) \Delta_t + o(\Delta_t)) + o(\Delta_t) \right] \\ &= \delta_{x_t, m} \left[\gamma_t^\top x_0 x_0^\top \log \mu_\theta(x_t, t) \Delta_t + \gamma_t^\top x_0 \Delta_t - \gamma_t^\top \mu_\theta(x_t, t) \Delta_t + o(\Delta_t) \right] \\ &= \delta_{x_t, m} \cdot \gamma_t^\top (x_0 x_0^\top \log \mu_\theta(x_t, t) + x_0 - \mu_\theta(x_t, t)) \Delta_t + o(\Delta_t). \end{aligned}$$

Therefore,

$$\begin{aligned}
& \lim_{T \rightarrow \infty} \sum_{i=2}^T \mathbb{E}_{q(x_{t(i)}|x_0)} [\text{KL}(q(x_{s(i)}|x_{t(i)}, x_0) \| p_\theta(x_{s(i)}|x_{t(i)}))] \\
&= \lim_{T \rightarrow \infty} \sum_{i=2}^T \mathbb{E}_{q(x_{t(i)}|x_0)} [\delta_{x_{t(i)}, m} \cdot \gamma_t^\top (x_0 x_0^\top \log \mu_\theta(x_{t(i)}, t(i)) + x_0 - \mu_\theta(x_{t(i)}, t(i))) \Delta t + o(\Delta t)] \\
&= \int_{t(1)}^1 \gamma_t^\top \mathbb{E}_{q(x_{t(i)}|x_0)} [\delta_{x_t, m} \cdot (x_0 x_0^\top \log \mu_\theta(x_t, t) + x_0 - \mu_\theta(x_t, t))] dt.
\end{aligned}$$

Letting $t(1) \rightarrow 0$ proves the result.

I.2 Training and gradient estimation

The model is applied to data consisted of N tokens where $x_0 = (x_0^1, \dots, x_0^N)$ and where each state in the masked diffusion is $x_t = (x_t^1, \dots, x_t^N)$. The reverse generated model has a factorizing transition conditional of the form $\prod_{n=1}^N p_\theta(x_s^{(n)}|x_t)$ where $p_\theta(x_s^{(n)}|x_t) = q(x_s^{(n)}|x_t^{(n)}, \mu_\theta^{(n)}(x_t, t))$ has a form that depends on whether $x_t^{(n)} = m$ or $x_t^{(n)} \neq m$. For the first case:

$$p_\theta(x_s^{(n)}|x_t^{(n)} = m, \{x_t^{(k)}\}_{k \neq n}) = \left(\frac{\mathbf{1} - \alpha_s}{\mathbf{1} - \alpha_t} \right)^\top \mu_\theta^{(n)}(x_t, t) e_m^\top x_s^{(n)} + \left(\frac{\alpha_s - \alpha_t}{\mathbf{1} - \alpha_t} \right)^\top \text{diag}(\mu_\theta^{(n)}(x_t, t)) x_s^{(n)},$$

where $\mu_\theta^{(n)}(x_t, t) = \text{softmax}(f_\theta(x_t))$ is a $m+1$ dimensional probability vector modelled by a NN (where the final value is constrained to be zero since $\mu_\theta^{(n)}(x_t, t)$ is a reconstruction of $x_0^{(n)}$ which cannot be mask, so in practice the NN classifier needs to have a softmax output only over the m actual token classes). Crucially, note that the NN classifier receives as input the full state x_t of all tokens, while additional time features to encode t are also included. When $x_t^{(n)} \neq m$ the reverse transition model is set to be $p_\theta(x_s|x_t^{(n)} \neq m, \{x_t^{(k)}\}_{k \neq n}) = (x_t^{(n)})^\top x_s^{(n)}$ which matches precisely $q(x_s^{(n)}|x_t^{(n)} = m, x_0^{(n)}) = (x_t^{(n)})^\top x_s^{(n)}$ from the forward process.

The full negative lower bound for state-dependent rates and assuming N tokens is given by

$$\mathcal{L}_\infty^{(N)} = \int_0^1 \left(\frac{\alpha'_t}{1 - \alpha_t} \right)^\top \mathbb{E}_{q(x_t|x_0)} \left[\sum_{n: x_t^{(n)} = m} (x_0^{(n)} - \mu_\theta^{(n)}(x_t, t) + x_0^{(n)} (x_0^{(n)})^\top \log \mu_\theta^{(n)}(x_t, t)) \right] dt. \quad (32)$$

Given that each $\alpha_{t,i} = 1 - t^{w_i}$, the reverse model becomes

$$p_\theta(x_s^{(n)}|x_t^{(n)} \neq m, \{x_t^{(k)}\}_{k \neq n}) = (e^{w \log \frac{s}{t}})^\top \mu_\theta^{(n)}(x_t, t) e_m^\top x_s^{(n)} + (1 - e^{w \log \frac{s}{t}})^\top \text{diag}(\mu_\theta^{(n)}(x_t, t)) x_s^{(n)},$$

where w is the $m+1$ dimensional vector of all w_i 's. Note that the probability of $x_s^{(n)}$ staying in the mask state, i.e., $x_s^{(n)} = m$ depends on the full x_t and it is given by $(e^{w \log \frac{s}{t}})^\top \mu_\theta^{(n)}(x_t, t) = \sum_{i=0}^{m-1} e^{w_i \log \frac{s}{t}} \mu_\theta^{(n)}(x_t, t)_i$ while the probability for $x_s^{(n)}$ to take a certain non-mask token value i is $(1 - e^{w_i \log \frac{s}{t}}) \mu_\theta^{(n)}(x_t, t)_i$. The gradient wrt t is $\alpha'_{t,i} = -w_i t^{w_i-1}$ and $\frac{\alpha'_{t,i}}{1 - \alpha_{t,i}} = -\frac{w_i}{t}$ the above loss is written as

$$\mathcal{L}_\infty^{(N)} = - \int_0^1 \frac{1}{t} w^\top \mathbb{E}_{q(x_t|x_0)} \left[\sum_{n: x_t^{(n)} = m} (x_0^{(n)} - \mu_\theta^{(n)}(x_t, t) + x_0^{(n)} (x_0^{(n)})^\top \log \mu_\theta^{(n)}(x_t, t)) \right] dt,$$

where w is the vector of all w_i 's. An unbiased gradient over the NN parameters θ is straightforward to obtain since we just need to sample one time point t and an $x_t \sim q(x_t|x_0)$ to approximate the integral and expectation and then use the gradient:

$$-\nabla_\theta \sum_{n: x_t^{(n)} = m} \frac{1}{t} w^\top \left(x_0^{(n)} - \mu_\theta^{(n)}(x_t, t) + x_0^{(n)} (x_0^{(n)})^\top \log \mu_\theta^{(n)}(x_t, t) \right).$$

The gradient wrt the w parameters is more complex since these parameters appear also in the discrete distribution $q(x_t|x_0)$ which is not reparametrizable. To deal with this we need REINFORCE

所以,

$$\begin{aligned}
& \lim_{T \rightarrow \infty} \sum_{i=2}^T \mathbb{E}_{q(x_{t(i)}|x_0)} [\text{KL}(q(x_{s(i)}|x_{t(i)}, x_0) \| p_\theta(x_{s(i)}|x_{t(i)}))] \\
&= \lim_{T \rightarrow \infty} \sum_{i=2}^T \mathbb{E}_{q(x_{t(i)}|x_0)} [\delta_{x_{t(i)}, m} \cdot \gamma_t^\top (x_0 x_0^\top \log \mu_\theta(x_{t(i)}, t(i)) + x_0 - \mu_\theta(x_{t(i)}, t(i))) \Delta t + o(\Delta t)] \\
&= \int_{t(1)}^1 \gamma_t^\top \mathbb{E}_{q(x_{t(i)}|x_0)} [\delta_{x_t, m} \cdot (x_0 x_0^\top \log \mu_\theta(x_t, t) + x_0 - \mu_\theta(x_t, t))] dt.
\end{aligned}$$

令 $t(1) \rightarrow 0$ 证明结果。

I.2 训练和梯度估计

该模型应用于由 N 标记组成的数据, 其中 $x_0 = (x_0^1, \dots, x_0^N)$ 且掩码扩散中的每个状态为 $x_t = (x_t^1, \dots, x_t^N)$ 。反向生成模型具有 $\prod_{n=1}^N p_\theta(x_s^{(n)}|x_t)$ 形式的因式分解转换条件, 其中 $p_\theta(x_s^{(n)}|x_t) = q(x_s^{(n)}|x_t^{(n)}, \mu_\theta^{(n)}(x_t, t))$ 的形式取决于 $x_t^{(n)} = m$ 还是 $x_t^{(n)} \neq m$ 。对于第一种情况:

$$p_\theta(x_s^{(n)}|x_t^{(n)} = m, \{x_t^{(k)}\}_{k \neq n}) = \left(\frac{\mathbf{1} - \alpha_s}{\mathbf{1} - \alpha_t} \right)^\top \mu_\theta^{(n)}(x_t, t) e_m^\top x_s^{(n)} + \left(\frac{\alpha_s - \alpha_t}{\mathbf{1} - \alpha_t} \right)^\top \text{diag}(\mu_\theta^{(n)}(x_t, t)) x_s^{(n)},$$

其中 $\mu_\theta^{(n)}(x_t, t) = \text{softmax}(f_\theta(x_t))$ 是由 NN 建模的 $m+1$ 维概率向量 (其中最终值被限制为零, 因为 $\mu_\theta^{(n)}(x_t, t)$ 是无法掩码的 $x_0^{(n)}$ 的重建, 因此在实践中, NN 分类器只需要在 m 实际令牌类上有一个 softmax 输出)。至关重要的是, 请注意, NN 分类器接收所有标记的完整状态 x_t 作为输入, 同时还包括用于编码 t 的附加时间特征。当 $x_t^{(n)} \neq m$ 时, 反向转换模型设置为 $p_\theta(x_s|x_t^{(n)} \neq m, \{x_t^{(k)}\}_{k \neq n}) = (x_t^{(n)})^\top x_s^{(n)}$, 它与前向过程中的 $q(x_s^{(n)}|x_t^{(n)} = m, x_0^{(n)}) = (x_t^{(n)})^\top x_s^{(n)}$ 精确匹配。

全负洛 我们受状态相关率的约束, 并假设 N 代币是 给出的

$$\mathcal{L}_\infty^{(N)} = \int_0^1 \left(\frac{\alpha'_t}{1 - \alpha_t} \right)^\top \mathbb{E}_{q(x_t|x_0)} \left[\sum_{n: x_t^{(n)} = m} (x_0^{(n)} - \mu_\theta^{(n)}(x_t, t) + x_0^{(n)} (x_0^{(n)})^\top \log \mu_\theta^{(n)}(x_t, t)) \right] dt. \quad (32)$$

鉴于每个 $\alpha_{t,i} = 1 - t^{w_i}$, 逆向模型变为

$$p_\theta(x_s^{(n)}|x_t^{(n)} \neq m, \{x_t^{(k)}\}_{k \neq n}) = (e^{w \log \frac{s}{t}})^\top \mu_\theta^{(n)}(x_t, t) e_m^\top x_s^{(n)} + (1 - e^{w \log \frac{s}{t}})^\top \text{diag}(\mu_\theta^{(n)}(x_t, t)) x_s^{(n)},$$

其中 w 是所有 w_i 的 $m+1$ 维向量。请注意, $x_s^{(n)}$ 保持在掩码状态 (即 $x_s^{(n)} = m$) 的概率取决于完整的 x_t , 它由 $(e^{w \log \frac{s}{t}})^\top \mu_\theta^{(n)}(x_t, t) = \sum_{i=0}^{m-1} e^{w_i \log \frac{s}{t}} \mu_\theta^{(n)}(x_t, t)_i$ 给出, 而 $x_s^{(n)}$ 取某个非掩码标记值 i 的概率为 $(1 - e^{w_i \log \frac{s}{t}}) \mu_\theta^{(n)}(x_t, t)_i$ 。 t 的梯度为 $\alpha'_{t,i} = -w_i t^{w_i-1}$, 而 $\frac{\alpha'_{t,i}}{1 - \alpha_{t,i}} = -\frac{w_i}{t}$ 上述损失可写为

$$\mathcal{L}_\infty^{(N)} = - \int_0^1 \frac{1}{t} w^\top \mathbb{E}_{q(x_t|x_0)} \left[\sum_{n: x_t^{(n)} = m} (x_0^{(n)} - \mu_\theta^{(n)}(x_t, t) + x_0^{(n)} (x_0^{(n)})^\top \log \mu_\theta^{(n)}(x_t, t)) \right] dt,$$

其中 w 是所有 w_i 的向量。神经网络参数 θ 上的无偏梯度很容易获得, 因为我们只需要采样一个时间点 t 和一个 $x_t \sim q(x_t|x_0)$ 来近似积分和期望, 然后使用梯度:

$$-\nabla_\theta \sum_{n: x_t^{(n)} = m} \frac{1}{t} w^\top \left(x_0^{(n)} - \mu_\theta^{(n)}(x_t, t) + x_0^{(n)} (x_0^{(n)})^\top \log \mu_\theta^{(n)}(x_t, t) \right).$$

w 参数的梯度更加复杂, 因为这些参数也出现在不可重新参数化的离散分布 $q(x_t|x_0)$ 中。为了解决这个问题, 我们需要加强

unbiased gradients [73, 74], and in our implementation we consider REINFORCE leave-one-out (RLOO) [53, 54] with two samples. Firstly, the exact gradient wrt w of the exact loss is written as

$$-\int_0^1 \frac{1}{t} \mathbb{E}_{q(x_t|x_0)} [g(x_t, x_0)] dt - \int_0^1 \frac{1}{t} \mathbb{E}_{q(x_t|x_0)} [f(x_t, x_0) \nabla_w \log q(x_t|x_0)] dt. \quad (33)$$

where

$$g(x_t, x_0) = \sum_{n: x_t^{(n)}=m} (x_0^{(n)} - \mu_\theta^{(n)}(x_t, t) + x_0^{(n)} (x_0^{(n)})^\top \log \mu_\theta^{(n)}(x_t, t)), \quad f(x_t, x_0) = w^\top g(x_t, x_0).$$

Note that $g(x_t, x_0)$ is a vector while $f(x_t, x_0)$ is a scalar. The left term in (33) is easy since it just requires sampling t and $x_t \sim q(x_t|x_0)$, while the right term is the REINFORCE term which could have high variance. For this second term we use RLOO with two samples x_t^1, x_t^2 and construct the unbiased estimate

$$-\frac{1}{2t} (\nabla_w \log q(x_t^1|x_0) - \nabla_w \log q(x_t^2|x_0)) [f(x_t^1, x_0) - f(x_t^2, x_0)].$$

Thus, the overall unbiased gradient for w we use is

$$-\frac{1}{2t} \{g(x_t^1, x_0) + g(x_t^2, x_0) + (\nabla_w \log q(x_t^1|x_0) - \nabla_w \log q(x_t^2|x_0)) [f(x_t^1, x_0) - f(x_t^2, x_0)]\}.$$

J Experimental Details

In all experiments, the model is trained with a continuous-time loss while samples are drawn from the discrete-time reverse model of 1000 timesteps unless otherwise noted. We used an exponential moving average factor 0.9999 for all evaluation including sample generation.

J.1 text8

We followed the standard dataset split as in Austin et al. [14], Lou et al. [32] and trained our models on text chunks of length 256 for 1 million steps with batch size 512. All models in the table used a standard 12-layer transformer architecture unless otherwise noted. Our transformer has also the same number of heads (12) and hidden dimension (784) as in Austin et al. [14], Lou et al. [32].

We used the continuous-time ELBO and drew one sample of t for each data to estimate the integral. To reduce the variance of training, we used the same antithetic sampling trick described in Kingma et al. [33] for continuous diffusion models. We used the linear masking schedule $\alpha_t = 1 - t$ and added a small shift $\epsilon = 10^{-4}$ when t is close to 0 and 1 to ensure numerical stability. The shifted schedule is $\alpha_t = (1 - 2\epsilon)(1 - t) + \epsilon$. The shift leads to a support mismatch between $q(x_1|x_0)$ and the prior $p(x_1)$, leading to an undefined KL divergence term. We explain in app. E how to modify the prior distribution to allow small uniform probabilities in non-mask states to mitigate this problem. The shift leads to a non-zero reconstruction term and KL divergence term for the prior distribution but both are of negligible scale so we can safely ignore them when reporting the ELBO.

We used a cosine learning rate schedule with a linear warm up of 2000 steps. We applied channel-wise dropout of rate 0.05 and used AdamW optimizer with learning rate 0.0003 and a weight decay factor of 0.03. Our model is trained on 16 TPU-v5 lite for less than a day.

J.2 OpenWebText

We kept 2% of the original training set for validation. Our small and medium transformer model have the same number of layers, heads, and hidden dimensions as in Lou et al. [32] and our tokenizer was also kept the same with a vocabulary size of around 50k. The training objective, masking schedule and other architectural choices were kept the same with the text8 experiment. We kept the training hyperparameters the same as text8 experiment except that we reduced the dropout rate to 0.02.

J.3 FineWeb-Edu

We kept the same training setup as the OpenWebText experiments. Our transformer models have the same number of layers, heads, and hidden dimensions as those of GPT-2 models. We use the same GPT-2 tokenizer.

无偏梯度 [73, 74], 在我们的实现中, 我们考虑使用两个样本强化留一法 (RLOO) [53, 54]。首先, 精确损失的精确梯度 wrt w 写为

$$-\int_0^1 \frac{1}{t} \mathbb{E}_{q(x_t|x_0)} [g(x_t, x_0)] dt - \int_0^1 \frac{1}{t} \mathbb{E}_{q(x_t|x_0)} [f(x_t, x_0) \nabla_w \log q(x_t|x_0)] dt. \quad (33)$$

在哪里

$$g(x_t, x_0) = \sum_{n: x_t^{(n)}=m} (x_0^{(n)} - \mu_\theta^{(n)}(x_t, t) + x_0^{(n)} (x_0^{(n)})^\top \log \mu_\theta^{(n)}(x_t, t)), \quad f(x_t, x_0) = w^\top g(x_t, x_0).$$

请注意, $g(x_t, x_0)$ 是向量, 而 $f(x_t, x_0)$ 是标量。(33) 中的左边项很简单, 因为它只需要采样 t 和 $x_t \sim q(x_t|x_0)$, 而右边项是可能具有高方差的 REINFORCE 项。对于第二项, 我们使用 RLOO 和两个样本 x_t^1, x_t^2 并构建无偏估计

$$-\frac{1}{2t} (\nabla_w \log q(x_t^1|x_0) - \nabla_w \log q(x_t^2|x_0)) [f(x_t^1, x_0) - f(x_t^2, x_0)].$$

因此, 我们使用的 w 的整体无偏梯度是

$$-\frac{1}{2t} \{g(x_t^1, x_0) + g(x_t^2, x_0) + (\nabla_w \log q(x_t^1|x_0) - \nabla_w \log q(x_t^2|x_0)) [f(x_t^1, x_0) - f(x_t^2, x_0)]\}.$$

J 实验细节

在所有实验中, 除非另有说明, 模型均采用连续时间损失进行训练, 而样本则从 1000 个时间步长的离散时间反向模型中抽取。我们使用指数移动平均因子 0.9999 进行所有评估, 包括样本生成。

J.1 文本8

我们遵循 Austin 等人中的标准数据集分割。[14], 卢等人。[32] 并在长度为 256 的文本块上训练我们的模型, 执行 100 万步, 批量大小为 512。除非另有说明, 表中的所有模型都使用标准的 12 层转换器架构。我们的变压器也具有与 Austin 等人相同的头数 (12) 和隐藏维度 (784)。[14], 卢等人。[32]。

我们使用连续时间 ELBO 并为每个数据抽取一个 t 样本来估计积分。为了减少训练的方差, 我们使用了 Kingma 等人中描述的相同的对偶采样技巧。[33] 连续扩散模型。我们使用线性掩码方案 $\alpha_t = 1 - t$ 并在 t 接近 0 和 1 时添加一个小偏移 $\epsilon = 10^{-4}$ 以确保数值稳定性。更改后的时间表为 $\alpha_t = (1 - 2\epsilon)(1 - t) + \epsilon$ 。这种转变导致 $q(x_1|x_0)$ 和先前的 $p(x_1)$ 之间的支持不匹配, 从而导致未定义的 KL 散度项。我们在应用程序中解释。E 如何修改先验分布以允许非掩模状态中的小均匀概率来缓解此问题。这种转变导致先验分布的非零重构项和 KL 散度项, 但两者的规模都可以忽略不计, 因此我们在报告 ELBO 时可以安全地忽略它们。

我们使用余弦学习率计划, 线性预热为 2000 步。我们应用了速率为 0.05 的通道级 dropout, 并使用学习率为 0.0003 和权重衰减因子为 0.03 的 AdamW 优化器。我们的模型在 16 TPU-v5 lite 上训练了不到一天。

J.2 开放网络文本

我们保留了 2% 的原始训练集用于验证。我们的中小型变压器模型具有与 Lou 等人相同的层数、头数和隐藏尺寸。[32] 我们的分词器也保持不变, 词汇量约为 50k。训练目标、掩蔽时间表和其他架构选择与 text8 实验保持相同。我们保持训练超参数与 text8 实验相同, 只是将 dropout 率降低到 0.02。

J.3 FineWeb-Edu

我们保留与 OpenWebText 实验相同的训练设置。我们的 Transformer 模型具有与 GPT-2 模型相同的层数、头数和隐藏维度。我们使用相同的 GPT-2 分词器。

For Hellaswag evaluation, we concatenate question with each answer option, tokenize the concatenated string, pad to the length of 1024. The padded token sequence gets fed to our MD4 model’s loss function for likelihood evaluation. We average 32 Monte Carlo samples to reduce variance. The answer with the highest likelihood estimate is the model’s prediction.

J.4 Images

We used the same linear masking schedule as in previous experiments in all likelihood results. We used the same U-Net plus self-attention architectures from the continuous diffusion model described in Kingma et al. [33] for CIFAR-10, except that we did not use Fourier feature inputs and added an additional input embedding layer with embedding size the same as the hidden dimension of the model. For ImageNet 64×64 , we reduced the number of residual blocks (in one side of the U-Net structure) from 64 to 48 and added a 12-layer diffusion transformer [75] with 768 hidden dimension and 12 heads in the middle.

For both datasets we used AdamW optimizer and trained for 2M iterations. We used learning rate 0.0004, batch size 256, weight decay factor 0.01 for CIFAR-10 and learning rate 0.0002, batch size 512, weight decay factor 0.03 for ImageNet 64×64 . The learning rate follows a cosine annealing after 100 warm up steps. Our CIFAR-10 model is trained on 32 TPU-v5 lite for 24 hours. Our ImageNet- 64×64 model is trained on 256 TPU-v5 lite for 3.5 days.

As explained in Sec. 4, we have observed that the cosine schedule leads to better sample quality so we used it to train a cheaper model for sample visualization. This model differs from the one that achieves best likelihood in that we used 8 residual blocks (in one side of the UNet structure) and a 20-layer diffusion transformer in the middle. All other configurations are kept the same.

K Additional Results

K.1 Sample quality evaluation by GPT-2

We use the GPT-2 large model to evaluate the perplexity of samples generated by our model, following Lou et al. [32]. Results are shown in Fig. 8.

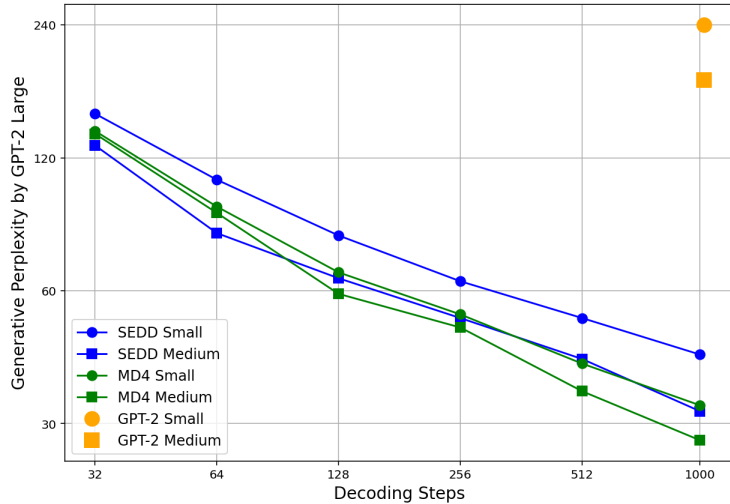


Figure 8: Generative perplexity evaluated by GPT-2 Large following Lou et al. [32]. We compare MD4 against the GPT-2 checkpoint (autoregressive baseline) and SEDD (the previous best discrete diffusion model on this task) in generating 1024-token text sequences. We investigate the effects of two orthogonal factors on sample quality: model size and decoding steps. The numbers for GPT-2 and SEDD are from Lou et al. [32].

对于 Hellaswag 评估，我们将问题与每个答案选项连接起来，对连接的字符串进行标记，填充到 1024 的长度。填充的标记序列被馈送到我们的 MD4 模型的损失函数中以进行似然评估。我们对 32 个蒙特卡洛样本进行平均以减少方差。具有最高似然估计的答案是模型的预测。

J.4 图像

我们在所有似然结果中使用了与之前实验相同的线性掩蔽方案。我们使用了与 Kingma 等人描述连续扩散模型相同的 U-Net 加上自注意力架构。[33]对于 CIFAR-10，不同之处在于我们没有使用傅立叶特征输入，而是添加了一个额外的输入嵌入层，其嵌入大小与模型的隐藏维度相同。对于 ImageNet 64×64 ，我们将残差块的数量（在 U-Net 结构的一侧）从 64 个减少到 48 个，并添加了一个 12 层扩散变换器 [75]，隐藏维度为 768，中间有 12 个头。

对于这两个数据集，我们使用 AdamW 优化器并进行 2M 迭代训练。对于 CIFAR-10，我们使用学习率 0.0004、批量大小 256、权重衰减因子 0.01；对于 ImageNet 64×64 ，我们使用学习率 0.0002、批量大小 512、权重衰减因子 0.03。学习率遵循 100 个预热步骤后的余弦退火。我们的 CIFAR-10 模型在 32 TPU-v5 lite 上训练了 24 小时。我们的 ImageNet- 64×64 模型在 256 TPU-v5 lite 上训练了 3.5 天。

正如第 2 节中所解释的。如图 4 所示，我们观察到余弦时间表可以带来更好的样本质量，因此我们用它来训练更便宜的样本可视化模型。该模型与实现最佳似然模型的不同之处在于，我们使用了 8 个残差块（位于 U-Net 结构的一侧）和中间的 20 层扩散变换器。所有其他配置保持不变。

K 附加结果

K.1 GPT-2 的样本质量评估

遵循 Lou 等人的研究，我们使用 GPT-2 大模型来评估我们的模型生成的样本的困惑度。[32]。结果如图 8 所示。

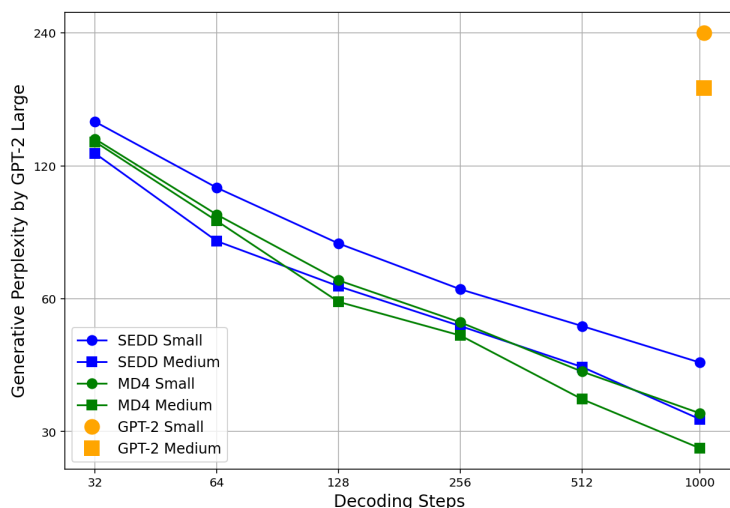


图 8：遵循 Lou 等人的 GPT-2 Large 评估生成困惑度[32]。我们在生成 1024 个标记的文本序列时将 MD4 与 GPT-2 检查点（自回归基线）和 SEDD（此任务之前最好的离散扩散模型）进行比较。我们研究了两个正交因素对样本质量的影响：模型大小和解码步骤。GPT-2 和 SEDD 的数字来自 Lou 等人。[32]。

K.2 Perplexity on OpenWebText validation set

Tab. 5 reports the final perplexity number achieved on OpenWebText validation set, corresponding to Fig. 4.

Table 5: Perplexity on OpenWebText validation set.

Size	Method	Perplexity ($\downarrow$)
Small	Gaussian Diffusion	≤ 27.28
	SEDD Absorb (reimpl.)	≤ 24.10
	MD4 (Ours)	≤ 22.13
	GenMD4 (Ours)	$\leq \mathbf{21.80}$
Medium	MD4 (Ours)	$\leq \mathbf{16.64}$

K.3 FID evaluation of MD4 trained on ImageNet 64×64

We provide the FID numbers corresponding to Fig. 2 in Tab. 6.

Table 6: FID of 50k samples generated by MD4 trained on ImageNet 64×64, corresponding to Fig. 2. Top three rows show results from an unconditional model, while the bottom row is from a model conditioned on class labels. Uniform discretization grid is used in Alg. 2 unless otherwise noted.

Method	Timesteps T			
	64	128	256	512
Linear α_t	193.81	128.18	72.94	50.21
Linear α_t , cosine grid	42.07	25.16	18.31	18.22
Cosine α_t	47.46	23.84	17.8	18.74
Cosine α_t , class conditional	30.75	11.39	7.13	7.8

K.4 Additional unconditional generation from MD4 trained on ImageNet 64×64

We provide more unconditional generation results from our pixel-level modeling experiments on ImageNet 64×64 in Fig. 9.

K.5 Additional unconditional generation from MD4 trained on OpenWebText

Below we include two unconditioned text samples generated by our MD4 Medium model trained on OpenWebText.

K.5.1 MD4-M unconditional sample 1: 1024 tokens

like, I don't have to be alive? Sometimes there are things that are too real and you're really supposed to experience them. So that's a good feeling. That is the scary thing. Not actually, being able to experience things, being able to do these things, when you're doing them, which, for most people having to wake in a dream is something that seems the most significant, and then you think about it the next day. It's like the hope of the future, and you wake up right now thinking about it. What happens is,, then you have to stop and think about it and then all of a sudden, somebody always says, "You're dreaming."

And sometimes I wonder if this is a good time to teach your gut instincts to your actors when you're doing a show like this. Because even on this particular show, it feels like everyone's been through this all the time before, if even a few years ago. I mean, if you're doing a show together, at least not on continuous development, you you're a vet. I mean, you should really be along.

K.2 OpenWebText 验证集的困惑

标签。图 5 报告了 OpenWebText 验证集上实现的最终困惑数，对应于图 4。

表 5: OpenWebText 验证集的困惑度。

Size	Method	Perplexity ($\downarrow$)
Small	Gaussian Diffusion	≤ 27.28
	SEDD Absorb (reimpl.)	≤ 24.10
	MD4 (Ours)	≤ 22.13
	GenMD4 (Ours)	$\leq \mathbf{21.80}$
Medium	MD4 (Ours)	$\leq \mathbf{16.64}$

K.3 在 ImageNet 64×64 上训练的 MD4 的 FID 评估

我们提供了与表 2 中对应的 FID 编号。6.

表 6: 在 ImageNet 64×64 上训练的 MD4 生成的 50k 样本的 FID，对应于图 2。顶部三行显示来自无条件模型的结果，而底部行来自以类标签为条件的模型。Alg中使用了均匀离散化网格。2 除非另有说明。

Method	Timesteps T			
	64	128	256	512
Linear α_t	193.81	128.18	72.94	50.21
Linear α_t , cosine grid	42.07	25.16	18.31	18.22
Cosine α_t	47.46	23.84	17.8	18.74
Cosine α_t , class conditional	30.75	11.39	7.13	7.8

K.4 在 ImageNet 64×64 上训练的 MD4 的附加无条件生成

我们在图 9 中的 ImageNet 64×64 上进行像素级建模实验，提供了更多无条件生成结果。

K.5 在 OpenWebText 上训练的 MD4 的附加无条件生成

下面我们提供了两个由我们在 OpenWebText 上训练的 MD4 Medium 模型生成的无条件文本样本。

K.5.1 MD4-M无条件样本1: 1024个令牌

比如，我不必活着？有时候有些事情太真实了，你真的应该去体验它们。所以这是一种很好的感觉。这就是可怕的事情。实际上并非如此，能够体验事物，能够做这些事情，当你在做这些事情时，对于大多数必须在梦中醒来的人来说，这似乎是最重要的事情，然后你第二天就会想到它。这就像未来的希望，你现在醒来就会想到它。发生的事情是，然后你必须停下来想一想，然后突然间，有人总是说，“你在做梦。”

有时我想知道，当你做这样的节目时，现在是否是向演员传授直觉的好时机。因为即使在这个特定的节目中，感觉每个人以前都经历过这种情况，甚至是几年前。我的意思是，如果你们一起做一个节目，至少不是持续发展，那么你就是一名兽医。我的意思是，你真的应该在一起。



Figure 9: More unconditional samples from MD4 trained on ImageNet 64×64 .

If you're not sure, well --

VS: I'm working on that one.

Did any of you guys feel that an instinct could work? I thought, "Well, because you didn't do 'Deadwood' you should stop doing this." But when I read the story for the first time, I thought, "I think this is going to work." What I can't picture is a way to hold this apart.



图 9：在 ImageNet 64×64 上训练的 MD4 的更多无条件样本。

如果你不确定，那么——

VS：我正在研究那个。

你们中有人觉得本能有用吗？我想，“好吧，因为你没有拍《戴德伍德》，所以你应该停止这样做。”但当我第一次读到这个故事时，我想：“我认为这会起作用。”我无法想象如何将其分开。

VS: That's me. It's what we have to do. So do we. When we wrote the first episode, we wrote a script that we felt like me and myself would want to see. I knew that I wanted to be able to be in something -- and I wanted to be able to take refuge in something that was real, that you could see and just really step out of yourself. And then I saw it. Then, you get rehearsing it and doing it. And then I actually started shooting. I think I knew I didn't think it was going to be good. But, I know it was good. And now people are talked about because it's not good enough.

Growing up, you say that you just completely hated the show, "Lost." Isn't that what you wish for at the end of the day?

VS: I don't like the concept.

And so there's a lot that you don't know about that, so I think for me to have had these ideas, if you didn't understand even that it was coming out of this world that doesn't exist, we might never get together.

It's so weird. This happened to happen at the same time?

VS: Yes. It happened to happen at basically the same time.

Nobody's even had a show or had a movie/come out of the movie, but ...

VS: If I'm going to pretend I'm definitely not you and have to live through that stuff, I don't think I'm going to swallow that. I didn't expect it to do quite that long.

There are always things now that happen with 'Deadwood' where you don't know where it's going to end up next time, but I think there are occasions now where we have to keep the fight, even if 'Lost' was pretty consistent in the mindset and the form.

VS: I'm glad that we did fight the odds, because we should have understood that there was a direct link. But there was almost a sense of not that we had showed up on the same day, we know we work in the same pieces, but a lot of stuff we don't know about. Some of it, we need to deal with. We also just have to accept the language, and there are a lot of things where we take from them and we do this what they did because we want to

K.5.2 MD4-M unconditional sample 2: 1024 tokens

the groups let recreational vehicles use the three roads that will stay open in the meantime of fighting off the permit. "The purpose of the permit is to make sure that we work with the NPS and made roadways and rest areas. We're not just scaring guys kind of messing around." Community plans to build an urban bike facility marched forward at the ongoing staff meeting of the King County Commission.

Trail will be finished just south of the Greenview 5.

Instead of continuing with a pedestrian and bike trail to the MBTA's campus, these two trails could bridle the areas from Market to 14 and carry communities closer.

"This project will provide a car-free path to King County," said Andrew Weed. It's been put the brakes on in the past several months, but there are those residents still skeptical.

"I've addressed some of the community concerns that've been raised. They've expressed some of their concerns. I don't think it's terribly reasonable from a

VS: 那就是我。这是我们必须做的。我们也是。当我们写第一集时，我们写了一个我们觉得我和我自己都想看的剧本。我知道我想要成为某种东西——我想要能够在真实的东西中寻求庇护，你可以看到并且真正走出自己。然后我看到了。然后，你就可以排练并付诸实践。然后我就真的开始拍摄了。我想我知道我认为这不会很好。但是，我知道那很好。现在人们谈论它是因为它还不够好。

在成长过程中，你说你完全讨厌《迷失》这部剧。这难道不是您最终所希望的吗？

VS: 我不喜欢这个概念。

所以有很多你不知道的事情，所以我认为对我来说，有了这些想法，如果你甚至不明白它来自这个不存在的世界，我们可能永远不会在一起。

太奇怪了。这竟然是同时发生的？

对比是的。事情发生的时间基本相同 时间。

甚至没有人看过演出或看过电影/从电影中出来，但是.....

VS: 如果我要假装我绝对不是你并且必须经历这些事情，我想我不会接受它。我没想到它会持续那么久。

现在《戴德伍德》总是会发生一些事情，你不知道下次它会在哪里结束，但我认为现在有些时候我们必须继续战斗，即使《迷失》在心态和形式上非常一致。

VS: 我很高兴我们确实克服了困难，因为我们应该明白两者之间存在直接联系。但几乎有一种感觉，我们不是在同一天出现的，我们知道我们在同样的地方工作，但有很多东西我们不知道。其中一些，我们需要处理。我们还必须接受这种语言，我们从他们那里学到了很多，我们做他们所做的事情，因为我们想要

K.5.2 MD4-M无条件样本2: 1024个令牌

这些团体允许休闲车使用在争取许可证期间保持开放的三条道路。“获得许可证的目的是确保我们与 NPS 合作，修建道路和休息区。我们不仅仅是吓唬那些乱搞的家伙。”在金县委员会正在进行的工作人员会议上，社区建设城市自行车设施的计划得到了推进。

步道将在 Greenview 5 南边结束。

这两条小路可以连接从 Market 到 14 的区域，拉近社区距离，而不是继续通过步行道和自行车道到达 MBTA 校园。

“这个项目将为金县提供一条无车道路，”安德鲁·威德说。在过去的几个月里，该政策已被搁置，但仍有居民持怀疑态度。

“我已经解决了社区提出的一些担忧。他们表达了一些担忧。我认为从一个角度来看这不是很合理。

transportation standpoint."

The trail had been set up to meet on for more than a year when the council approved funding for a different proposal.

Mayor Muriel Bowser said after meetings with Commissioner Bushell on Thursday that the new plan will be on board in December.

"There's enough of a finish for this project to roll out on time, and we're going to get it done," Bowser said.

For the public, the campaign appears over.

"There was one meeting that I feel like I lost at last night's meeting," said Shelley Potts, a local resident.

Local resident Joel Grimy, who lives on Uman Road, met residents there as well.

And in other groups that rode through Mayor assistant Stacey Land and even her son held fliers saying to look for light sign, and also met with Bowser's son, Deion Bowser, about a future plan to also have a dog park on the transit corridor.

Advocates at Brickley's event, many one waited at least 11 minutes in during the start of the public meeting, said they expect at least another month from the Board of Commissioners, even after a public hearing on Nov. 13.

"We've been trying to be a talkative board where we are meeting in advance, being respectful of folks," Bowser said.

He considered that the proposal for the section of trail between the Greenview 5 and 3 "has to move on a schedule. We have other historic preservation projects that would take over that."

But Chad Routledge, a local advocate of the project, spoke out against the mayor's plan.

"The mayor has sent a new meeting to the public using the same route that resulted from the loud criticism and onslaught of complaints from the community committee back during the public hearing," Routledge said.

The BDC doesn't have a particular plan-turns around for the end of the planned path, and says "nothing practical can happen right now." But, she said the agency still "looking to make investments in facilities along the route."

And still there is another part of the trail that might be just as much a wish for the dogs, as cars: the district wants to go west foot a couple blocks south, to make the trail safer for dogs.

"I feel that the accessibility of the trail is pretty important. I think the education of the trail, and the uses along different routes are very important pieces of a balanced outcome," said Bushell.

Trams coming off Route 1

K.6 Conditional generation from MD4 trained on OpenWebText

We share conditionally generated text samples by MD4 Medium in Fig. 10 and observe that slow unmasking near $t = 1$, enabled by the cosine schedule, tends to help produce more consist and meaningful samples than uniform unmasking counterpart.

交通运输的角度。”

当理事会批准为另一项提案提供资金时，这条路线已准备好持续一年多。

市长 Muriel Bowser 在周四与 Bushell 专员会面后表示，新计划将于 12 月实施。

鲍泽说：“这个项目已经完成了足够的工作，可以按时推出，我们将完成它。”

对于公众来说，这场运动似乎已经结束了。

“有一次会议我感觉自己在昨晚的会议上输了，”当地居民雪莉·波茨 (Shelley Potts) 说。

居住在乌曼路的当地居民乔尔·格里米也见到了那里的居民。

在其他骑行队伍中，市长助理史黛西·兰德 (Stacey Land) 甚至她的儿子都拿着传单，要求寻找灯光标志，并会见了鲍泽 (Bowser) 的儿子迪昂·鲍泽 (Deion Bowser)，讨论了未来在交通走廊上设立狗公园的计划。

布里克利活动的支持者中，许多人在公开会议开始时至少等待了 11 分钟，他们表示，即使在 11 月 13 日举行公开听证会之后，他们预计委员会至少还要等待一个月。

鲍泽说：“我们一直在努力成为一个健谈的董事会，我们提前开会，尊重人们。”

他认为 Greenview 5 和 3 之间的步道提案“必须按计划进行。我们还有其他历史保护项目可以接替这个项目。”

但该项目的当地倡导者查德·劳特利奇 (Chad Routledge) 公开反对市长的计划。

劳特利奇说：“市长已经向公众发送了一次新的会议，使用的路线与公众听证会期间社区委员会的大声批评和猛烈投诉相同。”

BDC 没有具体的计划，并表示“目前不会发生任何实际情况”。但是，她表示该机构仍在“寻求对沿线设施进行投资”。

这条小路的另一部分可能和汽车一样是狗的愿望：该地区希望向西走，向南走几个街区，以使这条小路对狗来说更安全。

“我觉得这条步道的可达性非常重要。我认为这条步道的教育以及不同路线的使用是平衡结果的非常重要的部分，”布谢尔说。

有轨电车从 1 号公路驶出

K.6 在 OpenWebText 上训练的 MD4 的条件生成

我们在图 10 中分享了由 MD4 Medium 有条件生成的文本样本，并观察到，由余弦时间表启用的 $t = 1$ 附近的缓慢解锁往往有助于生成比均匀解锁对应物更一致和更有意义的样本。

MD4-M linear schedule	skydiving is a fun sport, but it's pretty risky. You're getting is one to get last one for the season if something goes wrong and it can happen you know, we know about season, especially in Skydiving, but anybody that wins this year	Then some time on Saturday you should pretty much say: "This is what I am going to be doing right now." It's just the simplest thing—that is why I always shampoo twice a day and shower three times a day.
MD4-M linear schedule + cosine grid	skydiving is a fun sport. It is pure endurance and excitement for many people in the at many times we could have won or lost. So if something goes wrong and we can't make it, we have to help another friend as if we have come to our zoo	First, just keep your skin as healthy as possible,you never want to be oily,that is why I always shampoo twice a day and shower three times a day.
MD4-M cosine schedule	skydiving is a fun sport, but it's extremely risky. You can have so many injuries one time and then one next time. There are so many ways you can hurt, so, neuroconcussions, especially from Skydiving, are continuing to rise every year	Though antibacterial products are a poison, the skin needs a chemical solution that protects it from bacteria and spots that form within it —that is why I always shampoo twice a day and shower three times a day.

Figure 10: Conditionally generated text samples from MD4-M. Top: MD4-M trained with the linear schedule, sampled with a uniform grid; Middle: MD4-M trained with the linear schedule, sampled with the cosine grid; Bottom: MD4-M trained with the cosine schedule, sampled with a uniform grid. Context text shown in blue, model-generated text in black.

K.7 Effect of discretization on zero-shot perplexity

We carried out ablation study on the effect of discretization on zero-shot perplexity. Results are included in Tab. 7. Note that this is an inference ablation with the same trained model (MD4-S trained with the continuou-time objective).

Table 7: Effect of discretization on zero-shot perplexity.

Size	Timesteps	LAMBADA	WikiText2	PTB	WikiText103	IBW
Small	T = 100	≤ 49.8	≤ 36.1	≤ 105.2	≤ 36.1	≤ 70.3
	T = 1000	≤ 48.5	≤ 35.0	≤ 102.5	≤ 35.0	≤ 68.4
	T = 10000	≤ 48.4	≤ 34.9	≤ 102.4	≤ 34.9	≤ 68.2
	T = ∞ (continuous)	≤ 48.4	≤ 34.9	≤ 102.3	≤ 35.9	≤ 68.1

MD4-M linear schedule	skydiving is a fun sport, but it's pretty risky. You're getting is one to get last one for the season if something goes wrong and it can happen you know, we know about season, especially in Skydiving, but anybody that wins this year	Then some time on Saturday you should pretty much say: "This is what I am going to be doing right now." It's just the simplest thing—that is why I always shampoo twice a day and shower three times a day.
MD4-M linear schedule + cosine grid	skydiving is a fun sport. It is pure endurance and excitement for many people in the at many times we could have won or lost. So if something goes wrong and we can't make it, we have to help another friend as if we have come to our zoo	First, just keep your skin as healthy as possible,you never want to be oily,that is why I always shampoo twice a day and shower three times a day.
MD4-M cosine schedule	skydiving is a fun sport, but it's extremely risky. You can have so many injuries one time and then one next time. There are so many ways you can hurt, so, neuroconcussions, especially from Skydiving, are continuing to rise every year	Though antibacterial products are a poison, the skin needs a chemical solution that protects it from bacteria and spots that form within it —that is why I always shampoo twice a day and shower three times a day.

图 10: 从 MD4-M 有条件生成的文本样本。上图: MD4-M 使用线性计划进行训练, 使用统一网格进行采样; 中: MD4-M 使用线性计划进行训练, 使用余弦网格进行采样; 底部: MD4-M 使用余弦计划进行训练, 使用统一网格进行采样。上下文文本以蓝色显示, 模型生成的文本以黑色显示。

K.7 离散化对零样本困惑度的影响

我们对离散化对零样本困惑度的影响进行了消融研究。结果包含在选项卡中。7. 请注意, 这是使用相同训练模型的推理消融 (使用连续时间目标训练的 MD4-S)。

表 7: 离散化对零样本困惑度的影响。

Size	Timesteps	LAMBADA	WikiText2	PTB	WikiText103	IBW
Small	T = 100	≤ 49.8	≤ 36.1	≤ 105.2	≤ 36.1	≤ 70.3
	T = 1000	≤ 48.5	≤ 35.0	≤ 102.5	≤ 35.0	≤ 68.4
	T = 10000	≤ 48.4	≤ 34.9	≤ 102.4	≤ 34.9	≤ 68.2
	T = ∞ (continuous)	≤ 48.4	≤ 34.9	≤ 102.3	≤ 35.9	≤ 68.1